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CHAPTER

THREE

Electromagnetic Induction

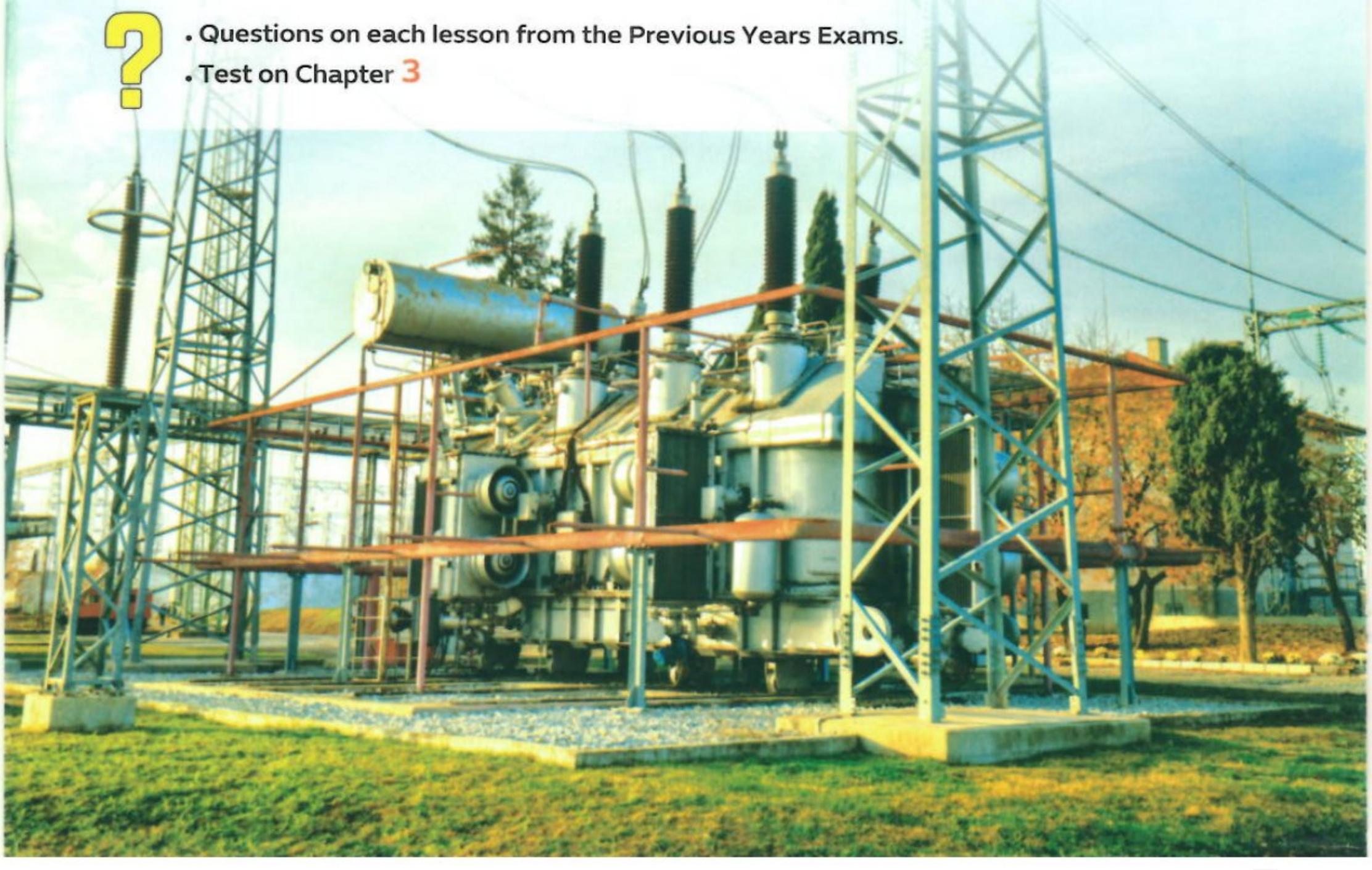
Lesson 1
- Faraday's Law.
- The Induced Electromotive Force in a Straight Wire.

Lesson 2
- Mutual Induction between Two Coils.
- Self Induction of the Coil.

Lesson 3
- Electric Generator (Dynamo).

Lesson 4
- The Electric Transformer.
- The Electric Motor.

- Questions on each lesson from the Previous Years Exams.
- Test on Chapter 3



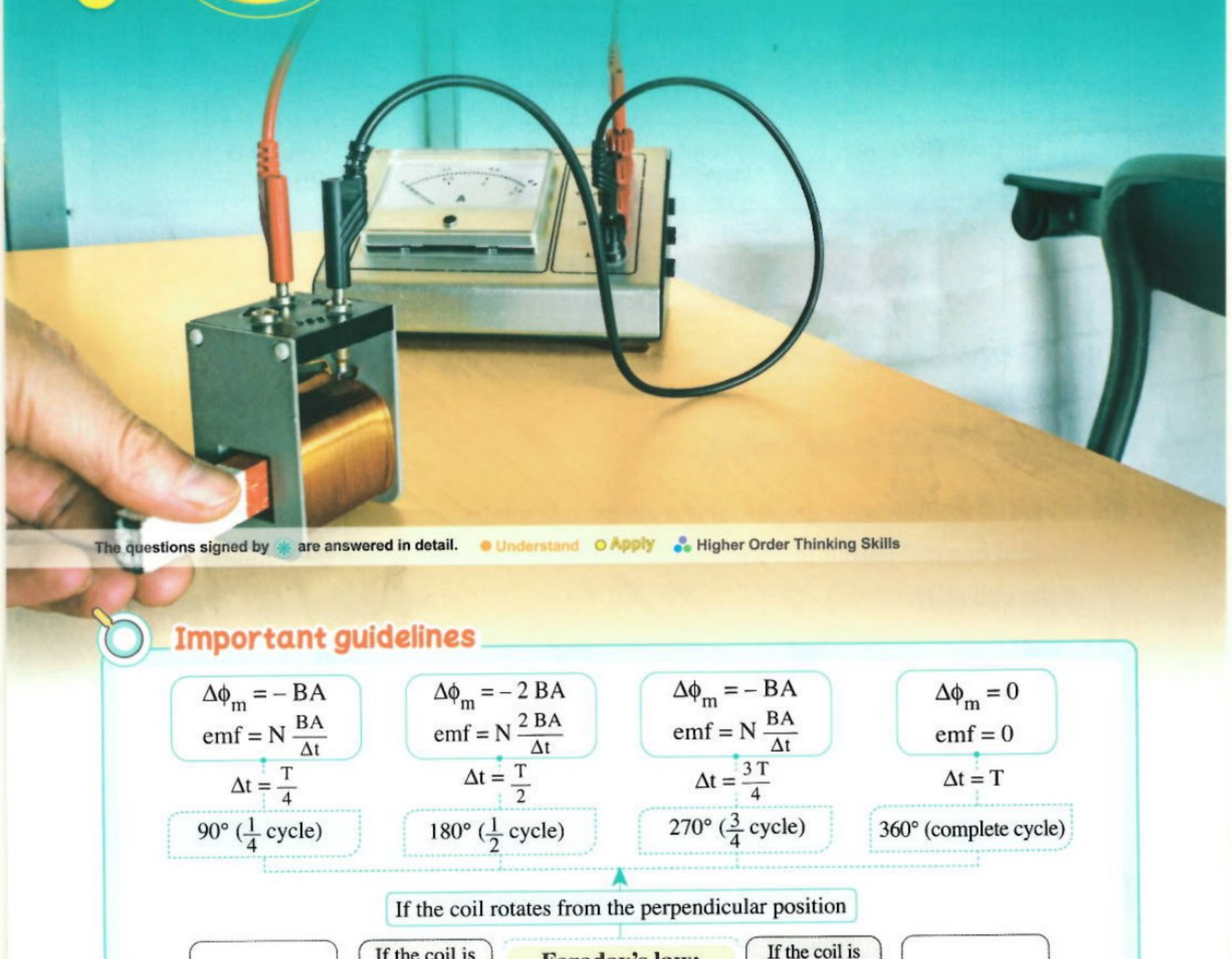
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Chapter 3

Questions on Lesson One

- Faraday's Law.
- The Induced Electromotive Force in a Straight Wire.

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

If the coil rotates from the perpendicular position

$\Delta t = \frac{T}{4}$	$\Delta t = \frac{T}{2}$	$\Delta t = \frac{3T}{4}$	$\Delta t = T$
$\Delta\phi_m = -BA$, $emf = N\frac{BA}{\Delta t}$	$\Delta\phi_m = -2BA$, $emf = N\frac{2BA}{\Delta t}$	$\Delta\phi_m = -BA$, $emf = N\frac{BA}{\Delta t}$	$\Delta\phi_m = 0$, $emf = 0$
90° ($\frac{1}{4}$ cycle)	180° ($\frac{1}{2}$ cycle)	270° ($\frac{3}{4}$ cycle)	360° (complete cycle)

Faraday's law: $emf = -N\frac{\Delta\phi_m}{\Delta t}$ **(Average induced emf)**

- If the coil is perpendicular to the field and the area changes: $\Delta\phi_m = B\Delta A$, $emf = -NB\frac{\Delta A}{\Delta t}$
- If the coil is perpendicular to the field and the magnetic flux density changes: $\Delta\phi_m = A\Delta B$, $emf = -NA\frac{\Delta B}{\Delta t}$

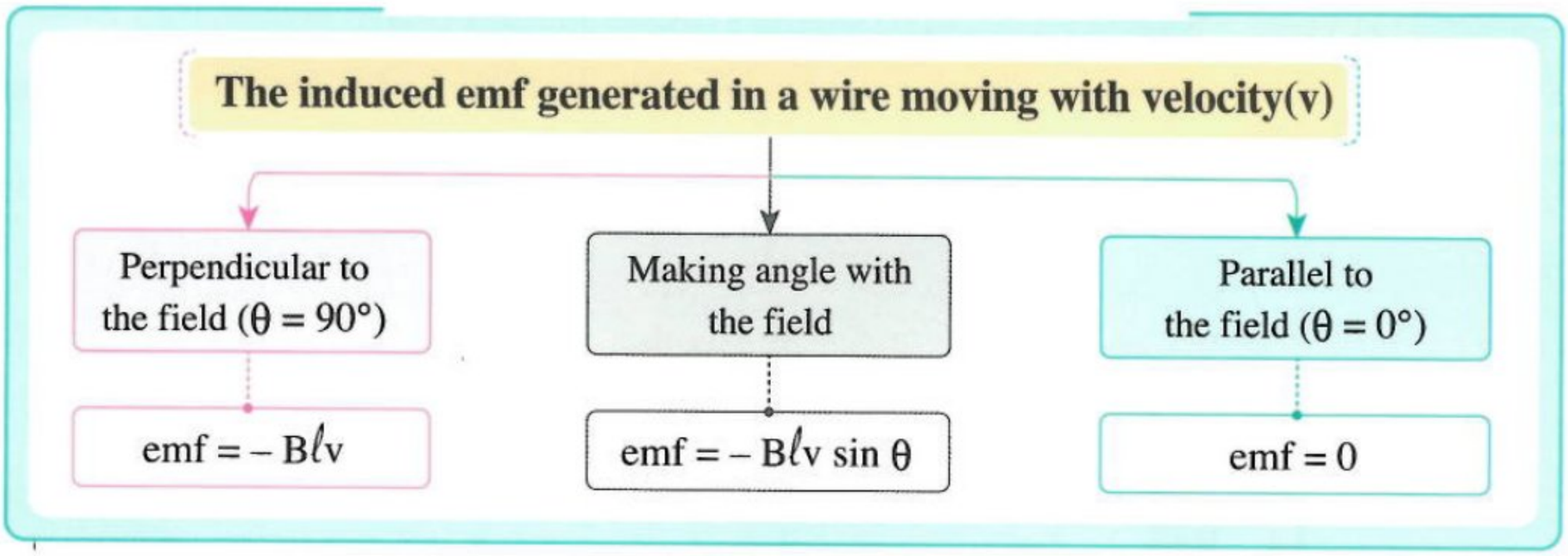
If the coil rotates from the parallel position

$\Delta t = \frac{T}{4}$	$\Delta t = \frac{T}{2}$	$\Delta t = \frac{3T}{4}$	$\Delta t = T$
90° ($\frac{1}{4}$ cycle)	180° ($\frac{1}{2}$ cycle)	270° ($\frac{3}{4}$ cycle)	360° (complete cycle)
$\Delta\phi_m = BA$, $emf = -N\frac{BA}{\Delta t}$	$\Delta\phi_m = 0$, $emf = 0$	$\Delta\phi_m = -BA$, $emf = N\frac{BA}{\Delta t}$	$\Delta\phi_m = 0$, $emf = 0$

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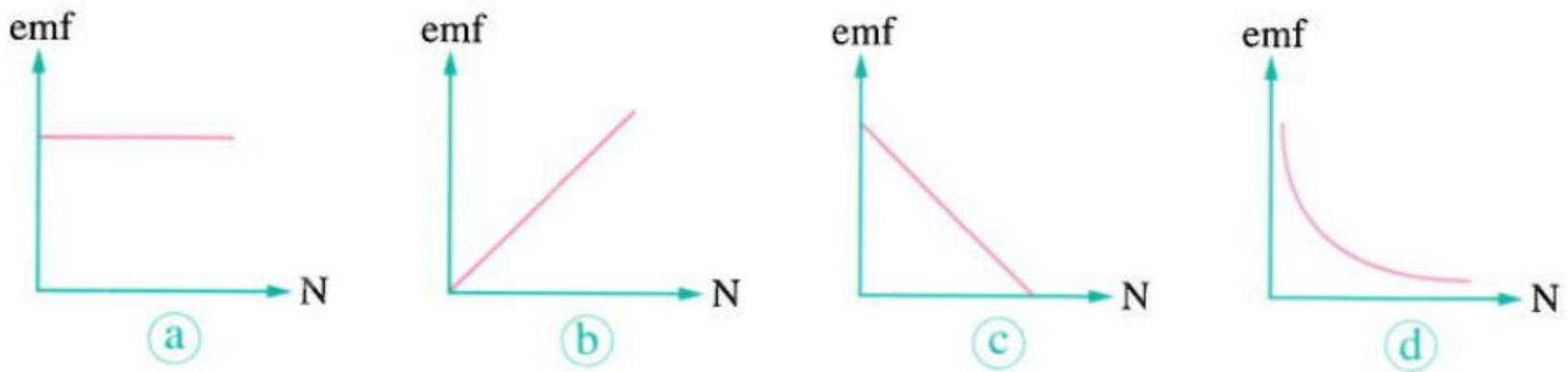


First Multiple choice questions

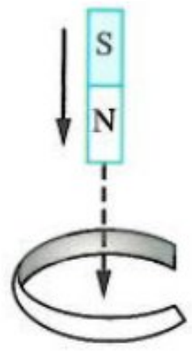
Use the following constant if it is needed: $\mu_{air} = 4\pi \times 10^{-7}$ Wb/A.m

Faraday's law and Lenz's rule

1 Which of the following graphs represents the relation between the magnitude of emf induced in a coil placed in a uniformly changing magnetic field and the number of turns of the coil (N)?



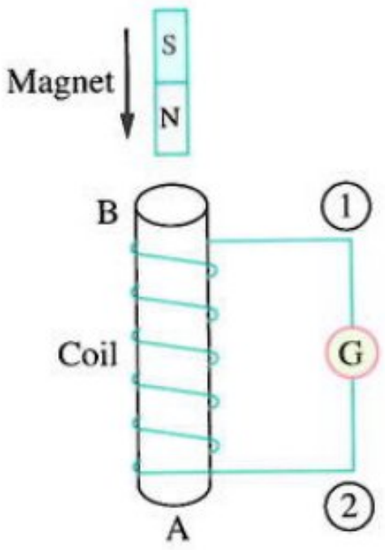
2 In the opposite figure, a magnet falls through an open ring of aluminum that is placed horizontally, so which of the following choices describes what happens between the magnet and the ring when the magnet approaches and moves away from the ring?



	When the magnet approaches the ring	When the magnet gets away from the ring
a	Repulsion force gets generated	Attraction force gets generated
b	Attraction force gets generated	Repulsion force gets generated
c	Repulsion force gets generated	Repulsion force gets generated
d	No magnetic force is generated	No magnetic force is generated

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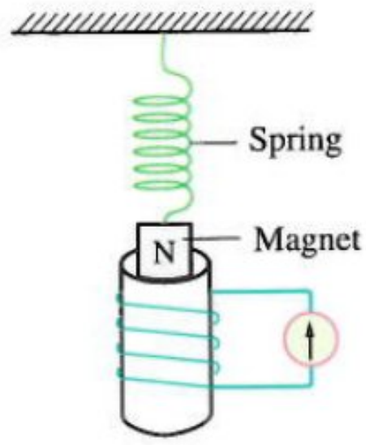
3 A bar magnet is dropped into a solenoid connected to a galvanometer as shown in the opposite figure. Which choice of the following is correct?



	Current direction through the galvanometer	The magnetic pole formed at A
a	From 1 to 2	North
b	From 1 to 2	South
c	From 2 to 1	North
d	From 2 to 1	South

4 In the opposite figure:

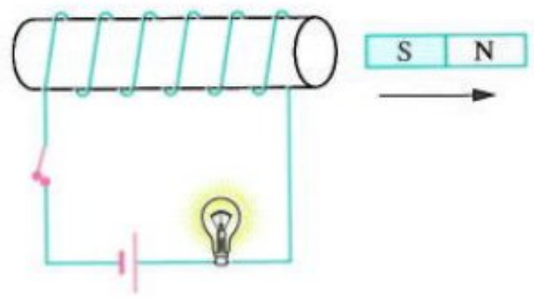
A magnet is suspended to a spring and it is free to move in and out of a coil that is connected to a galvanometer having its zero reading at the middle of its scale, so when the magnet vibrates up and down, the galvanometer's pointer ..... .



(a) swings right and left
(b) stops at left
(c) stops at right
(d) stops at zero

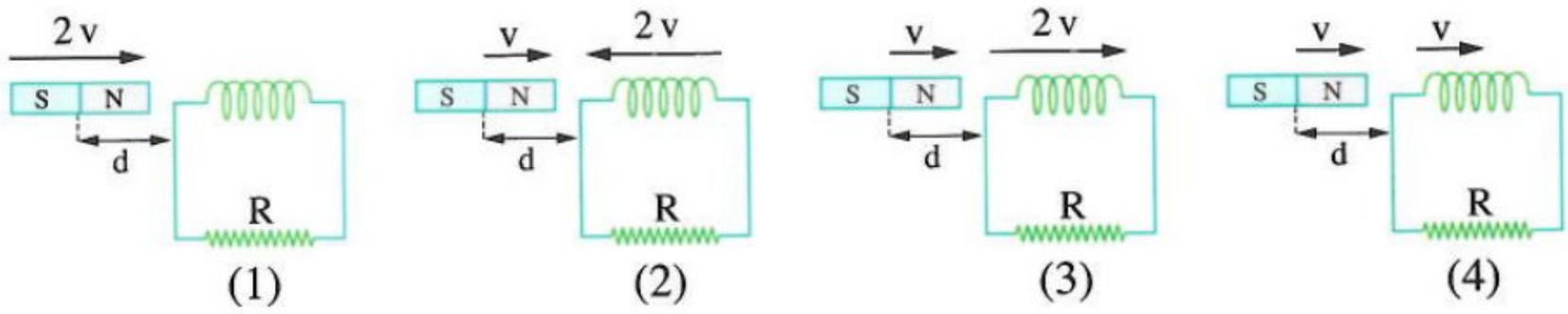
5 In the opposite figure:

During moving the magnet in the shown direction, the brightness of the bulb ..... .



(a) decreases momentarily
(b) increases momentarily
(c) vanishes
(d) remains unchanged

6 A student has carried out many trials to study the effect of direction and magnitude of the relative speed between a magnet and a solenoid on the value of the induced emf in the solenoid in Faraday's experiment:



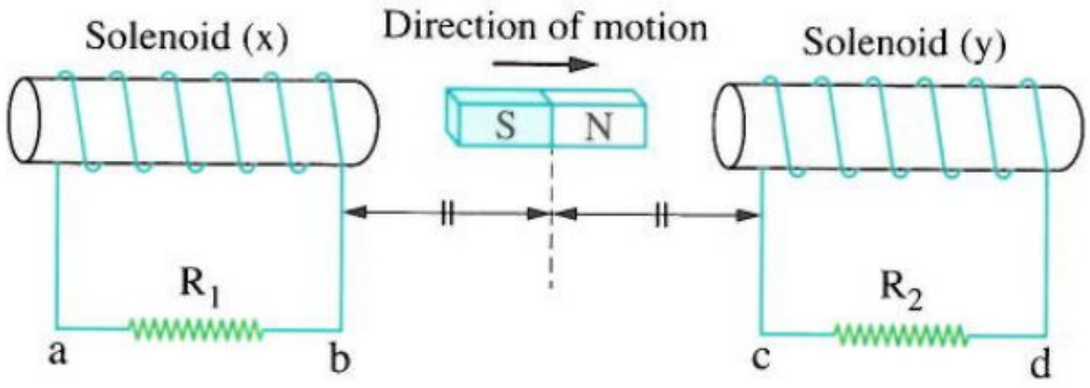
If the magnet moved in each trial with a uniform speed through the distance d, so the correct descending order for the values of the average induced emf in the solenoid in those trials is .....

(a) (3) ⟶ (2) ⟶ (4) ⟶ (1)
(b) (2) ⟶ (1) ⟶ (3) ⟶ (4)
(c) (1) ⟶ (3) ⟶ (2) ⟶ (4)
(d) (2) ⟶ (3) ⟶ (4) ⟶ (1)

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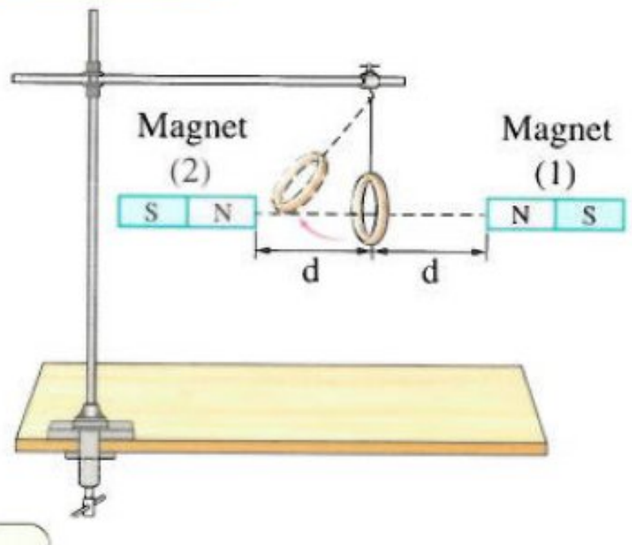
7 Two similar adjacent solenoids (x), (y) where the terminals of each are connected to a resistance R_1, R_2 respectively. A bar magnet is placed at the mid-distance between the coils on the extension of their axes as in the following figure.



When the magnet moves on the extension of the axes of the solenoids approaching the solenoid (y), an induced current passes through the resistors which its direction in:

	Resistor R_1	Resistor R_2
a	a ⟶ b	c ⟶ d
b	a ⟶ b	d ⟶ c
c	b ⟶ a	c ⟶ d
d	b ⟶ a	d ⟶ c

8 A metal ring is hung by a thread where its other end is fixed. When two similar magnets (1), (2) are moved with a uniform velocity (v) along the axis of the ring, the ring is displaced as in the opposite figure, so ..... .

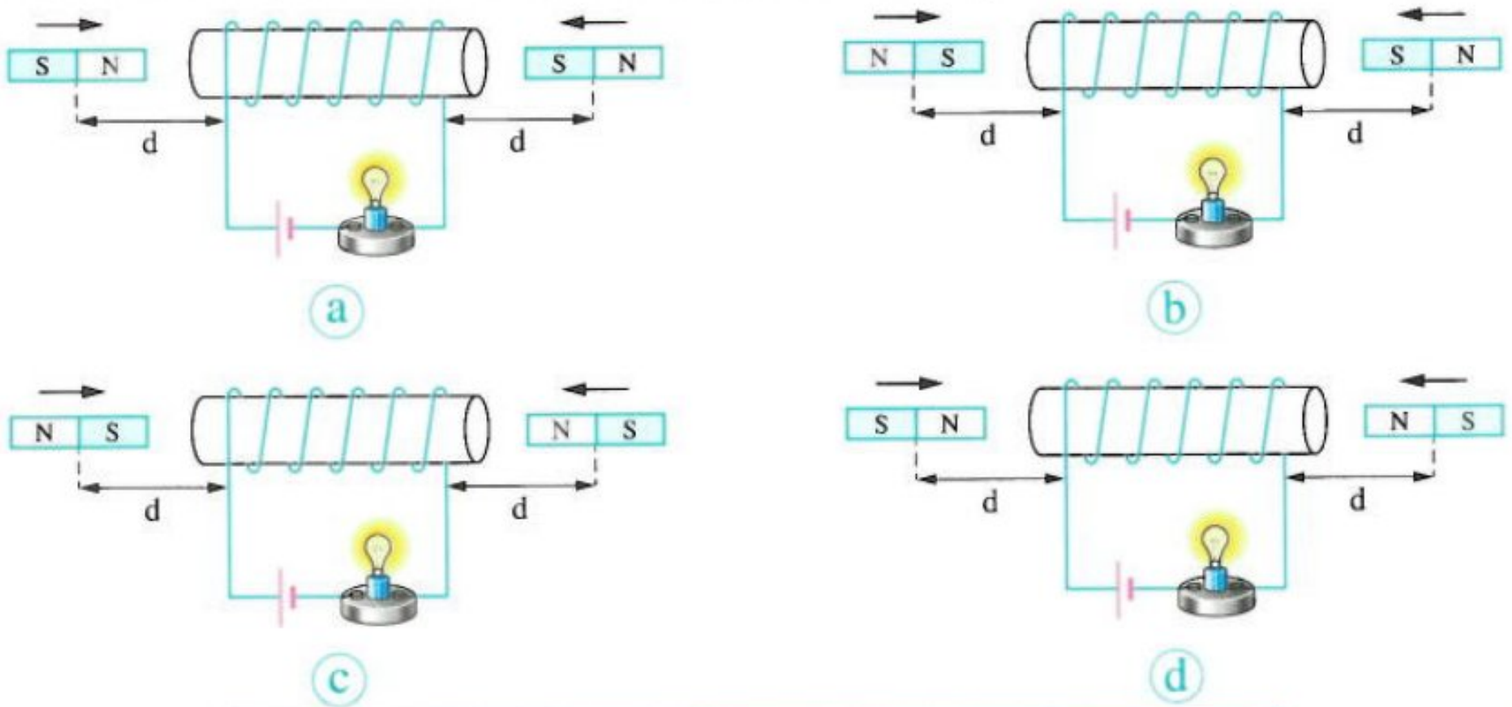


	Magnet (1)	Magnet (2)
a	gets close to the ring	gets close to the ring
b	gets close to the ring	moves away from the ring
c	moves away from the ring	gets close to the ring
d	moves away from the ring	moves away from the ring

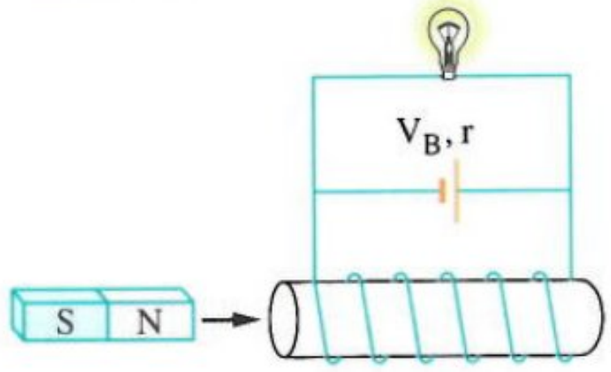
9 Two bar magnets are moving with the same uniform velocity towards a solenoid that is connected to an electric cell and a small electric bulb as in the following figures. In which of the following figures does the light intensity of the bulb increase during cutting the distance d?

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(Where: The resultant of the induced emf in the coil in all figures is less than the emf of the battery)

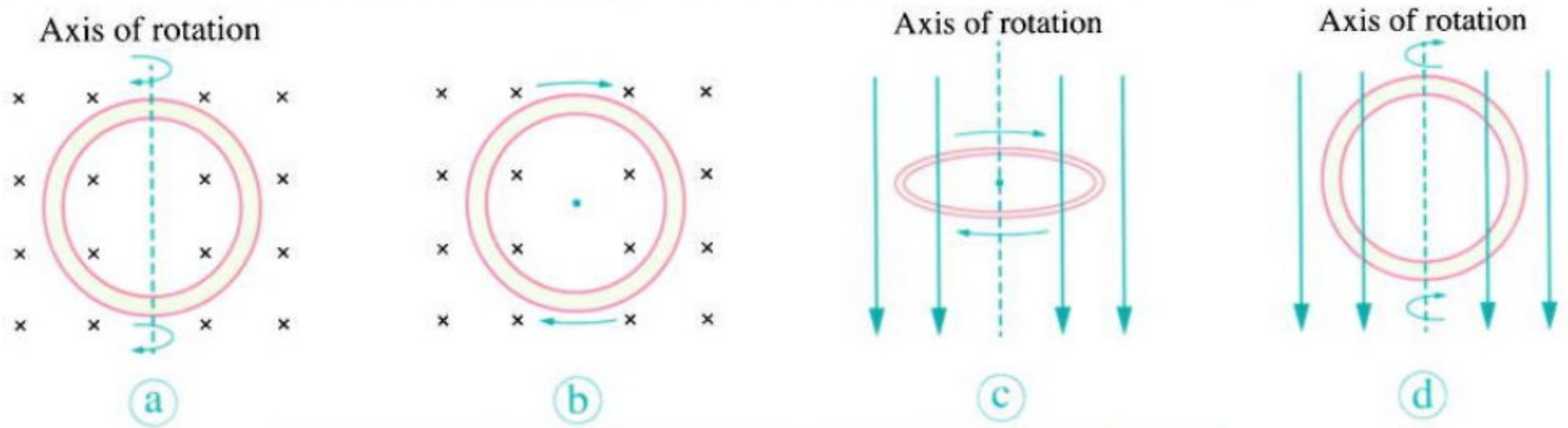


10 A battery of emf (V_B) and its internal resistance (r) is connected to an electric bulb and a solenoid of resistance R, so when a magnet gets approached to the coil with a uniform speed as in the opposite figure, an induced emf of value less than (V_B) is generated in the coil. What happens to bulb luminosity during the movement of the magnet?

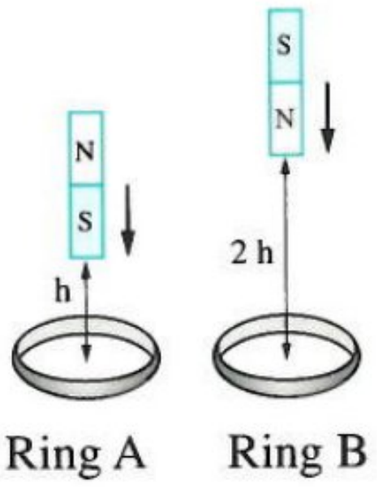


(a) Increases. (b) Decreases.
(c) Unchanged. (d) The answer is undeterminable.

11 The proper motion of a metallic ring in a magnetic field to produce an induced emf based on the rules of the electromagnetic induction is represented in figure ..... .



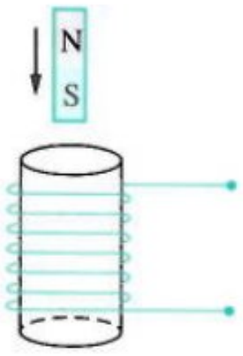
12 The opposite figure represents two identical bar magnets which are falling freely from two different heights h, 2 h along the axes of two identical metallic rings A, B respectively, so what is the statement that describes the induced current through the two rings at the moment when each of them reaches the ring's plane?

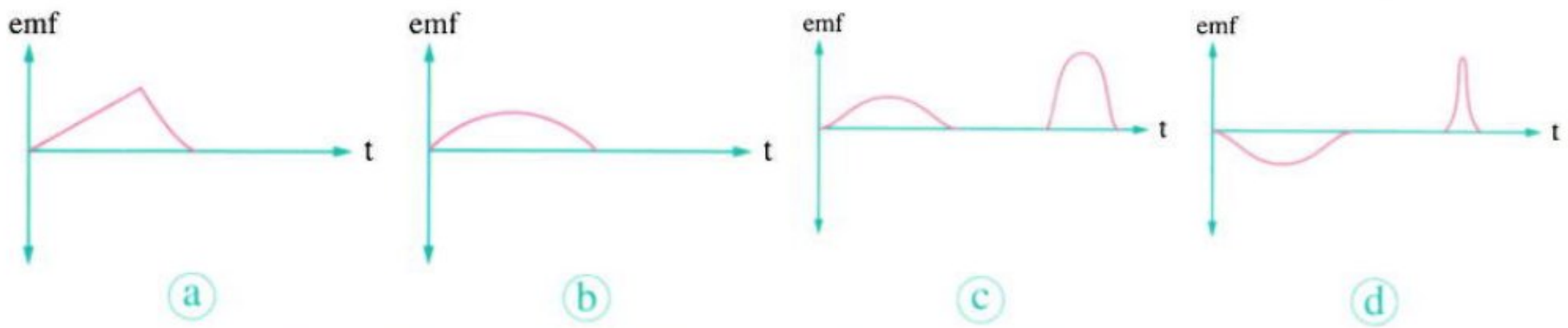


(a) The induced current intensity in ring A is greater.
(b) The induced current intensity in ring B is greater.
(c) The induced current intensities in the two rings are equal.
(d) The induced currents in the two rings are in the same direction.

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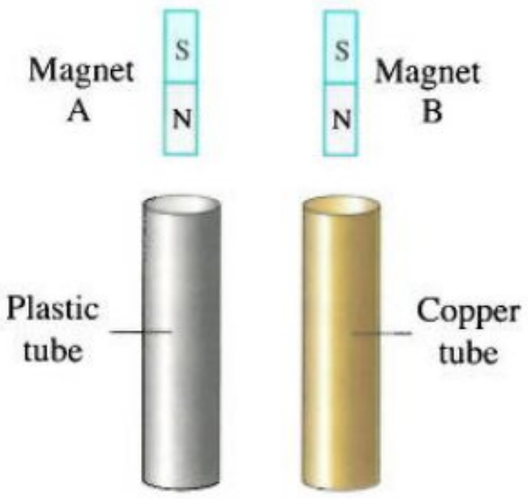
13 Which of the following graphs represents the change of the induced emf across the solenoid versus time (t) during the free fall of the magnet inside the solenoid until it gets out from the other side?





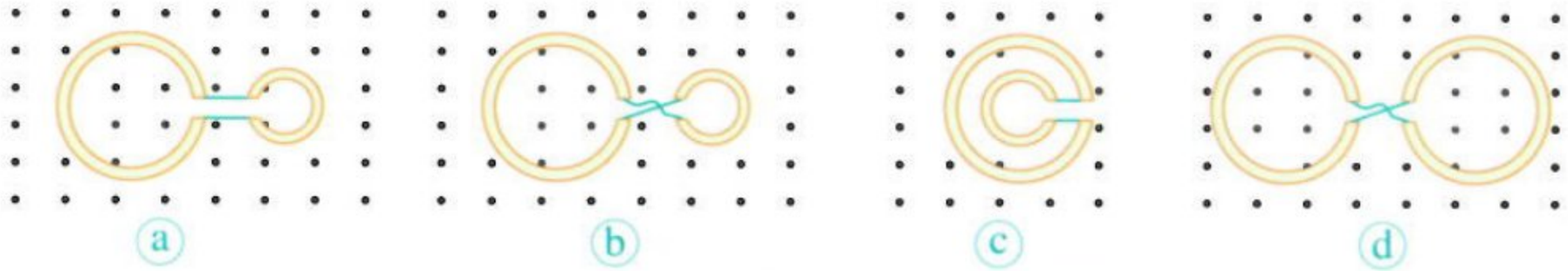
a b c d

14 The opposite figure shows two small similar magnets A, B. They have fallen at the same instant from the same level to pass through two vertical tubes having the same dimensions without touching their walls, one made of plastic while the other is made of copper. If the magnet (A) spent time t to pass through the tube, so the time spent by magnet (B) to pass through the tube is ..... .



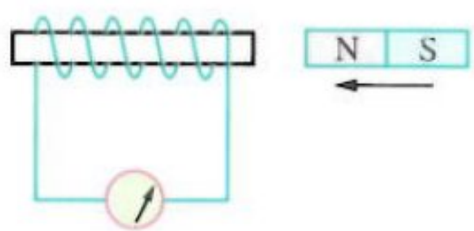
a greater than (t)
b less than (t)
c equals (t)
d the answer is indeterminable

15 Every figure of the following figures shows two incomplete metal rings in one plane which are connected together with two wires and placed in a uniform magnetic flux that is perpendicular to their planes. In which of the following figures, the most intense current passes in the rings when the magnetic flux density acting on them increases with a uniform rate?



a b c d

16 A solenoid consisting of 20 turns is connected to a galvanometer as shown in the opposite figure and when moving a magnet towards the solenoid, the magnetic flux linked with the solenoid gets increased by 0.2 Wb through 0.02 s, so the magnitude of the average induced emf in the circuit equals ..... .



a 0.2 V
b 1 V
c 20 V
d 200 V

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17 Two rings x, y of areas 2 A, A respectively are placed perpendicular to a magnetic field whose density is changing uniformly with time, so the ratio between the induced emf in each of the two rings $\left(\frac{emf_x}{emf_y}\right)$ within the same time interval is ..... .

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{1}$ (d) $\frac{4}{1}$

18 A metal wire of uniform cross-sectional area is divided into two parts to make two rings a, b of radii r, 2 r respectively. Each of the two rings is placed perpendicular to a magnetic field. If the magnetic flux density passing through them is changed in the same rate, the ratio between:

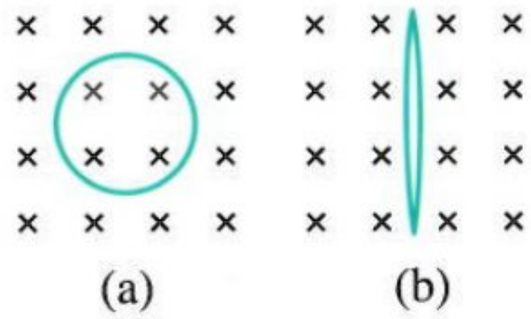
(i) Values of the induced emf in the two rings $\left(\frac{(emf)_a}{(emf)_b}\right)$ equals ..... .

(a) $\frac{1}{4}$ (b) $\frac{4}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{1}$

(ii) Intensities of the induced current in the two rings $\left(\frac{I_a}{I_b}\right)$ equals ..... .

(a) $\frac{1}{4}$ (b) $\frac{2}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{1}$

19 * A one turn coil of radius 0.12 m that is made of an elastic conducting wire is placed perpendicular to a uniform magnetic field of density 0.15 T as in figure (a). If the turn is pressed from its sides till its area becomes 3×10^{-3} m^2 as that in figure (b) through a time of 0.2 s, the average induced emf generated in the coil within this time interval equals ..... .



(a) (b)

(a) 0 (b) 1.12×10^{-3} V (c) 2.14×10^{-3} V (d) 31.7×10^{-3} V

20 A circular ring of radius 7 cm is placed perpendicular to a magnetic flux of density 0.1 T. If the ring is bent from its center through a time interval 2 s such that plane of one of the halves becomes parallel to the field, so magnitude of the average induced emf in the ring during that interval equals ..... .

(a) 0.1925 mV (b) 0.385 mV (c) 0.5775 mV (d) 0.77 mV

21 The following cases represent the change in flux through a circular coil of N turns, so which of them causes the greatest induced emf in the coil?

(a) A change of flux from 2 Wb to 2.1 Wb within 10^{-4} s.

(b) A change of flux from 0.2 Wb to 4 Wb within 0.2 s.

(c) A change of flux from 1 Wb to 20 Wb within 10 s.

(d) A change of flux from 0.01 Wb to 0.02 Wb within 0.2 s.

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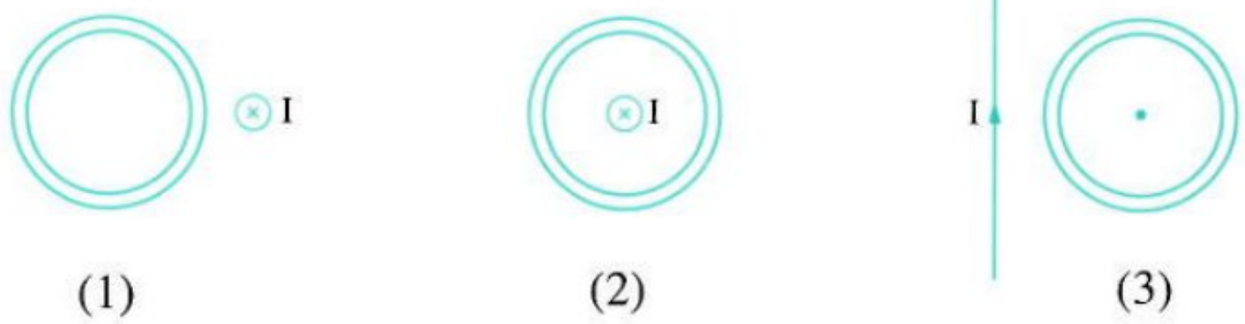
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22 A coil of 100 turns is penetrated by a magnetic flux that equals 0.02 Wb. If this magnetic flux is doubled within 0.01 s, the average induced emf across the terminals of the coil equals ..... .

(a) – 400 V (b) – 350 V (c) – 275 V (d) – 200 V

23 A straight wire that carries a current of intensity I is placed in many positions with respect to a vertical metal ring as in the following figures:



(1) (2) (3)

If the intensity of the electric current passing in the wire is increasing in a uniform rate, in which of the three cases an induced current is generated through the ring?

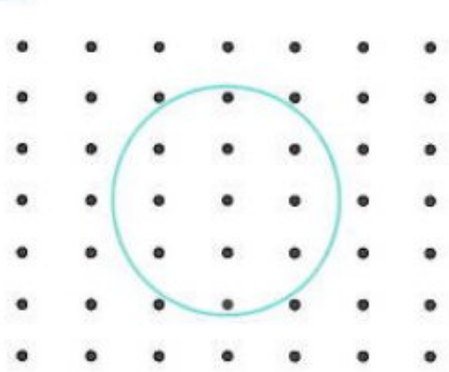
(a) Case (1) only. (b) Case (2) only. (c) Case (3) only. (d) Cases (1), (3) only.

24 Which of the following choices expresses the correct direction of the current induced in the metallic ring by the effect of the current intensity variation through the wire shown in each figure?



(a) (b) (c) (d)

25 The opposite figure shows a ring placed in the plane of the page and being affected by a uniform magnetic field whose direction is perpendicular to and out of the page, which of the following choices leads to generating an induced current in the clockwise direction through the closed metal ring?



(a) Moving the ring to the right through the magnetic field.

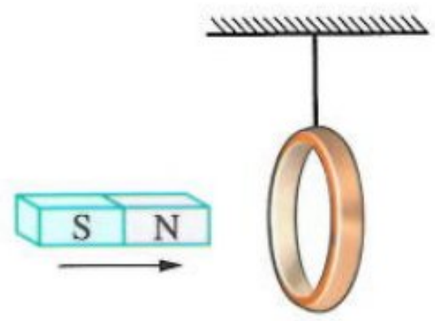
(b) Moving the ring upward through the magnetic field.

(c) Decreasing the density of the acting magnetic flux on the ring.

(d) Increasing the density of the acting magnetic flux on the ring.

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26 A copper ring is suspended freely to a thread. If a magnet is moved towards the ring as in the opposite figure, then ..... .



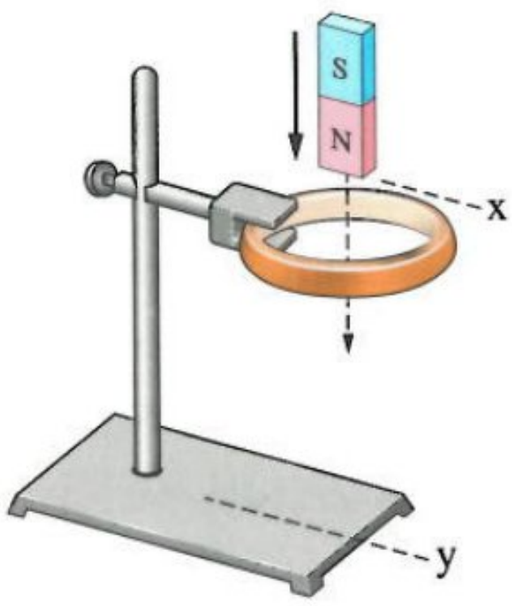
(a) the ring will be attracted to the magnet

(b) the ring's face that faces the magnet becomes a north pole

(c) the ring's face that faces the magnet becomes a south pole

(d) the ring won't be affected by a force because it's made of copper

27 A bar magnet fell from level x to level y through an aluminum ring that is fixed horizontally to a stand as shown in the figure, so if the ring was viewed from above at the time in which the magnet was falling, the induced current in the ring would be ..... .



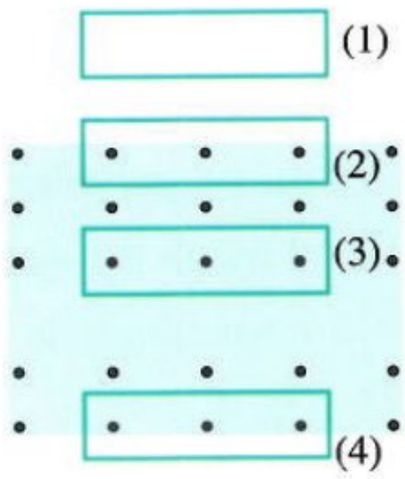
(a) always in clockwise direction

(b) always in anti-clockwise direction

(c) in clockwise direction then reverses its direction

(d) in anti-clockwise direction then reverses its direction

28 A rectangular metal frame is left to fall with its vertical plane to pass through a uniform magnetic field as in the opposite figure, in which of the shown positions in the opposite figure the induced emf in the frame is maximum value?



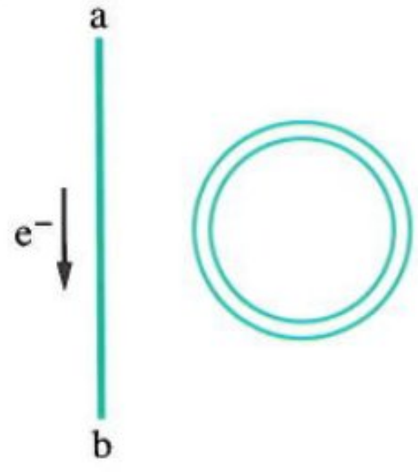
(a) Position (1).

(b) Position (2).

(c) Position (3).

(d) Position (4).

29 A straight wire ab that carries a DC electric current is placed beside and in the same plane of a fixed metallic ring as in the opposite figure, if the direction of the free electrons movement in the wire is from terminal a to terminal b:



(i) In the shown position, ...

(a) an induced emf of constant value and variable direction is generated in the ring

(b) an induced emf that is constant in value and direction is generated in the ring

(c) an induced emf that of variable value and variable direction is generated in the ring

(d) no induced emf is generated in the ring

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(ii) When moving the wire with a uniform velocity towards the center of the ring in a direction parallel to its plane, ...

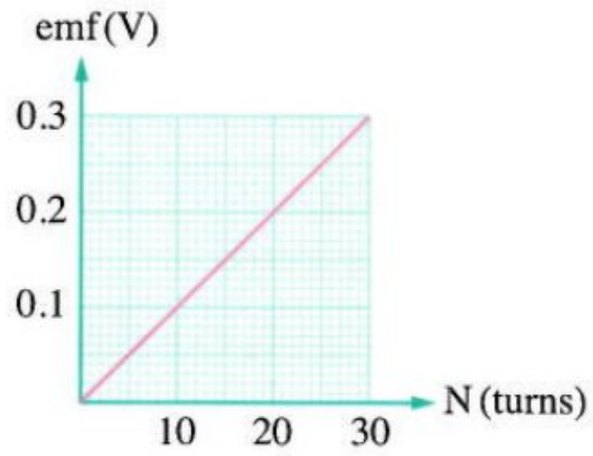
(a) an induced current is generated in the ring in the clockwise direction

(b) an induced current is generated in the ring in the counter-clockwise direction

(c) an induced current of constant intensity and variable direction is generated in the ring

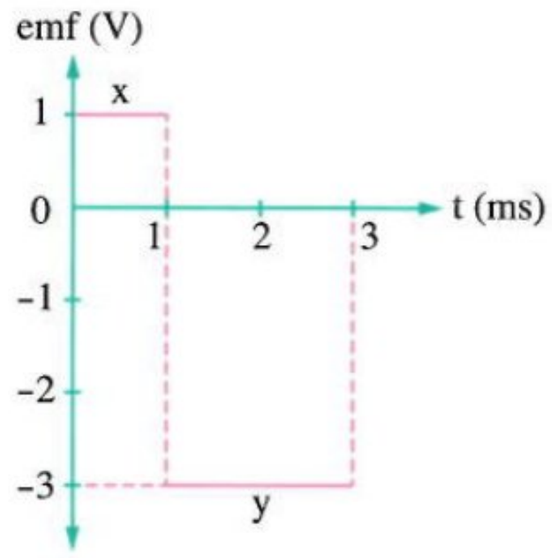
(d) no induced current is generated in the ring

30 The opposite graph represents the relation between the average induced electromotive force (emf) in multiple coils through which a magnetic field is uniformly changing with the same rate and the number of turns of each coil (N), so the time rate of change in the magnetic flux that passes through the coils is ..... .



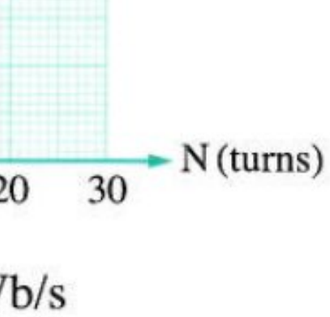
(a) 10 Wb/s (b) 1 Wb/s (c) 0.1 Wb/s (d) 0.01 Wb/s

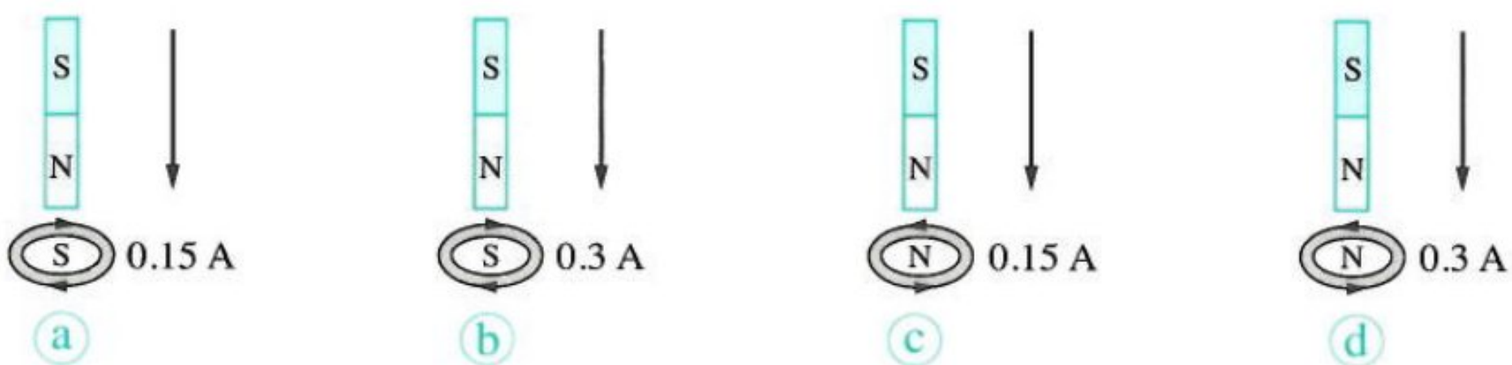
31 The opposite graph represents the relation between the average induced electromotive force (emf) in a metallic ring and the time (t), then the ratio between the magnitudes of the change in the magnetic flux through the ring in the two intervals (x) and (y) respectively is ..... .



(a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{2}{3}$ (d) $\frac{1}{6}$

32 The opposite figure represents a bar magnet moving perpendicular to a horizontal circular coil of 200 turns that has resistance 10 Ω, so the magnetic flux through the coil changes from 2.5×10^{-3} Wb to 8.5×10^{-3} Wb during a time interval of 0.4 s, so the figure that illustrates correctly the intensity and the direction of the induced current in the coil as viewed from above is ..... .





33 ✳ A solenoid coil is formed of 400 turns, the cross-sectional area of each is 4 cm^2. If its axis is parallel to a uniform magnetic field of flux density 0.3 T, then the average induced emf generated in the coil when:

(i) the magnetic flux density increases to 0.5 T within 2 ms is equal to ..... .

(a) – 16 V (b) – 8 V (c) 8 V (d) 16 V

(ii) the magnetic flux density decreases to 0.2 T within 2 ms is equal to ..... .

(a) 6 V (b) 8 V (c) 10 V (d) 12 V

---

34 A coil of 400 turns, where the area of each turn is 50 cm^2, is being penetrated by a perpendicular magnetic field of flux density 0.2 T, then the average value of the induced emf between its terminals if the magnetic flux intercepting the coil has vanished through 0.01 s is equal to ..... .

(a) 20 V (b) 40 V (c) 60 V (d) 80 V

---

35 ✳ A rectangular coil of dimensions 20 cm, 10 cm is formed of 100 turns, its plane is perpendicular to a magnetic field. When this coil is turned $\frac{1}{4}$ cycle about an axis in the same plane of the coil through a time interval of 0.2 s, an induced emf of 0.4 V is generated. So, the magnetic flux density equals ..... .

(a) 0.01 T (b) 0.03 T (c) 0.04 T (d) 0.05 T

---

36 ✳ A rectangular coil of area 75 cm^2 that consists of 150 turns is placed parallel to a magnetic flux of density 0.65 T. Then the average induced electromotive force in the coil if:

(i) the coil is turned to become perpendicular to the field for the first time within 0.02 s is equal to ..... .

(a) – 36.56 V (b) – 40.2 V (c) – 50.3 V (d) 0

(ii) the coil is inverted from the first position in the field within 0.01 s is equal to ..... .

(a) 36.56 V (b) 40.3 V (c) 73.12 V (d) 0

(iii) the coil is turned $\frac{3}{4}$ cycle from the first position to become perpendicular to the field within 0.02 s is equal to ..... .

(a) 0 (b) 36.56 V (c) 50.4 V (d) 73.12 V

(iv) the coil made a complete cycle within 0.02 s equals ..... .

(a) 0 (b) 18.28 V (c) 36.56 V (d) 73.12 V

(v) the magnetic field is vanished within 0.1 s is equal to ..... .

(a) 0 (b) 20 V (c) 40 V (d) 60 V

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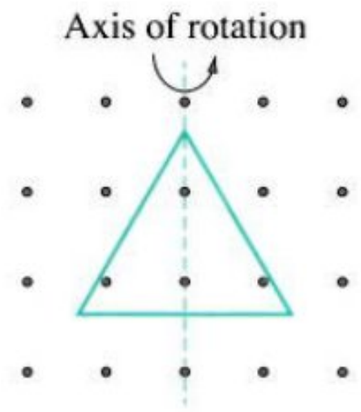
37 ✱ A coil of face area 1.8 cm^2 has 25 turns coiled on a hollow cylinder. The coil is connected to a closed circuit of resistance 3 Ω while being affected by a uniform magnetic field parallel to its axis. If the magnetic flux density has increased from zero to 0.55 tesla within a time of 0.75 s, the induced electric current intensity in the coil equals ..... .

(a) 1.1×10^{-3} A (b) 2.2×10^{-3} A (c) 3.3×10^{-3} A (d) 6.6×10^{-3} A

38 ✱ A coil of 100 turns whose face area is 20 cm^2 is placed perpendicular to a uniform magnetic field of flux density 0.2 T, if the direction of the magnetic flux is reversed through 0.2 s, then the average induced emf will be equal to ..... .

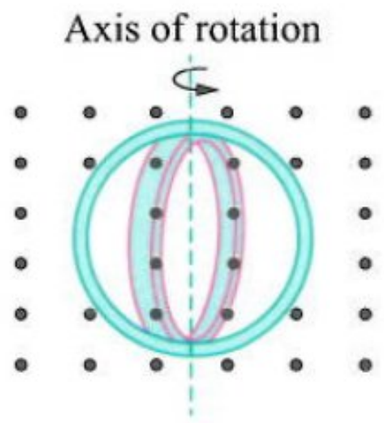
(a) 0.2 V (b) 0.4 V (c) 0.6 V (d) 0.8 V

39 ✱ A triangular frame rotates about an axis as shown in the opposite figure in a rate of 1800 turn/minute in a uniform magnetic field. If the induced emf in the frame through $\frac{1}{60}$ s of its rotation starting from the perpendicular position is 4.8 V, so the maximum flux passing through the frame equals ..... .



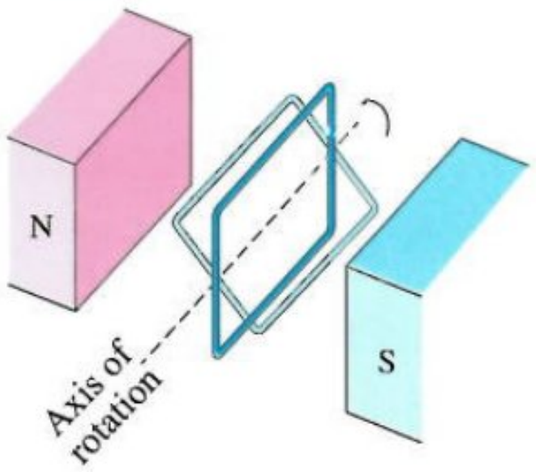
(a) 0.01 Wb (b) 0.02 Wb
(c) 0.03 Wb (d) 0.04 Wb

40 The opposite figure represents a circular metallic ring of cross-sectional area 2×10^{-3} m^2 whose plane is perpendicular to a uniform magnetic field of flux density 0.1 T, if it is rotated by an angle of 45° about an axis perpendicular to the direction of the field within 0.25 s, what is the magnitude of the average induced emf through this period?



(a) 2.34×10^{-4} V (b) 5.75×10^{-4} V (c) 8×10^{-4} V (d) 8.25×10^{-4} V

41 The opposite figure represents a rectangular metal frame of cross-sectional area 0.02 m^2 which is placed perpendicular to a uniform magnetic field of flux density 0.1 T, so if the frame is rotated by an angle θ about an axis perpendicular to the direction of the field within 0.25 s, an average emf of 4 mV is generated, what is the angle (θ) with which the coil's plane is rotated?



(a) 30° (b) 45° (c) 60° (d) 75°

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42 A rectangular coil of area 4 cm^2, resistance 50 Ω and number of turns N is placed perpendicular to a uniform magnetic field of density 0.2 T. When the coil is rotated 180° to be inverted, an electric charge of 1.6×10^{-3} C flows through a cross-section of the coil's wire, then the number of turns of the coil N equals ..... .

(a) 100 turns (b) 200 turns (c) 500 turns (d) 750 turns

43 A metal ring is rotating about an axis in its plane with a uniform rate in a uniform magnetic flux. During the rotation of the ring from the parallel position to the field by an angle 60° an average induced current of intensity I_1 passes through the ring, and during its rotation from the perpendicular position to the field by an angle 60° an average induced current of intensity I_2 passes through the ring, so the ratio $\left(\frac{I_1}{I_2}\right)$ equals ..... .

(a) $\frac{1}{2}$ (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

44 * A circular coil of face area 0.045 m^2, number of turns 150 turns is connected to a closed circuit of total resistance 0.9 Ω. If the plane of that coil is perpendicular to a uniform magnetic flux of density 8×10^{-5} T. Then, the amount of electric charge passing through a cross-section of the coil's wire if the coil has been removed away completely from the field equals ..... .

(a) 6×10^{-4} C (b) 8×10^{-4} C (c) 9×10^{-4} C (d) 9.5×10^{-4} C

45 A static solenoid is connected to the terminals of a galvanometer in which the zero point of its scale is at the middle where a static bar magnet is placed beside it. In figure (1), the bar magnet is moved with a uniform speed (v) towards the static coil, while in figure (2), the coil is moved towards the static bar magnet with the same uniform speed (v), what do you expect for the deflection of the galvanometer's pointer in figure (2) compared to figure (1)?

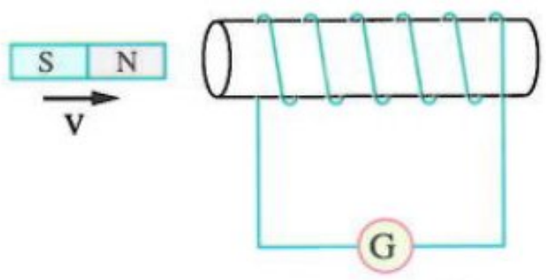


Figure (1)

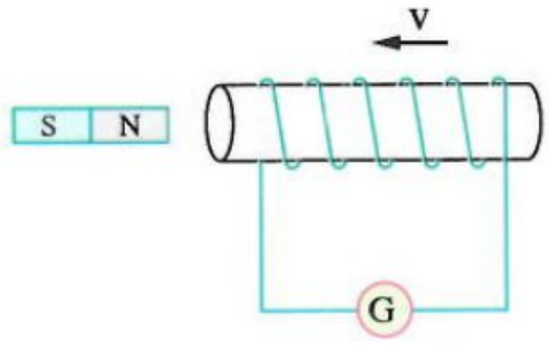


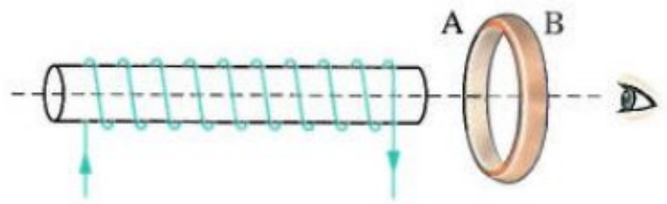
Figure (2)

(a) The pointer doesn't deflect because the magnet is stationary.
(b) It gives the same deflection in the opposite direction.
(c) It gives less deflection in the same direction.
(d) It gives the same deflection in the same direction.

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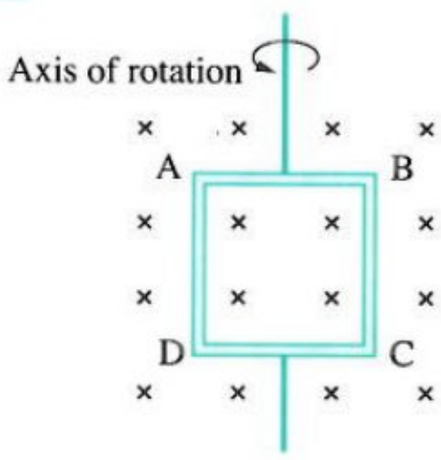
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46 The opposite figure shows a solenoid that carries an electric current while being placed adjacent to a metallic ring such that the axis of the coil is perpendicular to the plane of the ring, so an electric current gets induced in the ring in the anticlockwise direction as the ring is viewed from face (B) when ..... .



(a) decreasing the current intensity in the coil
(b) increasing the current intensity in the coil
(c) moving the solenoid toward the ring
(d) rotating the ring about an axis normal to its plane

47 ✳ The opposite figure shows a square metallic frame of side length 20 cm that is placed where its plane is perpendicular to a uniform magnetic field of flux density 0.4 T, so when the frame rotates in the direction that is shown in the figure until its plane becomes parallel to the field within a period of 0.08 s, the average induced electromotive force in the frame becomes ..... and the direction of the induced current in one side of it becomes ..... .



(a) 0.1 V, from A to B
(b) 0.1 V, from A to D
(c) 0.2 V, from A to B
(d) 0.2 V, from A to D

48 Two coils are moving in the magnetic fields that are produced due to the electric currents (I) in two very long wires as shown in figures A and B, so the induced currents in ..... .

	Figure A	Figure B
(a)	is in anticlockwise direction	is in clockwise direction
(b)	zero	zero
(c)	is in clockwise direction	is in clockwise direction
(d)	is in clockwise direction	zero

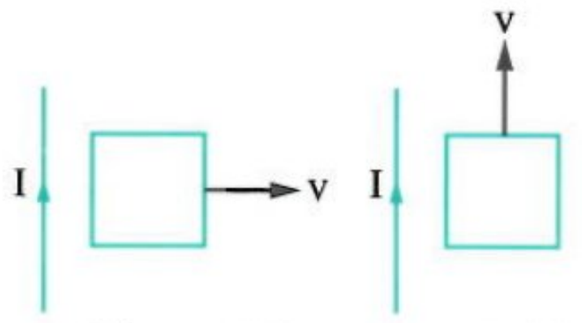
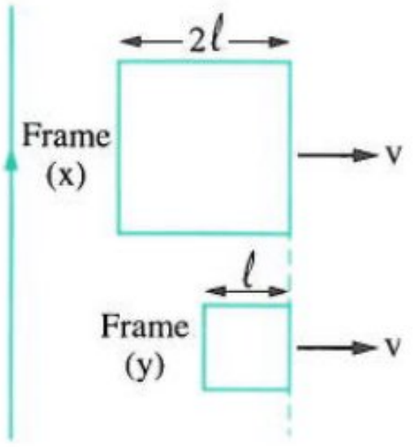


Figure (A) Figure (B)

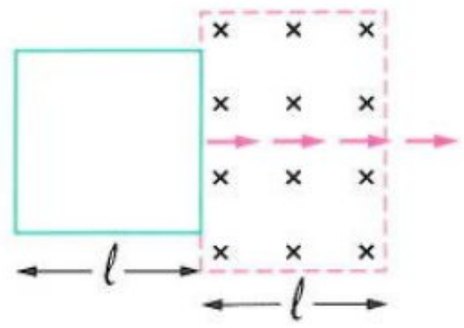
49 Two square metallic frames x, y in the same plane, where a straight wire that is carrying electric current is placed in their plane as in the opposite figure. If the frames moved away from the wire with a speed (v), in which of the two frames the biggest induced emf will be generated?



(a) Frame x.
(b) Frame y.
(c) Equal in both frames.
(d) No emf is generated in any of the two frames.

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50 The opposite figure shows a square coil moving in the plane of the page and entering a region of a uniform magnetic field that is perpendicular into the plane of the page till gets out from it. So, the direction of the induced current in the coil is ..... .



(a) always in the clockwise direction

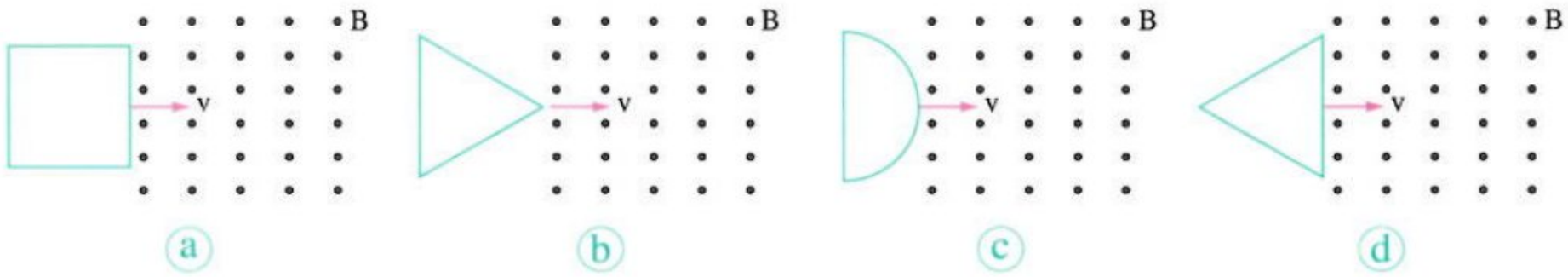
(b) always in the anti-clockwise direction

(c) in the clockwise direction then in the anti-clockwise direction

(d) in the anti-clockwise direction then in the clockwise direction

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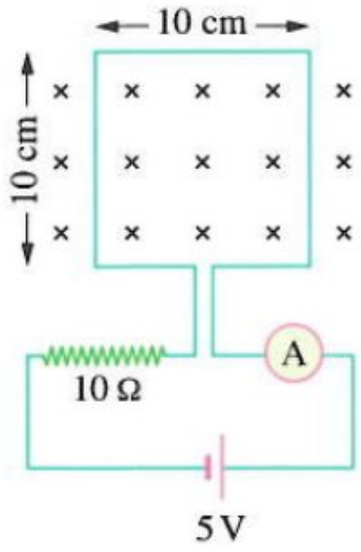
51 In each of the following figures, a metal frame is moving with a uniform velocity v perpendicular to a uniform magnetic flux of density B, then the frame in which the induced electric current intensity gets decreasing during its motion while entering the field region is ..... .



(a) (b) (c) (d)

---

52 ✱ In the circuit illustrated in the figure, a coil of one turn whose resistance is negligible is placed in a magnetic field that is directed perpendicular to the coil into the page. If the magnetic flux density gets decreasing at a uniform rate of 150 T/s, then the reading of the ammeter in the interval of decreasing the flux will be ..... .

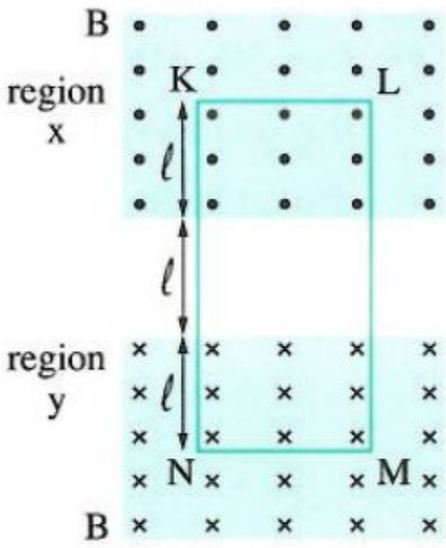


(a) 0.15 A (b) 0.35 A

(c) 0.5 A (d) 0.65 A

---

53 A rectangular coil KLMN is affected by two magnetic fields having the same magnetic flux density but in opposite directions as in the opposite figure, so during:



(i) increasing of the flux density in the region x in a certain rate and decreasing that in the region y in the same rate ..... .

(a) no induced current is generated in the coil

(b) an induced AC current of variable intensity and direction is generated in the coil

(c) an induced current in the clockwise direction is generated in the coil

(d) an induced current in the counter-clockwise direction is generated in the coil

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(ii) increasing of the flux density in the two regions x, y in the same rate ..... .

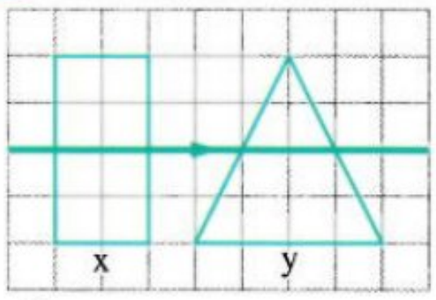
(a) no induced current is generated in the coil

(b) an induced AC current of variable intensity and direction is generated in the coil

(c) an induced current in the clockwise direction is generated in the coil

(d) an induced current in the counter-clockwise direction is generated in the coil

54 Two metallic frames x, y on a horizontal table and above them a long straight wire carrying an electric current that its intensity is increasing in a uniform rate as in the opposite figure, so induced emf ..... .



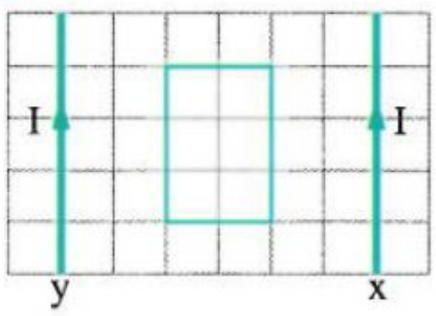
(a) is generated in frame x only

(b) is generated in frame y only

(c) is generated in both frames

(d) is generated in none of the two frames

55 A rectangular frame is placed in the same plane between two parallel straight wires x, y which are carrying electric currents, each of intensity I in the same direction as in the opposite figure, so during:



(i) increasing the currents of the two wires in the same rate ..... .

(a) induced current in the clockwise direction is generated in the metal frame

(b) induced current in the counter-clockwise direction is generated in the metal frame

(c) AC induced current is generated in the metal frame

(d) no induced current is generated in the metal frame

(ii) increasing the current of wire x with a certain uniform rate and decreasing that of wire y in the same uniform rate ..... .

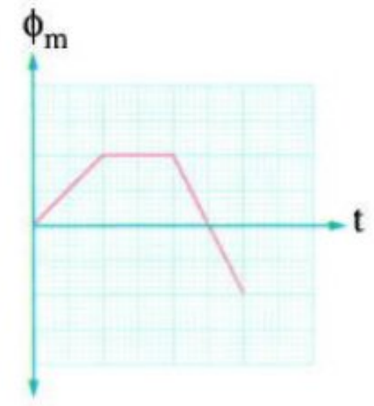
(a) induced current in the clockwise direction is generated in the metal frame

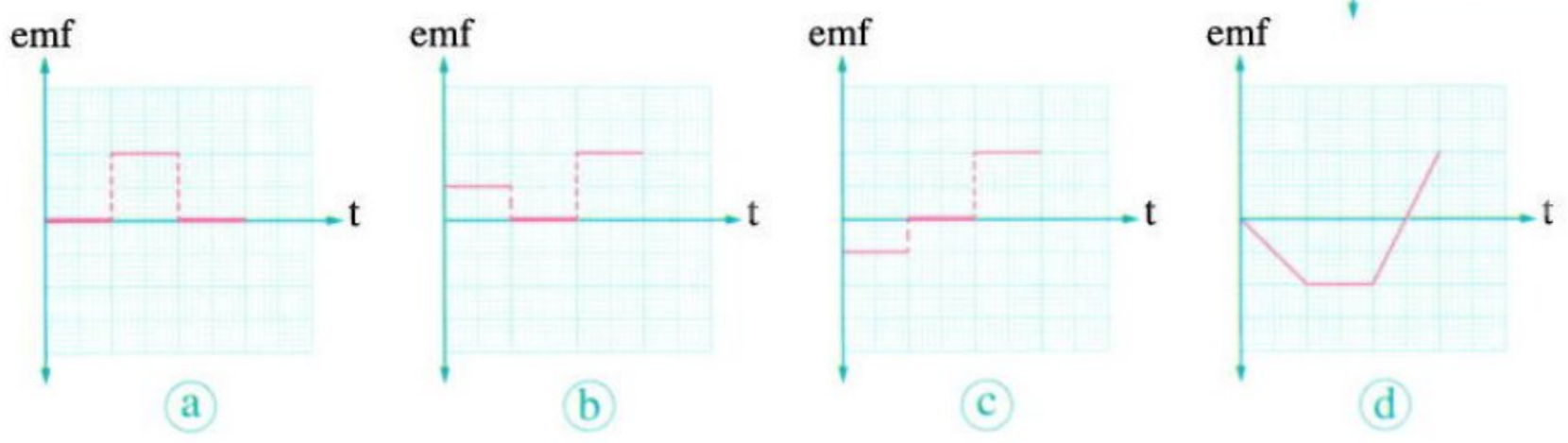
(b) induced current in the counter-clockwise direction is generated in the metal frame

(c) AC induced current is generated in the metal frame

(d) no induced current is generated in the metal frame

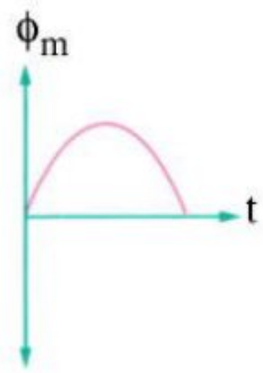
56 The opposite graph shows the relation between the magnetic flux (ϕ_m) that is penetrating a coil and time (t), so which of the following graphs represents the induced emf in the coil through the same period of time?

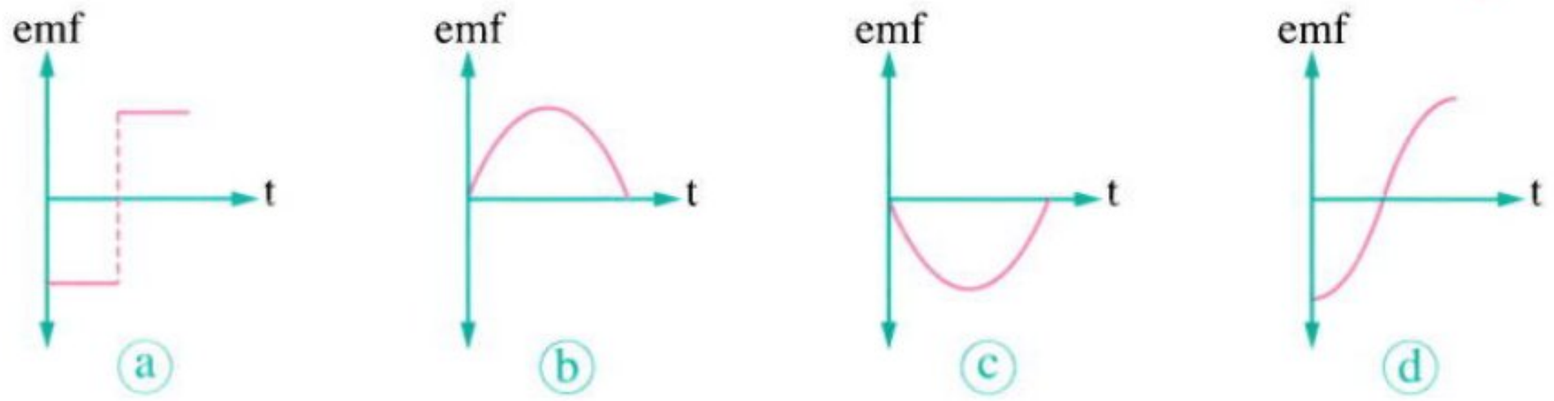




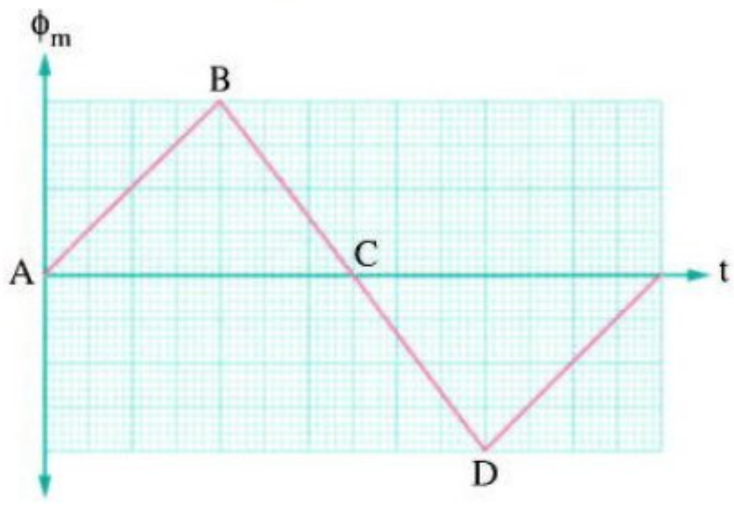
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57 A circular coil is placed parallel to the direction of a magnetic field. When the coil is inverted in a uniform speed, the passing magnetic flux through the coil is changed as in the opposite graph, so which of the following graphs represents the induced emf in the coil through the same time interval?





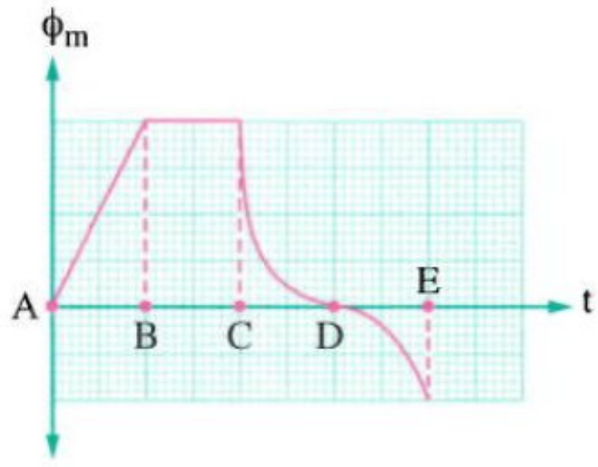
58 The opposite graph represents the variation of the magnetic flux that penetrates a circular coil when the coil is connected to a closed circuit versus time, so at which two points the induced current reverses its direction in the coil?



(a) B, A (b) C, D

(c) B, C (d) B, D

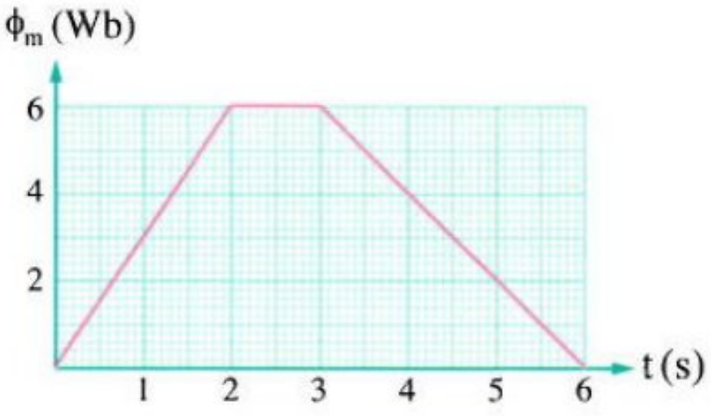
59 ✳ The opposite graph represents the change in the magnetic flux (ϕ_m) through a coil during four equal intervals of time AB, BC, CD, DE. So, the time interval in which the magnitude of the induced emf in the coil increases is ..... .



(a) AB (b) BC

(c) CD (d) DE

60 ✳ The opposite graph represents the change of the magnetic flux through a coil of 200 turns during 6 seconds, so the average induced emf through:



(i) the first two seconds equals ..... .

(a) – 300 V (b) 300 V

(c) – 600 V (d) 600 V

(ii) the third second equals ..... .

(a) 1200 V (b) 600 V (c) 400 V (d) 0

(iii) the last three seconds equals ..... .

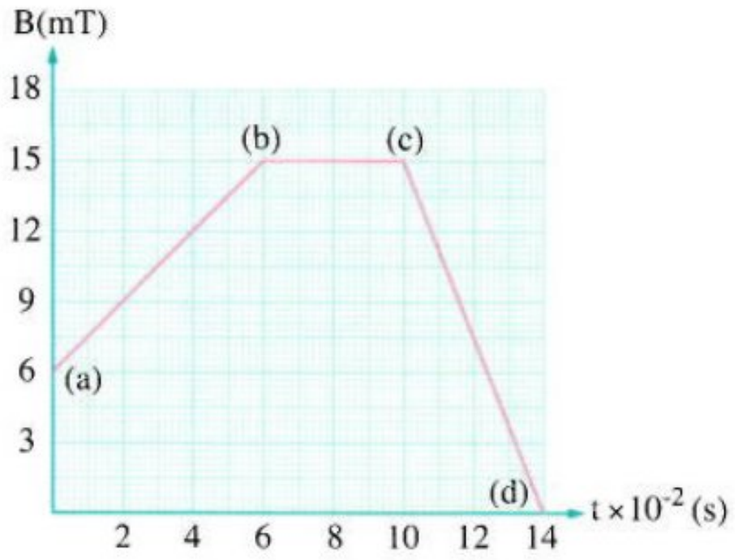
(a) 200 V (b) – 200 V (c) 400 V (d) – 400 V

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61 A coil of area 0.04 m^2 and number of turns 150 turns is placed in such a way where its plane is perpendicular to a magnetic field of flux density that changes with time according to the graph illustrated in the opposite figure, then:



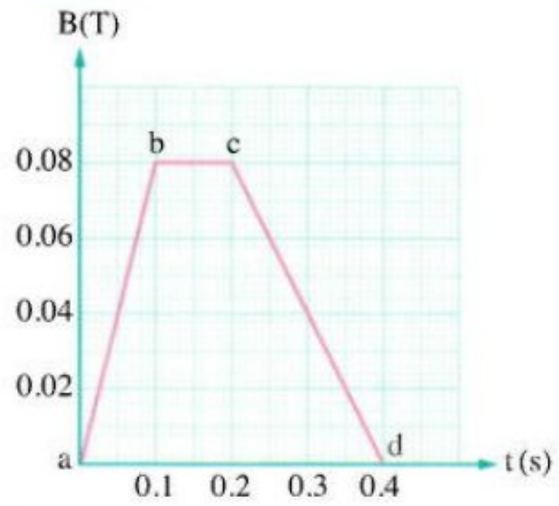
(i) The ratio between the magnitudes of the average induced emf in the coil through the two stages ab and cd respectively is ..... .

(a) $\frac{2}{5}$ (b) $\frac{5}{2}$ (c) $\frac{1}{3}$ (d) $\frac{3}{1}$

(ii) The value of the average induced emf in the coil through stage bc equals ..... .

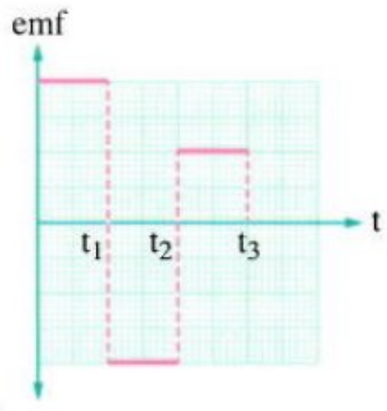
(a) 0 (b) 1 V (c) 1.25 V (d) 2.5 V

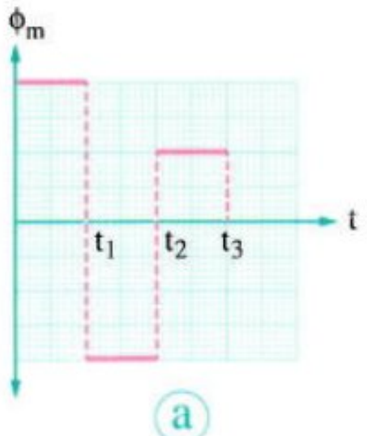
62 A magnetic flux is perpendicular into the plane of the page where a metallic ring of area 45 cm^2 is placed perpendicular to that field. If the density of the magnetic flux changes with time as shown in the opposite graph, then the magnitude of the induced emf in the ring during interval ab and the direction of the induced current in the ring during interval cd respectively are ..... .



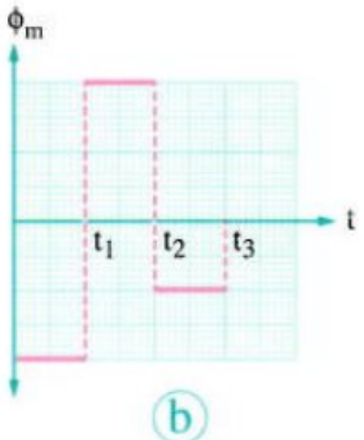
	The induced emf in the ring during interval ab	The direction of the induced current in the ring during interval cd
(a)	2.4×10^{-3} V	in the clockwise direction
(b)	2.4×10^{-3} V	in the anticlockwise direction
(c)	3.6×10^{-3} V	in the clockwise direction
(d)	3.6×10^{-3} V	in the anticlockwise direction

63 The opposite graph represents the induced emf in a coil when the magnetic flux passing through it changes through a certain interval of time, so which of the following graphs represents the magnetic flux (ϕ_m) passing through the coil during this interval?

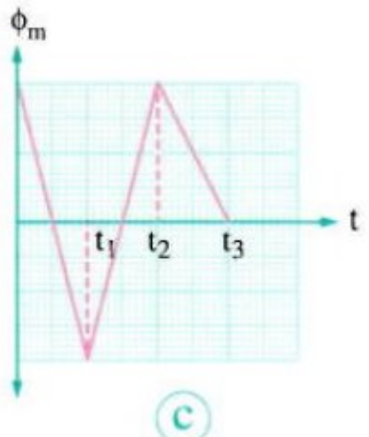




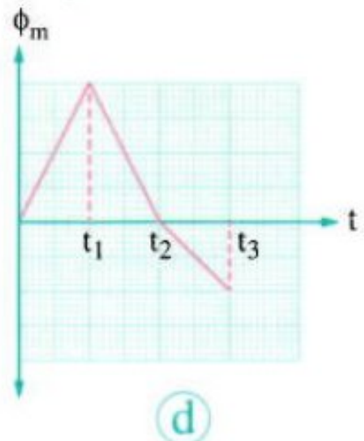
(a)



(b)



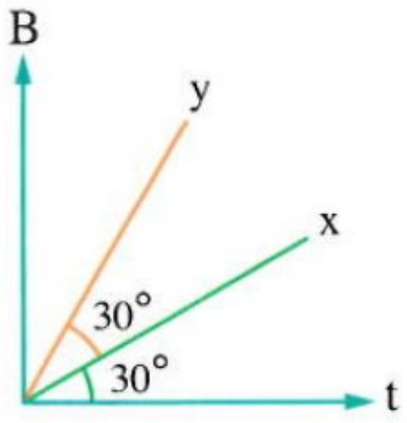
(c)



(d)

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64 ✳ A metal wire of uniform cross-section is used to make two rings (x), (y) which are placed perpendicular to a magnetic flux of constant direction but its density changes with time as represented by the opposite graph. If the ratio between the intensities of the induced current in the rings $\left(\frac{I_x}{I_y}\right)$ equals $\frac{2}{3}$, the ratio between the areas of the two rings $\left(\frac{A_x}{A_y}\right)$ equals ..... .



(a) $\sqrt{2}$ (b) $\frac{3}{2}$ (c) 2 (d) 4

65 ✳ It is noticed that a potential difference of 5.5×10^{-3} V is generated between the terminals of the second hand in a turret clock due to being exposed to a magnetic field perpendicular to it, given that the change in area that intercepts the magnetic flux due to the rotation of the second hand for one complete revolution is $\frac{11}{14}$ m^2. Then the affecting magnetic flux density is equal to ..... .

(a) 0.26 T (b) 0.42 T (c) 0.53 T (d) 0.62 T

66 The order of energy conversions in induction furnaces is .....

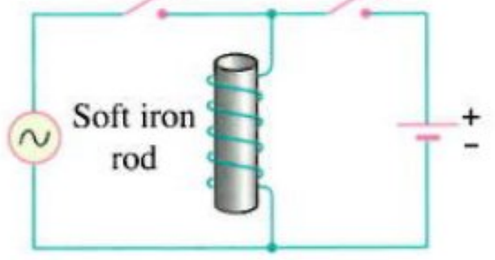
(a) thermal ⟶ electrical ⟶ magnetic ⟶ electrical

(b) electrical ⟶ thermal ⟶ electrical ⟶ magnetic

(c) magnetic ⟶ electrical ⟶ magnetic ⟶ thermal

(d) electrical ⟶ magnetic ⟶ electrical ⟶ thermal

67 An isolated rod of soft iron is placed inside a solenoid that may get connected to AC source of variable direction and intensity or DC source through one of the switches K_1, K_2 as in the opposite figure, so which of the following choices represents what happens to the soft iron rod when closing:

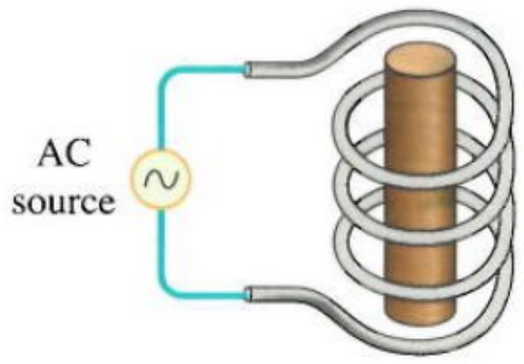


	Switch K_1?	Switch K_2?
(a)	Gets magnetized	Gets heated
(b)	Gets magnetized	Gets magnetized
(c)	Gets heated	Gets heated
(d)	Gets heated	Gets magnetized

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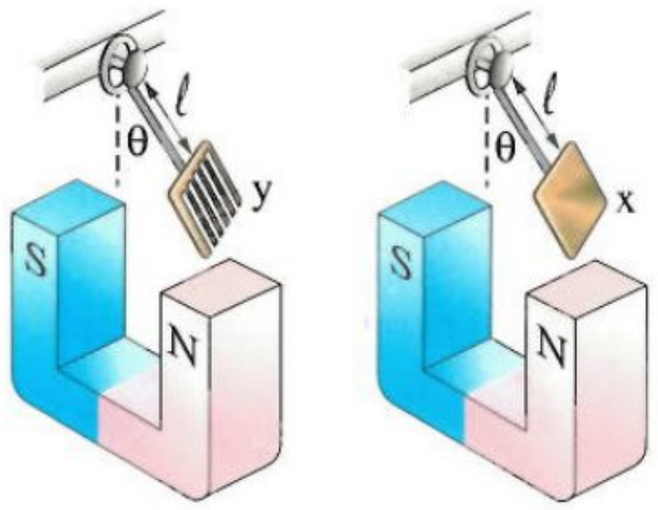
CamScanner

68 The opposite figure shows a coil that is coiled on a solid metal cylinder and connected to an AC source of variable direction and intensity, so if you know that the resistivity of aluminum is greater than that of copper, the produced eddy currents in the cylinder will be of higher intensity when the cylinder is ..... .



(a) thick and made of aluminum
(b) thin and made of aluminum
(c) thick and made of copper
(d) thin and made of copper

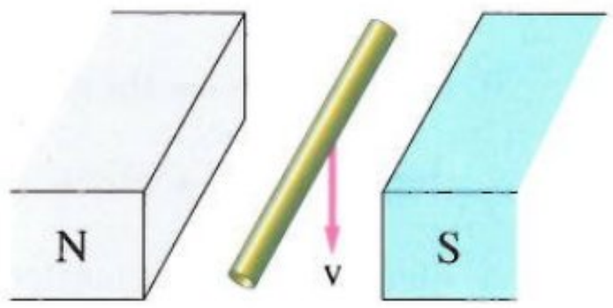
69 The opposite figure shows two aluminum slides x, y each is displaced by an angle θ from its rest position, then left at the same moment to swing in a magnetic field, which of the slides gets to rest first?



(a) Slide (x).
(b) Slide (y).
(c) The two slides get to rest together.
(d) The slides continue swinging and don't rest.

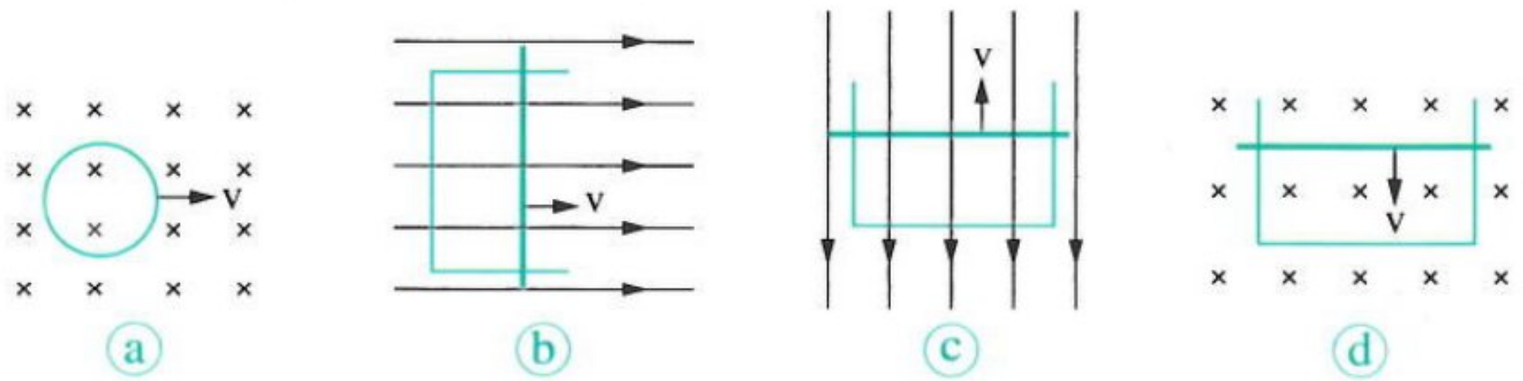
The emf induced in a straight wire

70 The opposite figure shows a metallic rod of resistance R cutting perpendicularly the magnetic flux lines at constant velocity v, so an induced electromotive force (emf) is produced across its terminals. If another rod of resistance greater than R that has the same length is used with the same velocity (v), then the value of the induced electromotive force will be ..... .



(a) greater than emf
(b) smaller than emf
(c) equal to emf
(d) zero

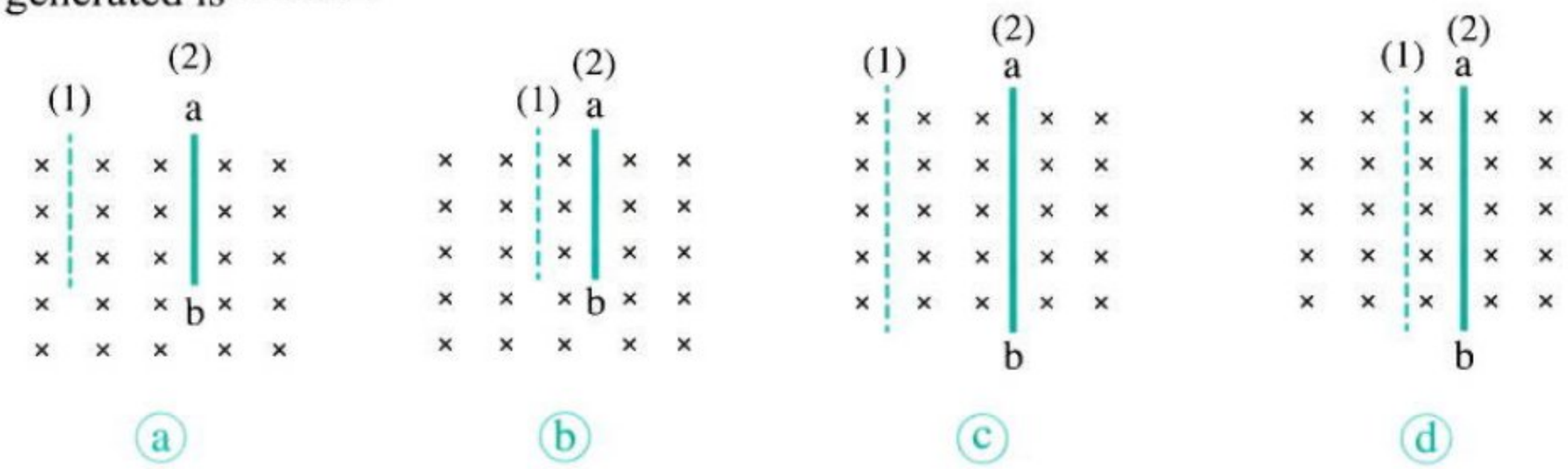
71 In which of the following figures an induced current is generated due to the motion of a conductor through a uniform magnetic field?



(a) (b) (c) (d)

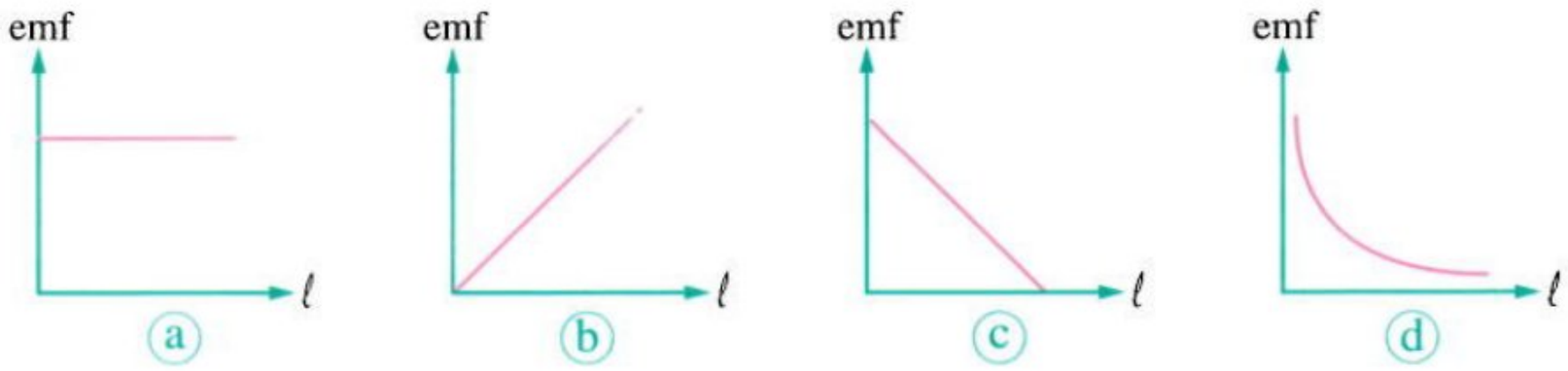
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72 The following figures show four straight wires moving perpendicular to a uniform magnetic field from position (1) to position (2) during the same time interval, so the figure that illustrates the case in which the greatest induced electromotive force gets generated is ..... .



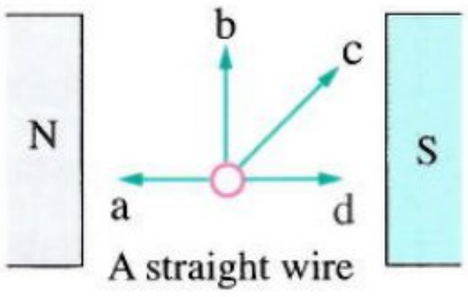
(a) (b) (c) (d)

73 Which of the following graphs represents the relation between the magnitude of the induced emf produced across the ends of a group of wires and their length (ℓ) if they are having the same cross-sectional area, made of the same material and moving at the same uniform velocity perpendicular to a uniform magnetic field?



(a) (b) (c) (d)

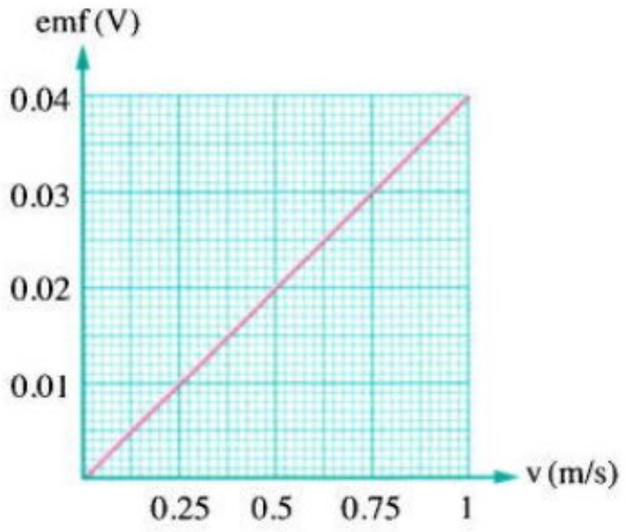
74 The opposite figure represents a straight wire moving perpendicular to the page with a uniform velocity (v) between the two poles of a magnet, which of the directions shown in the figure represents the direction of the motion of the wire that produces a greater induced emf between its ends (terminals)?



A straight wire

(a) The direction (a) (b) The direction (b) (c) The direction (c) (d) The direction (d)

75 A straight wire of length 20 cm moves many times perpendicular to a uniform magnetic flux with a different uniform velocity (v) in each time. The opposite graph shows the relation between the induced emf in the wire and the velocity of the wire (v), then the magnetic flux density equals ..... .

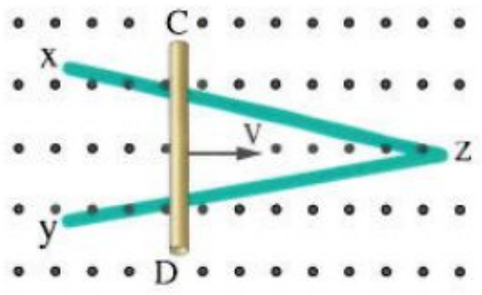


(a) 0.1 T (b) 0.2 T
(c) 0.3 T (d) 0.4 T

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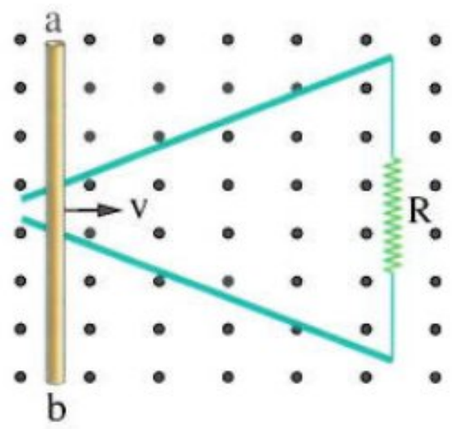
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76 A metal rod (CD) of uniform cross-section and resistance R moves with constant velocity v perpendicular to a uniform magnetic field touching wire xzy of negligible resistance as in the opposite figure, then the intensity of the current which is passing in the rod during its motion ..... .



(a) equals zero
(b) remains constant
(c) increases gradually
(d) decreases gradually

77 The following figure shows a metal rod (ab) of negligible resistance sliding perpendicular to a uniform magnetic field on two metal rails of negligible resistance with a constant velocity v. The two rails are connected together through a resistor R, so magnitude of the induced current in the resistor (R) during the movement of the rod ..... .



(a) gets increased
(b) gets decreased
(c) remains constant
(d) the answer is indeterminable

78 A straight wire xy passes with a uniform velocity through two magnetic field regions a, b such that its motion direction is perpendicular to each of them as in the opposite figures:

So, the value of the induced emf between the terminals of the wire xy in ..... .

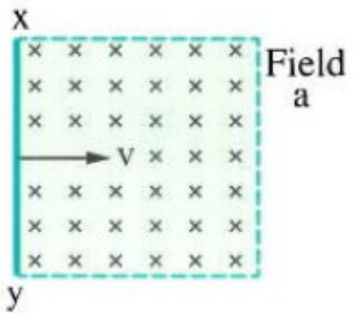


Figure (1)

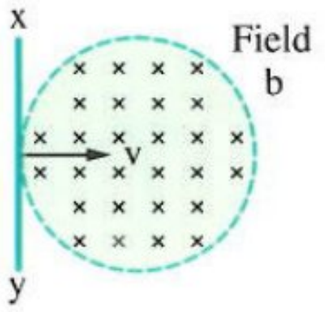
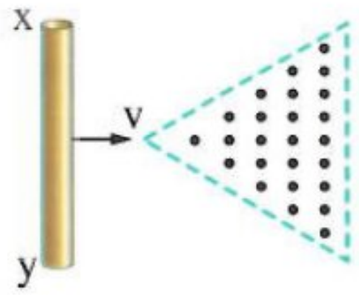
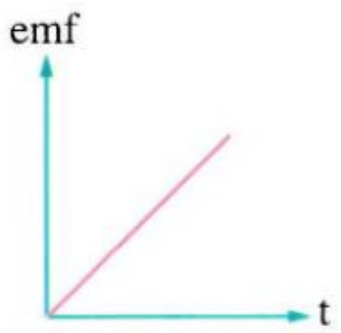


Figure (2)

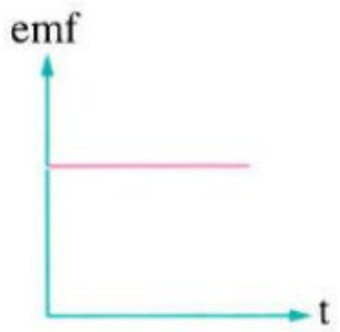
	Figure (1)	Figure (2)
(a)	constant	constant
(b)	variable	constant
(c)	constant	variable
(d)	variable	variable

79 Wire xy is moving with a uniform velocity (v) through a uniform magnetic field that is perpendicular out of the plane of the page and confined in a triangle as shown in the opposite figure, then the graph that represents the relation between the induced emf in the wire and the time from the moment of entering the magnetic field to the moment it exits from it is ..... .

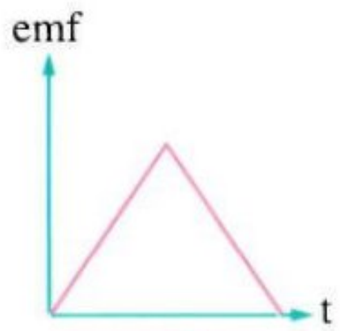




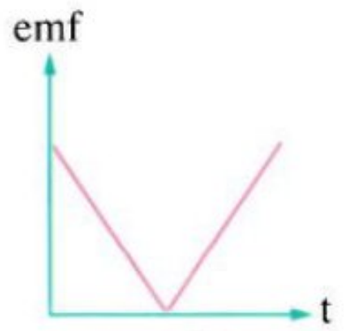
(a)



(b)



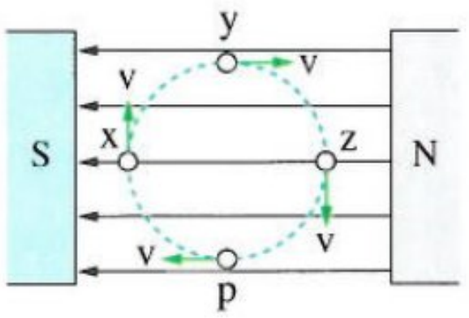
(c)

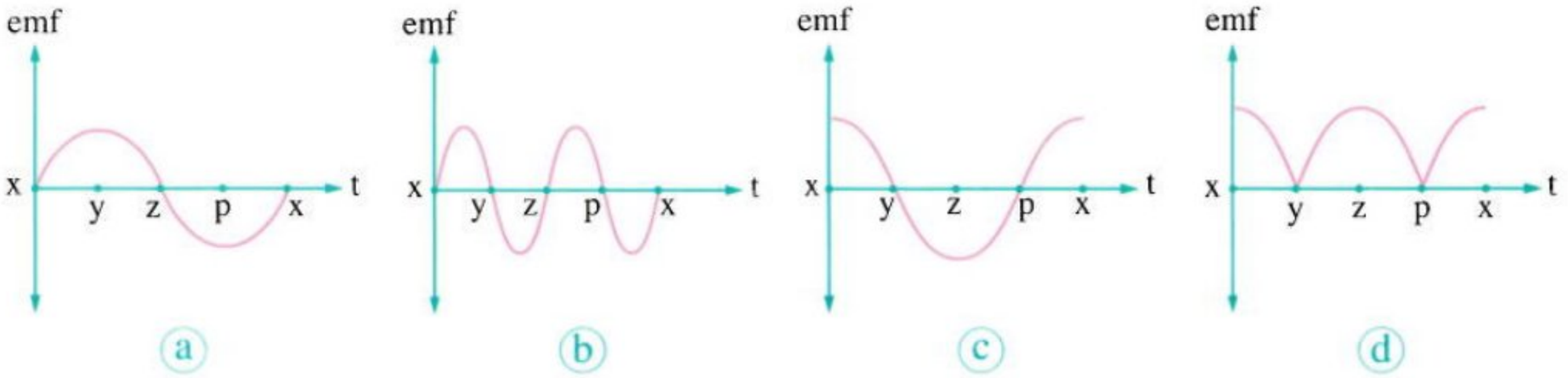


(d)

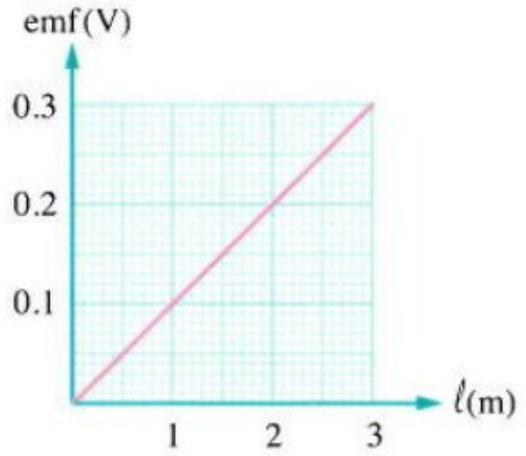
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80 In the opposite figure, a straight copper wire is moving perpendicular to a uniform magnetic field with a uniform speed (v) in a circular path from point x to y to z to p to x again, so which of the following graphs represents the relation between the induced emf across the terminals of the wire during its motion and the time of motion?





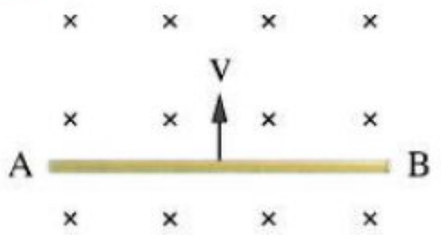
81 The opposite graph represents the relation between the induced electromotive force (emf) in a straight wire that moves with a uniform velocity v perpendicular to a magnetic flux of density 0.1 T and the length of that wire (ℓ), so the magnitude of its velocity v is ..... .



(a) 0.1 m/s (b) 0.2 m/s

(c) 1 m/s (d) 2 m/s

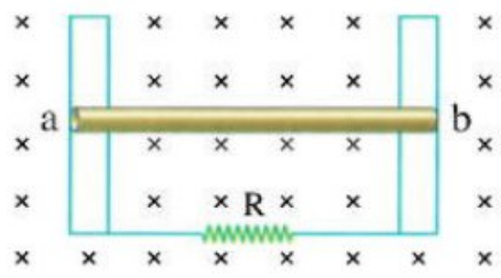
82 In the opposite figure, if the wire is moving perpendicular to the magnetic flux lines in the shown direction, then the potential of point A is ..... .



(a) greater than the potential of point B (b) less than the potential of point B

(c) equal to the potential of point B (d) can't determine the answer

83 In the opposite figure, a straight wire ab is moving in the plane of the page perpendicular to a magnetic field sliding on two metallic rails, hence an electric current is induced in the wire from terminal a to terminal b, so ..... .



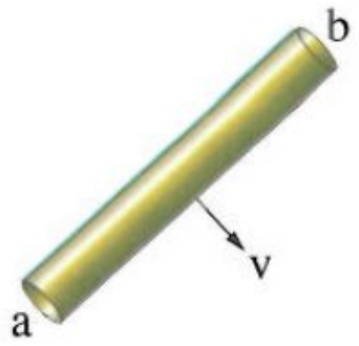
	The direction of motion of the wire is	The relation between the potentials of the points a, b is
(a)	upwards of the page	$V_a < V_b$
(b)	upwards of the page	$V_a > V_b$
(c)	downwards of the page	$V_a < V_b$
(d)	downwards of the page	$V_a > V_b$

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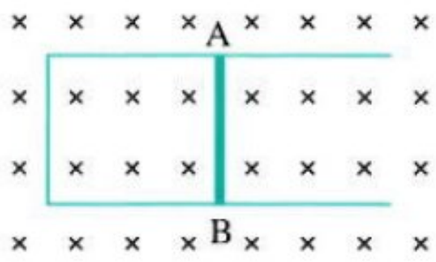
CamScanner

84 ✱ The opposite figure shows a metal wire (ab) of length 2 m in the plane of the page moving at a uniform velocity of 5 m/s perpendicular to a uniform magnetic field, so that the electric potential of terminal (a) becomes greater than that of terminal (b) by a value of 0.4 V, then the magnitude and the direction of the magnetic flux density acting on the wire are ..... .



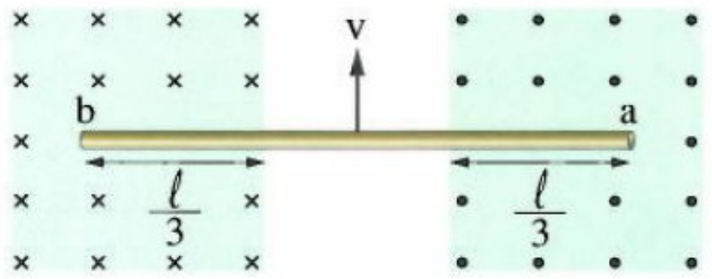
(a) 0.02 T, into the page
(b) 0.02 T, out of the page
(c) 0.04 T, into the page
(d) 0.04 T, out of the page

85 The opposite figure shows a metallic wire AB of length 0.15 m that is moved with uniform velocity (v) perpendicular to a magnetic flux of density 0.4 T so that an induced emf of 0.03 V is generated across its terminals leading to the flow of an electric current in the wire from terminal A to terminal B, then the velocity of the wire and its direction are ..... .



(a) 0.5 m/s, to the right of the page
(b) 0.5 m/s, to the left of the page
(c) 1 m/s, to the right of the page
(d) 1 m/s, to the left of the page

86 A cylindrical metal rod ab of length ℓ moves with uniform velocity (v) such that at each of its ends, part of it passes perpendicularly through a different magnetic field of flux density B as in the opposite figure, which terminal of the rod has greater electric potential?



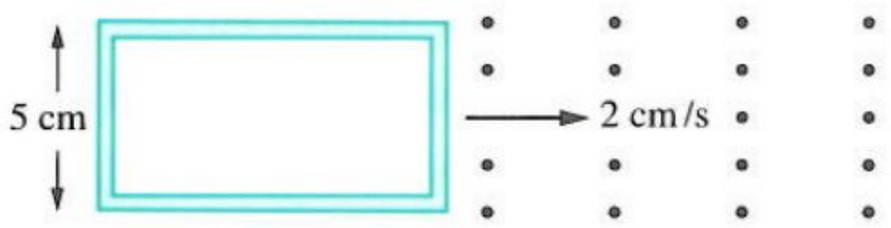
(a) Terminal a
(b) Terminal b
(c) The two terminals have the same potential.
(d) The answer is indeterminable.

87 ✱ A rod of copper of length 30 cm moves with a velocity of 0.5 m/s in a magnetic field of density 0.8 tesla. Then the value of the induced emf produced across the terminals of that rod if it is moved:

	Perpendicular to the field	Parallel to the field
(a)	0.12 V	0.12 V
(b)	0.12 V	0
(c)	0	0.12 V
(d)	0	0

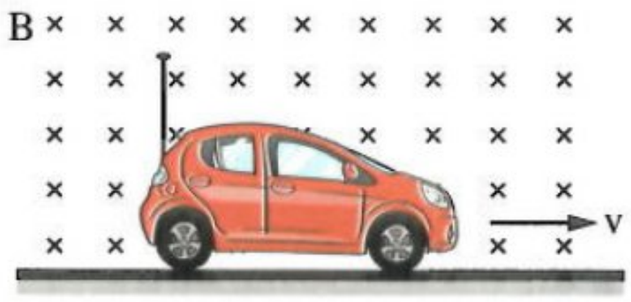
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88 In the opposite figure, a rectangular metal frame of width 5 cm and electric resistance 1 Ω, enters with a uniform velocity of 2 cm/s a region of a magnetic field of density 0.5 T that its direction is perpendicular to the frame. The magnitude of the induced current in the frame during entering the field equals ..... .



(a) 0.5 mA (b) 1 mA (c) 2 mA (d) 5 mA

89 ✱ In the opposite figure, an antenna of length one meter is fixed on a car that moves at velocity 80 km/h in a direction perpendicular to the horizontal component of the Earth's magnetic field. An emf of 4×10^{-4} V is induced in the antenna. In such case the horizontal component of the magnetic field of the Earth is equal to ..... .

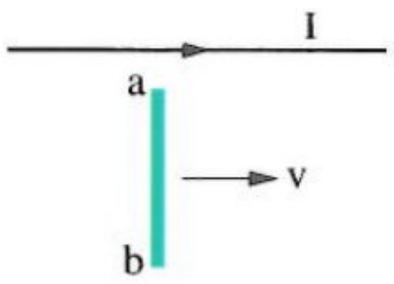


(a) 12×10^{-6} T (b) 14×10^{-6} T (c) 16×10^{-6} T (d) 18×10^{-6} T

90 When a wire of length 0.5 m is moved in a uniform magnetic field of flux density 0.4 T with a velocity of 2 m/s in a direction perpendicular to its length, an induced emf of 0.336 V is produced across its terminals, so the angle between the direction of the wire's velocity and the direction of the magnetic field is approximately ..... .

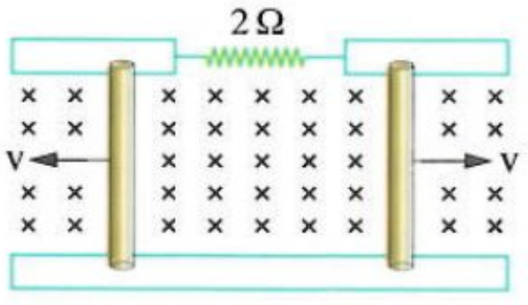
(a) 36° (b) 57° (c) 64° (d) 82°

91 The opposite figure represents a long current carrying wire and a metal rod ab that are placed in the plane of the page, so if the rod is moved at a uniform velocity v in the direction which is shown in the figure, then the relation between the electric potentials of the two points a and b is .....



(a) $V_a = V_b = 0$ (b) $V_a > V_b$ (c) $V_a < V_b$ (d) $V_a = V_b \neq 0$

92 ✱ The opposite figure shows two cylinderical conductors of negligible resistances where the length of each is 20 cm moving on a metallic track of negligible resistance with constant speeds of 5 m/s perpendicular to a uniform magnetic field of flux density 0.4 T, then the induced current intensity passing through the 2 Ω resistance equals ..... .

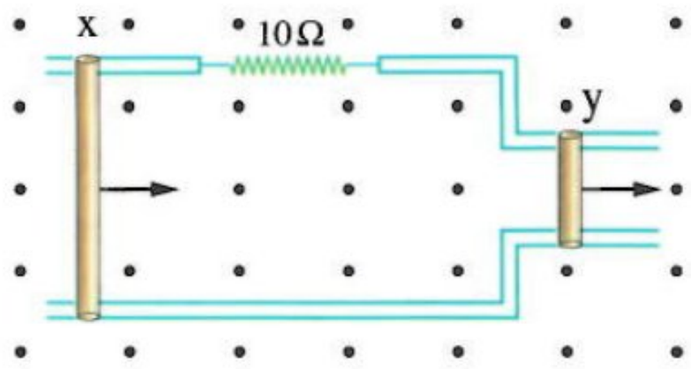


(a) zero (b) 0.8 A (c) 0.6 A (d) 0.4 A

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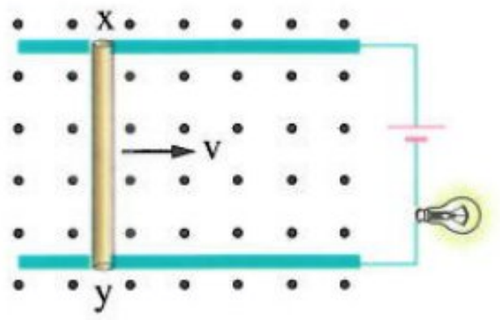
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93 ✳ In the opposite figure two conductors x, y are moving in the same direction with a uniform velocity 0.5 m/s on two metal rods of negligible resistance and perpendicular to a uniform magnetic field of density 0.2 T. If the magnitude of the induced current in the resistor 10 Ω is 2.5 mA, so length of the conductor x is greater than that of y by ..... .



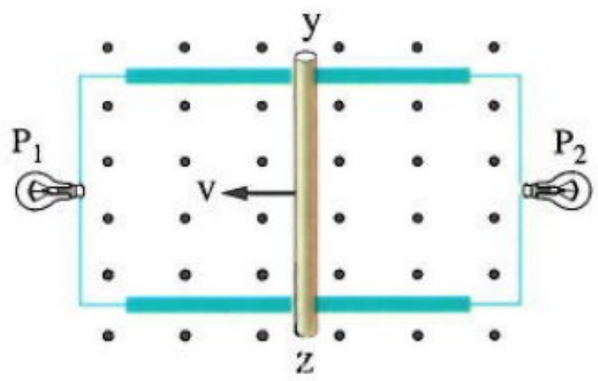
(a) 10 cm (b) 15 cm (c) 25 cm (d) 35 cm

94 In the opposite figure, what happens to the brightness of the electric lightbulb during the motion of rod xy with uniform velocity v in the shown direction?



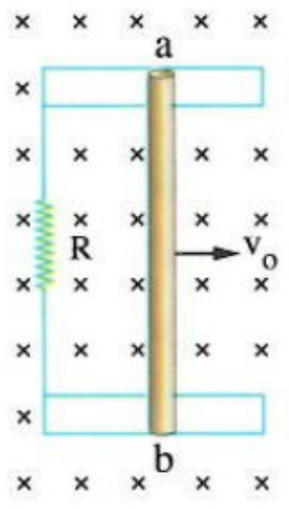
(a) Remains constant. (b) Vanishes.
(c) Decreases but does not vanish. (d) Increases.

95 The opposite figure represents a metal rod yz of resistance R placed on two frictionless rails, each of resistance 2 R, where two similar electric bulbs P_1 and P_2 are connected to the terminals of the two rails from each side. This group is placed perpendicular to a uniform magnetic field of flux density B. What happens to the brightness of each of the two bulbs during the motion of the rod with uniform velocity v in the shown direction?



	Brightness of bulb P_1	Brightness of bulb P_2
(a)	Decreases	Decreases
(b)	Decreases	Increases
(c)	Increases	Decreases
(d)	Increases	Increases

96 ✳ A conductor ab moves horizontally with a uniform velocity v_o perpendicular to a uniform magnetic field while touching two metal rods of negligible resistance and connected to each other by a resistor (R) as in the opposite figure. If the consumed power in the resistor (R) during the motion of the conductor is $(P_w)_o$, so the required force (F) to pull the conductor with a uniform velocity v_o equals ..... .

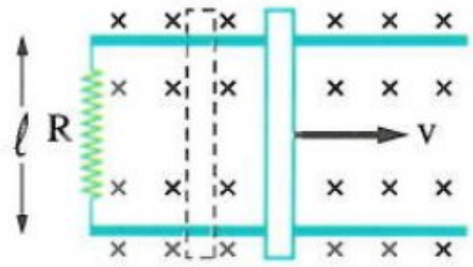


(a) $\frac{(P_w)_o}{v_o}$ (b) $(P_w)_o\, v_o$ (c) $\frac{(P_w)_o^2}{v_o}$ (d) $(P_w)_o^2\, v_o$

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97 ✳ In the opposite figure:

if ℓ = 15 cm, R = 25 Ω, B = 0.6 T, v = 8 m/s, ignoring the resistances of the sliding copper rod and the two rails, then:



(i) The magnitude of the induced emf generated in the rod equals ..... .

(a) 0.72 V (b) 0.85 V (c) 1.44 V (d) 2.82 V

(ii) The electric current intensity through the resistance (R) equals ..... .

(a) 0.0288 A (b) 0.0576 A (c) 0.0864 A (d) 0.1152 A

(iii) The force that is required to keep the rod moving at the same uniform velocity (v) equals ..... .

(a) 1.43×10^{-3} N (b) 1.87×10^{-3} N (c) 2.59×10^{-3} N (d) 4.32×10^{-3} N

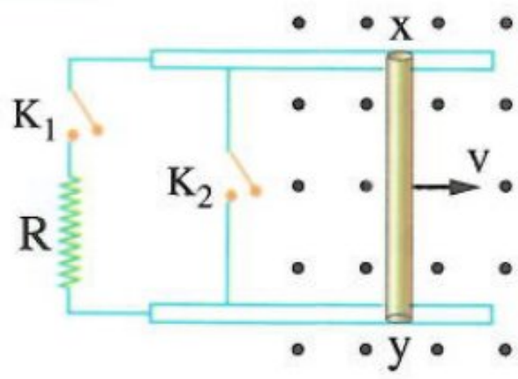
(iv) The power consumed in the resistance (R) during the rod's motion equals ..... .

(a) 0.72 W (b) 0.88 W (c) 15.6×10^{-3} W (d) 20.7×10^{-3} W

98 ✳ An electric circuit consists of a resistor of 3 Ω and two parallel thick wires at 50 cm apart. A metal rod is placed perpendicular to the wires to close the circuit. If the confined area between the wires was perpendicular to a magnetic flux of density 0.15 T, so the force which is required to move the metal rod to acquire a uniform velocity of 200 cm/s equals ..... .

(a) 1.87×10^{-3} N (b) 2.55×10^{-3} N (c) 3.75×10^{-3} N (d) 7.5×10^{-3} N

99 A cylindrical rod xy of resistance R is placed in a magnetic field and slides on two smooth rods as in the opposite figure, so the required force to move the rod (xy) with a uniform velocity v while touching the two other rods and perpendicularly to the direction of the magnetic field is less but not equal zero when ..... .



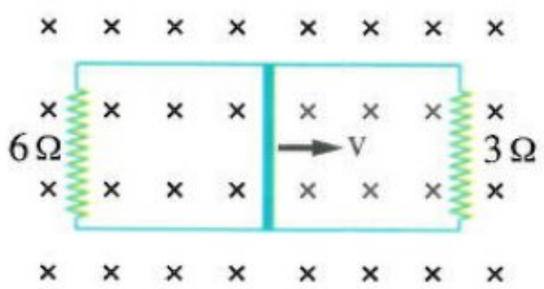
(a) the switches K_1, K_2 are opened

(b) the switches K_1, K_2 are closed

(c) the switch K_1 is opened and K_2 is closed

(d) the switch K_1 is closed and K_2 is opened

100 ✳ In the opposite figure, a closed electric circuit in the form of a rectangle where a conducting rod of 1 m length is sliding on it. If the circuit is placed in a uniform magnetic field of density 2 T perpendicular to the plane of the circuit and if the resistance of the conducting rod is 2 Ω, so the required force to slide the conducting rod with a constant velocity of 2 m/s equals ..... .



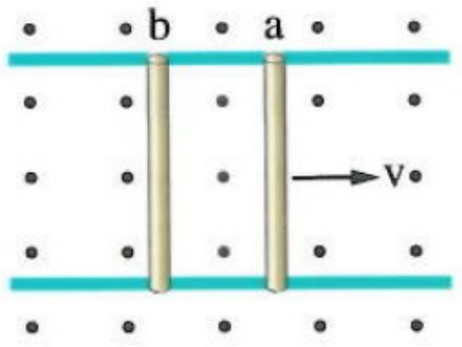
(a) 2 N (b) 4 N (c) 6 N (d) 8 N

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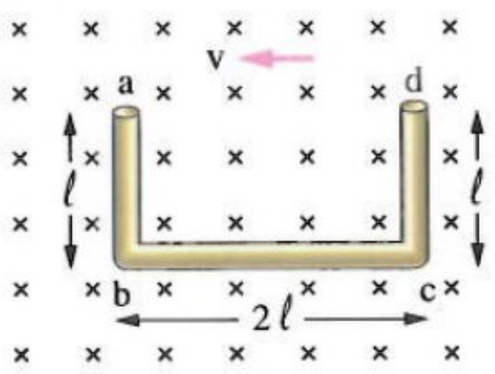
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101 The opposite figure shows two similar metal cylindrical rods a, b that are movable on two frictionless metal rails in the plane of the page while being affected by a uniform magnetic field perpendicular to the plane of the page, when rod a is dragged with a uniform velocity v to the right of the page, the direction of the force acting on rod b due to the external magnetic field will be ..... .



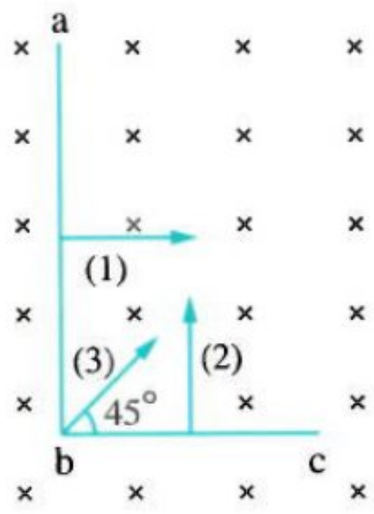
(a) in the plane of the page to the right (b) in the plane of the page to the left
(c) perpendicular into the page (d) perpendicular out of the page

102 Wire abcd is bent to be in the shape of letter U such that the length of segment ab is ℓ and the length of bc is 2ℓ, if the rod is moved with a uniform velocity v in the direction shown in the opposite figure through a uniform magnetic field of flux density B that is perpendicular to the plane of the wire, the magnitude of the induced emf across the wire between terminals a and d equals ..... .



(a) 0 (b) $B\ell v$ (c) $2B\ell v$ (d) $4B\ell v$

103 A wire abc is shaped in the form of letter L where the lengths of its parts are ℓ, 2ℓ. It is placed in a uniform magnetic field such that its plane is perpendicular to the field as in the opposite figure. When the wire moves with a uniform velocity (v) in the shown directions (1), (2), (3) each separately the potential difference between the ends of the wire becomes V_1, V_2, V_3 respectively, so which of the following relations is correct?



(a) $V_1 = V_2 = V_3$ (b) $V_3 > V_1 > V_2$ (c) $V_1 > V_3 > V_2$ (d) $V_3 > V_2 > V_1$

Second Essay questions

Faraday's law and Lenz's rule

1 Figure (a) shows a copper wire carrying an electric current (I) while being in the same plane with a copper ring and figure (b) shows the copper wire when it coincides with the axis of the copper ring. **In which** case an induced current passes through the ring when the electric current passing in the wire increases? **Explain your answer.**

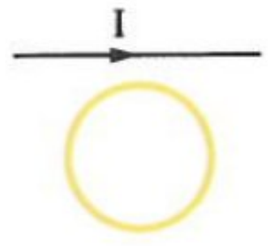


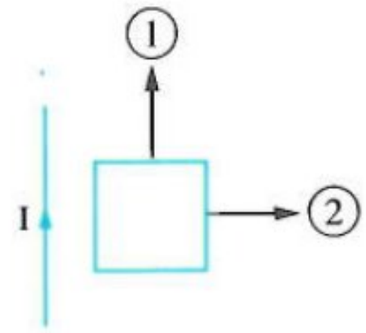
Figure (a)



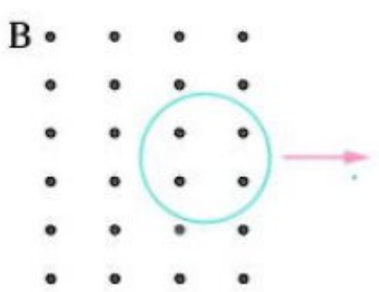
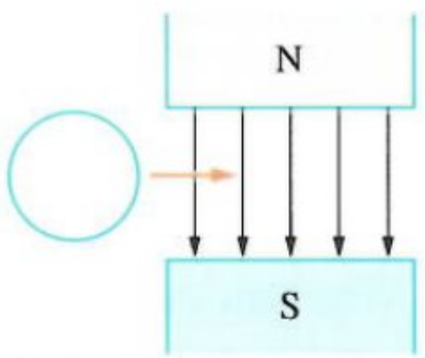
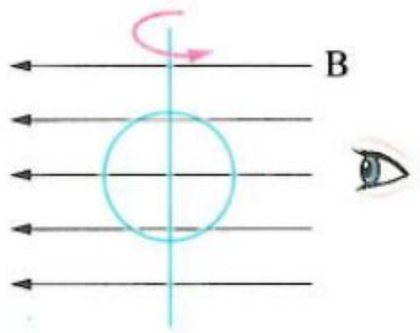
Figure (b)

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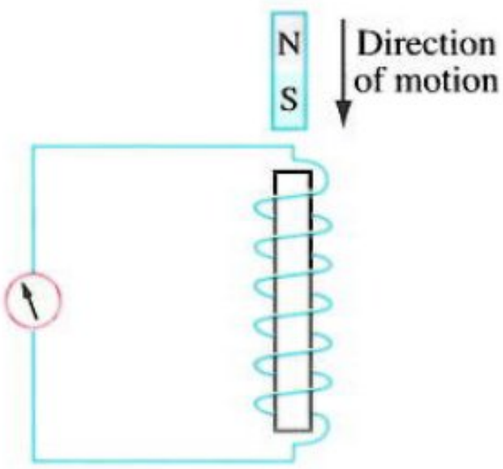
2 A square metal frame is placed in the same plane with a long straight current-carrying wire, **determine giving reasons** the direction of the induced current in the frame (if there is any) when the frame is moved once in direction (1) and another in direction (2) as shown in the opposite figure.



3 **Determine with explanation the direction of the induced current in the shown face of the ring in each of the following cases if there is any:**

		
(a) Moving the ring rightwards the page to get out of the magnetic field.	(b) Passing the ring between the poles of the magnet.	(c) Rotating the ring for a quarter cycle about an axis in its plane in the shown direction.

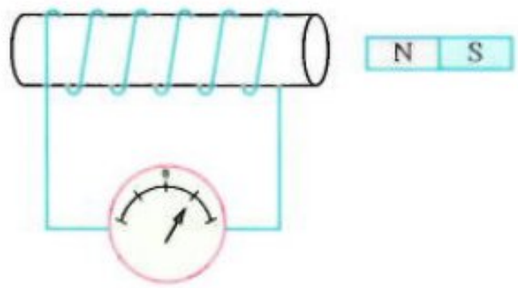
4 In the opposite figure, a magnet is being moved towards a solenoid making the galvanometer pointer to deflect instantly.



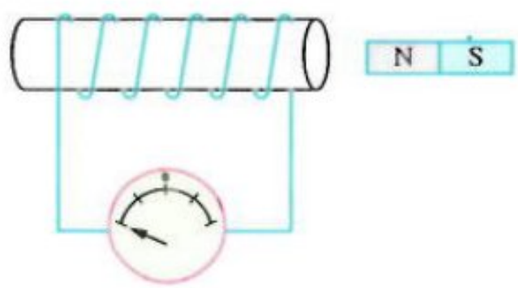
(a) **Determine** on the figure the direction of the induced electric current in the coil, its polarity **and what** is the used rule?

(b) **What** is the effect of inserting a soft iron cylinder inside the coil on the instantaneous deflection of the galvanometer pointer? **Explain.**

5 A student has moved a magnet once with a uniform velocity (v_1) for a given distance along the axis of a solenoid whose terminals are connected to a galvanometer having its zero at the middle of its scale and another time he has moved the magnet with a uniform velocity (v_2) for the same distance along the axis of the solenoid where the student observation of the maximum deflection of the galvanometer pointer in both cases was as in the opposite figure. **Compare with explanation between** the magnitudes and the directions of the magnet velocity in both cases.



Case (1)



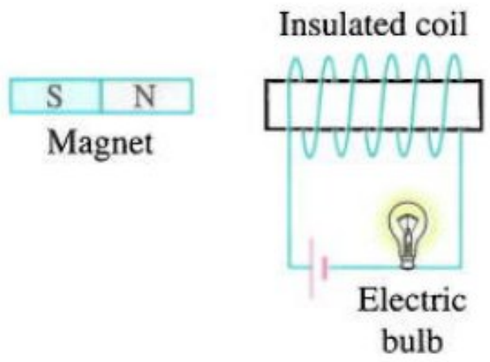
Case (2)

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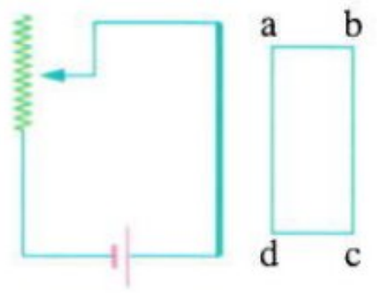
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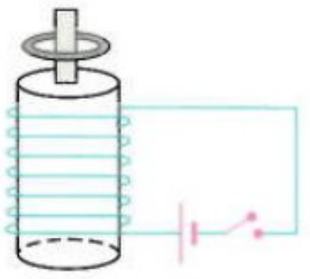
6 In the opposite figure, if the induced emf in the case of its generation is less than the emf of the battery then:
What happens to the intensity of the bulb luminosity while:
(a) moving the magnet towards the coil?
(b) the magnet is settled inside the coil?
(c) moving the magnet away from the coil?



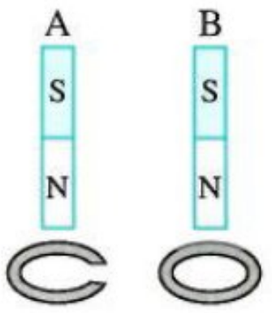
7 In the opposite figure, when the value of the resistance taken from the rheostat increases gradually, **what** will be the direction of the induced current in the coil abcd?
Explain your answer.



8 A metallic ring of aluminum was placed around the shaft of a strong electromagnet's core. When the circuit of the electromagnet was closed, the ring jumped up, **explain.**



9 In the opposite figure:
Two similar magnets A, B are falling freely from the same height through two copper rings, one is opened and the other is closed, **which** of the two magnets reaches the ground first?
Explain your answer.



10 A circular coil of radius 10 cm that has number of turns N, where the resistance of each turn is 1 Ω, is positioned perpendicular to a magnetic flux of density 0.1 T, so if this flux has vanished through 0.01 s, **calculate** the average induced current intensity through the coil.

11 A ring of radius 22 cm is placed perpendicular to a uniform magnetic field of flux density 0.05 T, then if the ring is rotated 80° through a time interval of 0.25 s about an axis in its plane, **calculate** the average induced emf in the ring.

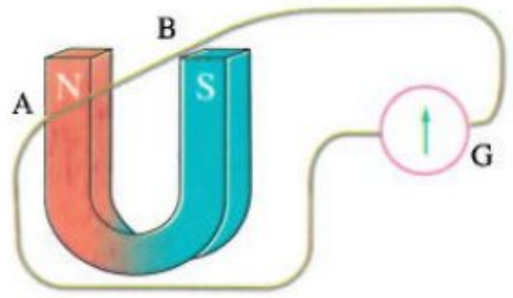
12 A metal ring of resistance R is penetrated by a magnetic flux ϕ_m, if the flux gets vanished through a time interval t, **prove that** the quantity of electric charge that flows through a cross-section of the ring wire is calculated by the relation; $Q = \frac{\phi_m}{R}$

13 A copper sheet is vibrating as a pendulum.
Explain why its vibration gets damped when it vibrates between the poles of a strong magnet as in the opposite figure.



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The emf induced in a straight wire



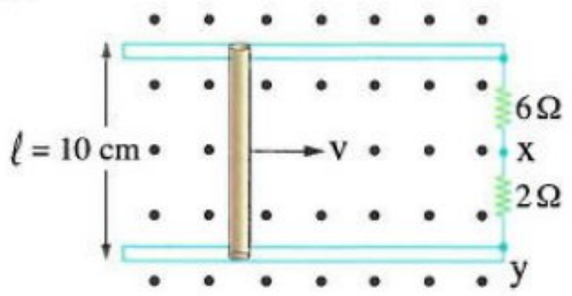
14 In the opposite figure:
The wire AB moves with a uniform speed downwards between the poles of the magnet:

(a) **What does happen to** the pointer of the sensitive galvanometer?

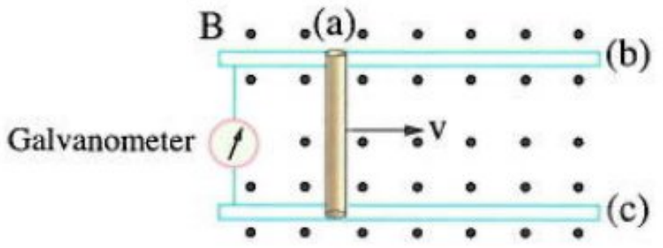
(b) **What is the change that happen to** the pointer of the galvanometer when the wire AB moves with a uniform speed upwards?

(c) **How** can the wire AB move through the magnetic field without affecting the pointer of the galvanometer?

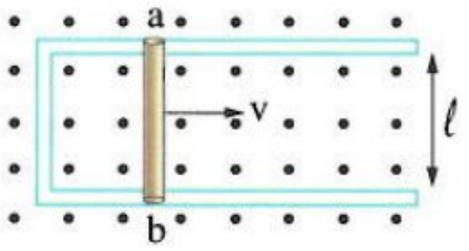
15 The opposite figure illustrates a cylindrical rod of length 10 cm that has a negligible resistance. The rod is moving with a uniform velocity of 4 m/s on two parallel metal rails of negligible resistance perpendicular to a uniform magnetic field of flux density 1 T. **Calculate** the potential difference between the two points x, y during the motion of the rod.



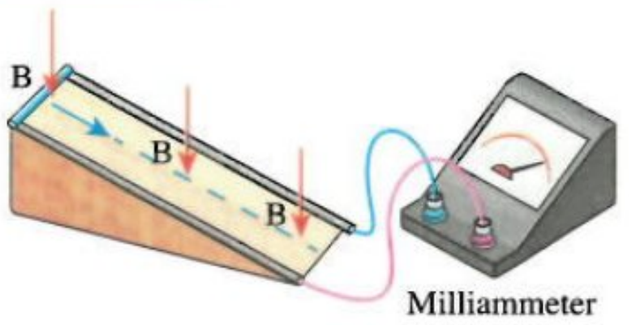
16 A metal rod (a) is moving perpendicular to a uniform magnetic field B with a uniform velocity (v) on two metal rails (b), (c) where the resistance of each of them is R. **Explain what happens to** the reading of the galvanometer during the motion of rod (a) in the direction shown in the opposite figure.



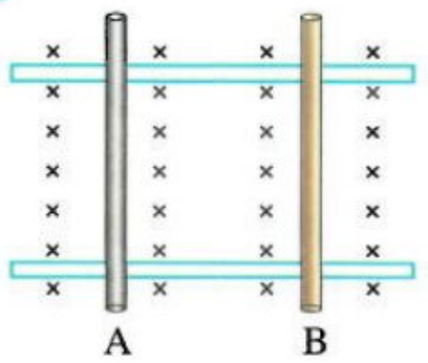
17 The opposite figure shows a rod (ab) of length ℓ and resistance R being dragged with a uniform velocity v perpendicular to a uniform magnetic field of flux density B, **prove that** the magnitude of the required force to drag the rod ab and keep it moving with the same uniform velocity v is given by relation; $F = \frac{B^2 \ell^2 v}{R}$



18 A metal cylindrical rod of resistance R slides freely down an inclined plane through a vertical magnetic field (B) touching two smooth metal rails of negligible resistance that are connected to a milliammeter as shown in the opposite figure, **explain why** the milliammeter reading doesn't remain constant while the rod is sliding.



19 In the opposite figure:
The two metal rods (A) and (B) can slide on two parallel rails perpendicular to a uniform magnetic field. If the magnetic field started to decrease gradually, **then describe with explanation** how the two rods will move.



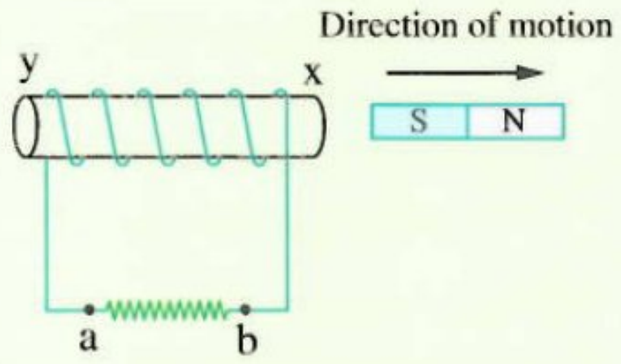
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Questions of the Previous Years Exams on Lesson 1

Faraday's law and Lenz's rule

1 In the opposite figure, as the magnet moves in the shown direction, which of the following choices becomes correct? *(Exp. May 21)*



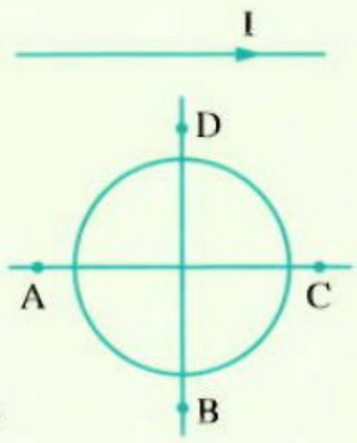
(a) End (y) of the coil acts as a north pole and point (a) has a negative potential.

(b) End (x) of the coil acts as a north pole and point (b) has a positive potential.

(c) End (x) of the coil acts as a south pole and point (a) has a positive potential.

(d) End (y) of the coil acts as a south pole and point (b) has a negative potential.

2 A metallic ring is placed in the same plane with a straight wire that carries an electric current I as shown in the figure. If the ring is moved and consequently a current is induced through it in the anticlockwise direction, the direction in which the ring has been moved is towards point ..... . *(Exp. May 21)*

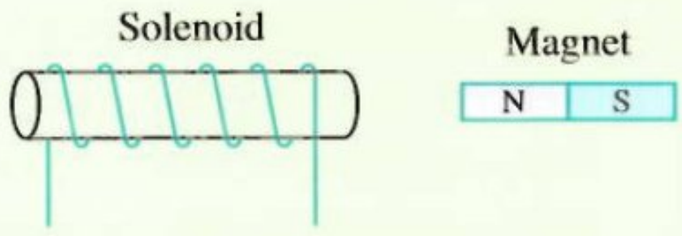


(a) A (b) B (c) C (d) D

3 A magnetic flux whose density is changing by a constant rate is acting normally to a circular coil so that an induced electromotive force (E) is produced in the coil. If the number of turns of the coil gets increased to its double and its area gets decreased to its half while the rate of change of the magnetic flux density remains the same, the induced electromotive force in the coil will be ..... . *(Exp. June 21)*

(a) E (b) 4 E (c) $\frac{1}{2}$ E (d) $\frac{1}{4}$ E

4 A student conducts the following steps by using the tools shown in the figure. *(Exp. June 21)*



Step (I): moving the magnet towards the stationary solenoid while the solenoid is kept settled.

Step (II): moving both the solenoid and the magnet with the same velocity and in the same direction.

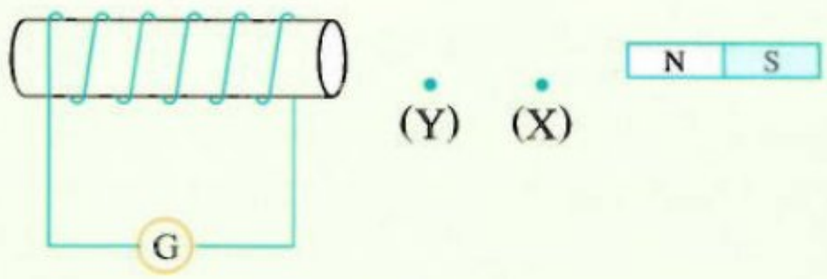
Step (III): moving both the solenoid and the magnet with the same speed but in two opposite directions.

Which of the above steps does not produce an induced electromotive force in the coil?

(a) Step (I) only. (b) Step (II) only. (c) Step (III) only. (d) All steps.

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5 The figure illustrates an experiment so that when the magnet is moved towards the coil with velocity v from point (X) to point (Y), the pointer of galvanometer gets deflected with 2 divisions from zero to the right of its scale.



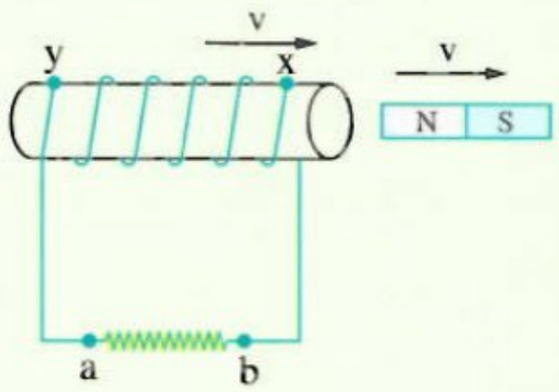
If the experiment is repeated such that the south pole of the magnet is facing the coil and is moving towards the coil with velocity 2 v from point (X) to point (Y), so the pointer of galvanometer deflects with ..... .

(Exp. June 21)

(a) 4 divisions to the left
(b) 4 divisions to the right
(c) 2 divisions to the left
(d) 2 divisions to the right

6 A magnet is moving as shown in the figure, if the coil is moved with the same velocity and in the same direction as the magnet, so ..... .

(1st session 21)



(a) the potential of point (a) is greater than the potential of point (b)
(b) the potential of point (x) is less than the potential of point (y)
(c) the potential of point (x) is greater than the potential of point (y)
(d) the potential of point (a) equals the potential of point (b)

7 Two circular coils (1), (2) of cross-sectional areas A_1, A_2 respectively, having the same number of turns are placed in a magnetic field perpendicular to their plane. When the magnetic flux density is changed through them in the same rate, it is noticed that the average induced emf in coil (1) equals double its value in coil (2), so ..... .

(1st session 21)

(a) $A_1 = 2\,A_2$ (b) $A_1 = 4\,A_2$ (c) $A_1 = \frac{1}{2}\,A_2$ (d) $A_1 = \frac{1}{4}\,A_2$

8 Two coils (x), (y) where cross-section of (x) is double that of (y), are placed in a magnetic field of flux density B where plane of each coil is perpendicular to the direction of the flux lines. When reversing the direction of the magnetic field acting on the coils through a time interval of 2 ms, the ratio between $\frac{\text{The average induced emf in coil x}}{\text{The average induced emf in coil y}} = \frac{3}{1}$, so the ratio between $\frac{\text{Number of turns of coil x}}{\text{Number of turns of coil y}}$ = ...

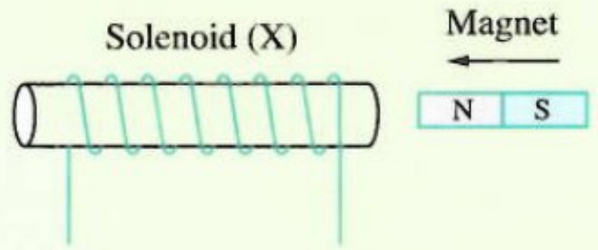
(1st session 21)

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{4}{3}$

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9 A student conducted Faraday's experiment to generate an induced electromotive force in a coil. He conducted the following steps to increase the average induced electromotive force in the coil (X):



(I) replacing the coil (X) by another greater in cross-sectional area.

(II) replacing the coil (X) by another higher in number of turns.

(III) increasing the time of magnet motion.

Which of the above steps lead to achieve the purpose of the student? *(2nd session 21)*

(a) (I) and (III). (b) (I) and (II). (c) (II) and (III). (d) (I), (II) and (III).

10 When the magnetic flux interacting with a circular coil changes, an induced electromotive force (E) is produced in the coil. If the number of turns of the coil is increased four times with the same area and the rate of change of the magnetic flux interacting with the coil is decreased to its half, then the induced electromotive force will equal ..... . *(2nd session 21)*

(a) 2 E (b) 4 E (c) $\frac{1}{2}$ E (d) $\frac{1}{4}$ E

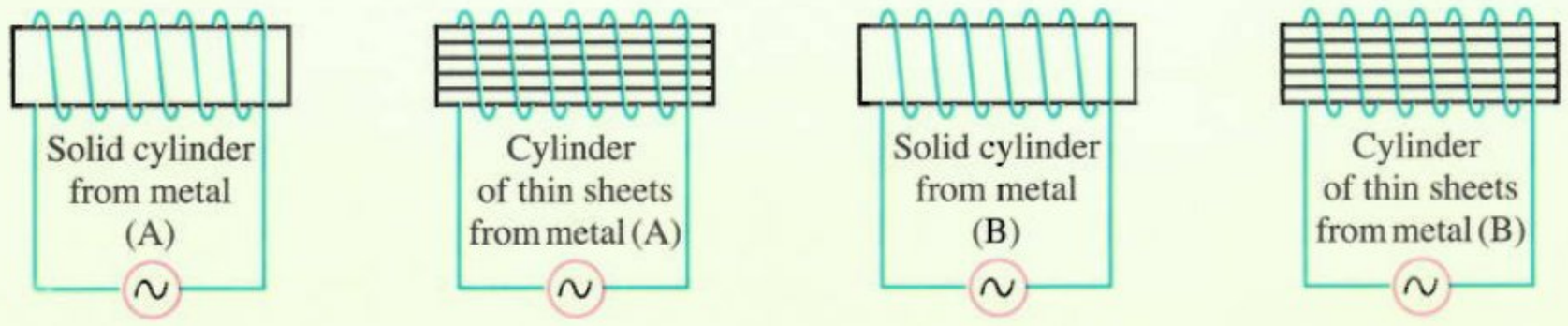
11 Two coils (X) and (Y), the area of coil (X) is double that coil (Y) while the number of turns of (X) is $\frac{1}{3}$ that of coil (Y). When the two coils are placed inside a magnetic field where the coils' planes are perpendicular to the flux lines direction and the flux density that affects the two coils changes by the same rate, then the ratio of

$$\frac{\text{The average induced e.m.f in coil (X)}}{\text{The average induced e.m.f in coil (Y)}} = \text{.....}$$

(1st session 22)

(a) $\frac{1}{6}$ (b) $\frac{3}{4}$ (c) $\frac{2}{3}$ (d) $\frac{2}{5}$

12 The figure shows 4 similar AC circuits, if you know that the resistivity of metal (A) is greater than the resistivity of metal (B). *(1st session 22)*



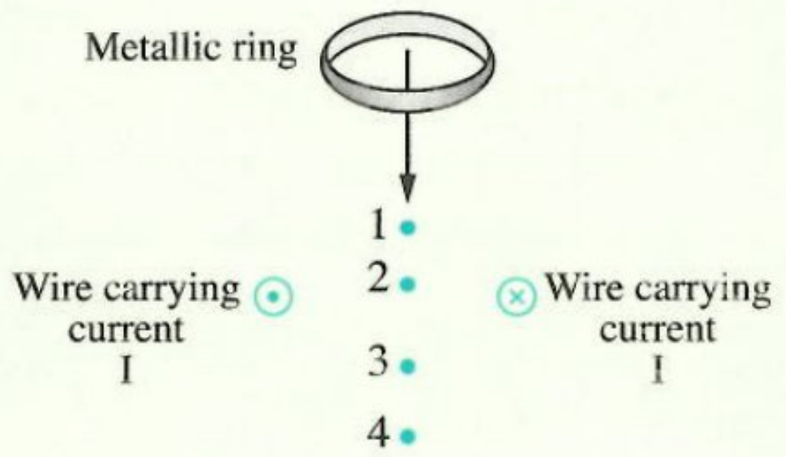
Circuit (1) Circuit (2) Circuit (3) Circuit (4)

In which electric circuit, the metal cylinder has the greatest generated eddy current?

(a) Circuit (3) (b) Circuit (1) (c) Circuit (2) (d) Circuit (4)

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13 The figure shows two straight current-carrying wires placed perpendicular to the page.
If a metallic ring moves in the plane of the page downwards to cut the resulted magnetic field due to the currents in the two wires such that the plane of the ring remains perpendicular to the page. At which points (1, 2, 3, 4) an induced electric current is generated in the ring and produces a magnetic field opposite to the original magnetic field at the mid-point between the two wires? *(1st session 22)*



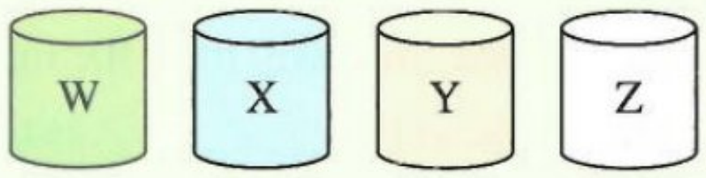
(a) 1, 3 (b) 2, 3 (c) 1, 2 (d) 1, 4

14 Two circular coils (1) and (2) have number of turns (N_1) and (N_2) respectively. If they have the same area and are placed in a magnetic flux perpendicular to their plane. When the magnetic flux density linked with the two coils changes with the same rate, it is noticed that the average induced electromotive force in coil (2) equals quarter that induced in coil (1). So, ...
(2nd session 22)

(a) $N_1 = \frac{1}{4} N_2$ (b) $N_1 = 8 N_2$

(c) $N_1 = 4 N_2$ (d) $N_1 = \frac{1}{8} N_2$

15 The figure shows four metallic cylinders of the same dimensions but of different metals. The table illustrates the values of the electrical conductivity for these metals. When the cylinders are exposed to the same variable magnetic flux resulting from an alternating current source, and by neglecting the differences in the magnetic permeability of these metals, then in which cylinder the amount of heat energy generated due to the eddy currents will be the least? *(2nd session 22)*



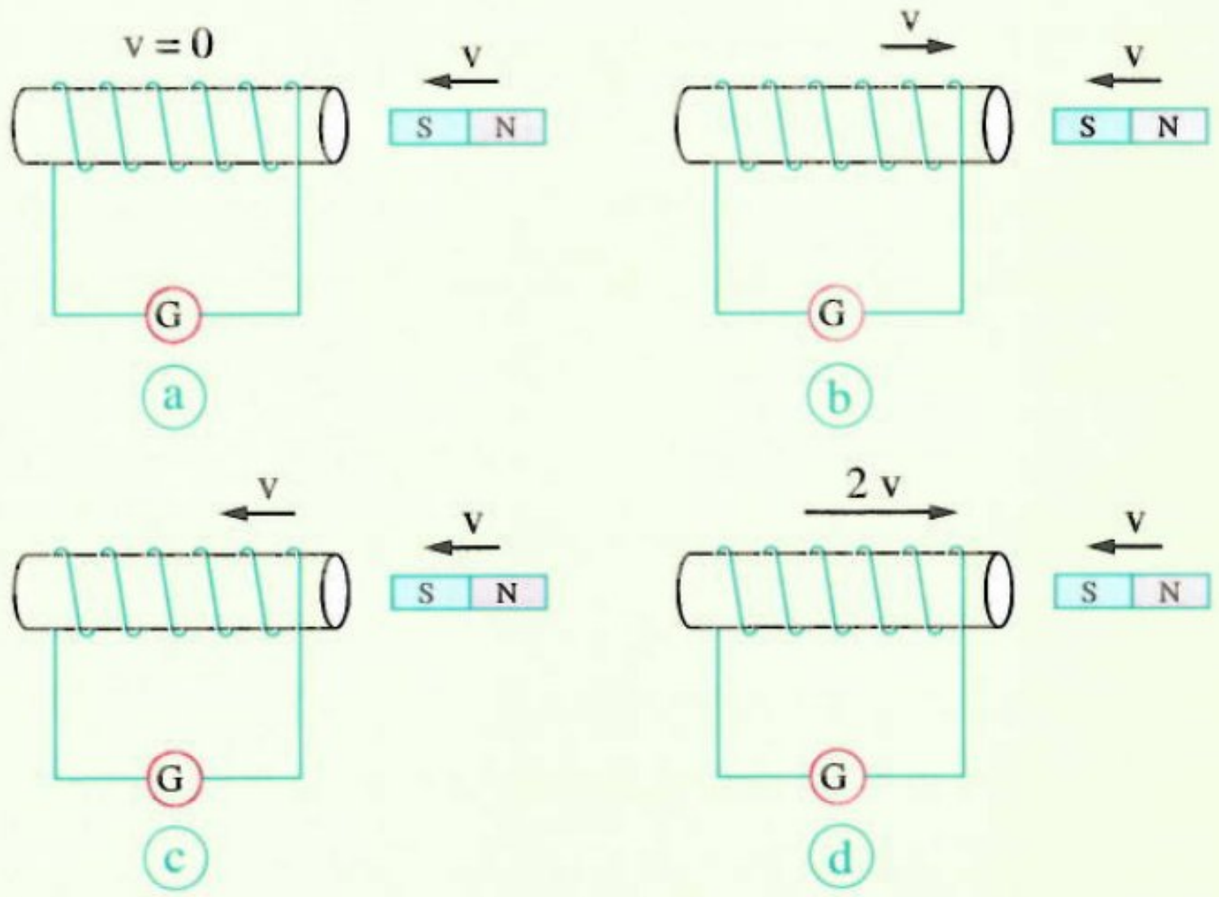
Metal	Electrical conductivity
W	$5.96 \times 10^7\ \Omega^{-1}.m^{-1}$
X	$3.5 \times 10^7\ \Omega^{-1}.m^{-1}$
Y	$2.98 \times 10^7\ \Omega^{-1}.m^{-1}$
Z	$0.217 \times 10^7\ \Omega^{-1}.m^{-1}$

(a) W (b) X (c) Y (d) Z

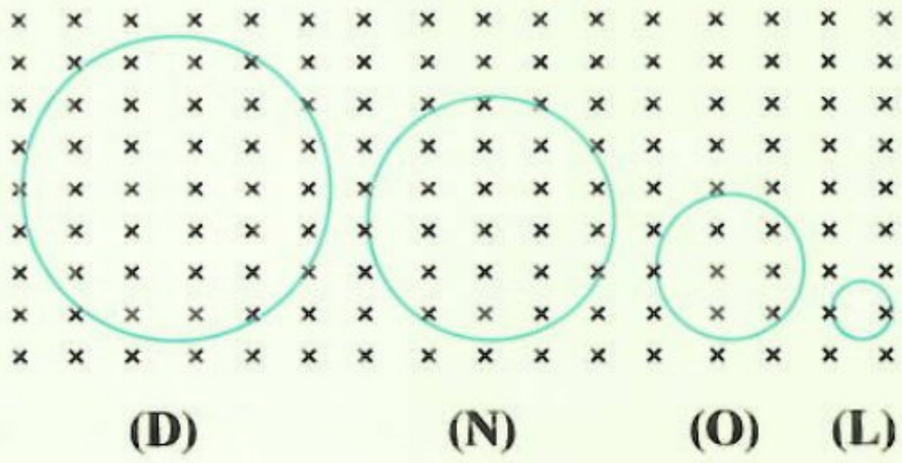
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16 A magnet, a solenoid and a galvanometer are used to verify Faraday's law for electromagnetic induction. If the experiment is repeated four times such as the magnet and the solenoid were moved with the speeds shown in the four figures. So, the pointer of the galvanometer has the greatest deflection in the experiment ..... . *(2nd session 22)*

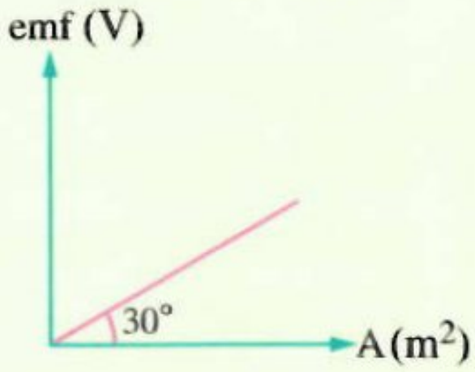


17 Four rings of different radii are made of a copper wire of uniform cross-section. All rings lie in the plane of the page and are exposed to a uniform magnetic field as in the figure. If the magnetic field vanished, which of the rings will have the highest induced current? *(Exp. 23)*



(a) D (b) L (c) O (d) N

18 A group of coils of different areas, the number of turns of each coil is 100 turns, are placed in the same plane and get exposed to a variable magnetic flux for the same time. The opposite graph represents the relation between the average induced electromotive force (emf) in each coil and the area (A) of the coil, so the rate of change in the magnetic flux density is ..... . *(Exp. 23)*

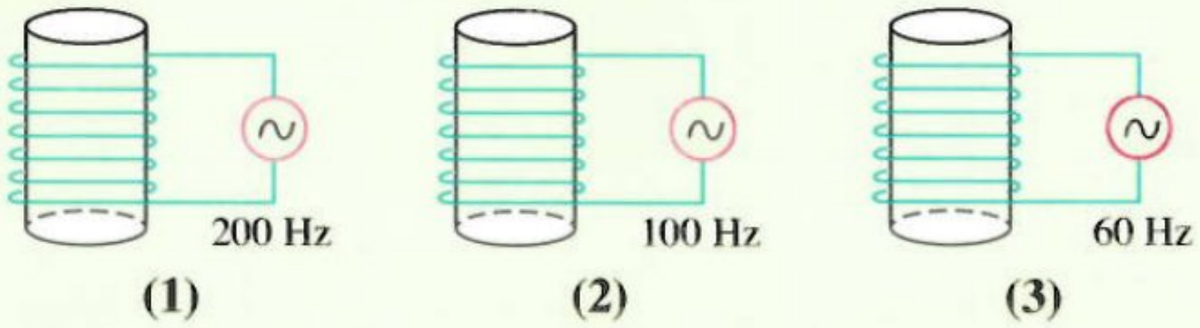


(a) 5.77×10^{-4} T/s (b) 5.77×10^{-2} T/s

(c) 0.577 T/s (d) 5.77×10^{-3} T/s

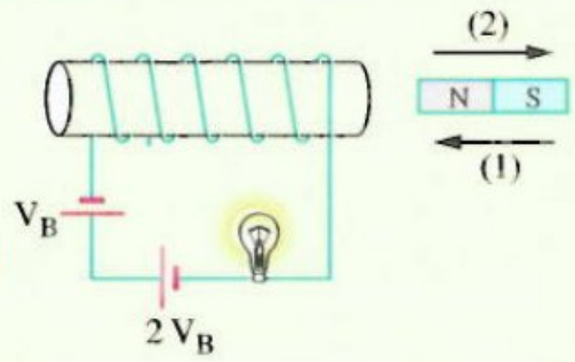
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19 The figure shows three identical metallic blocks inside three identical coils, if an alternating current with the same voltage (but with different frequencies) passes through the three coils for the same time, then the temperature of the three blocks increased, which of the following choices is correct concerning the temperature (T) of the three blocks? *(Exp. 23)*



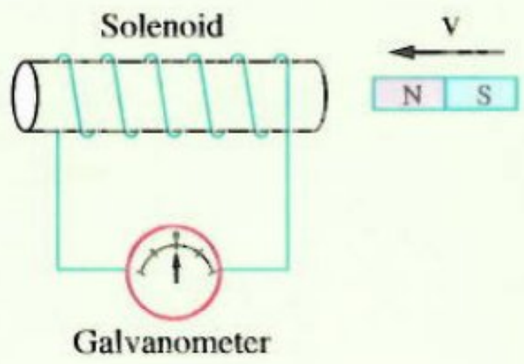
(a) $T_1 > T_2 > T_3$ (b) $T_2 > T_1 > T_3$ (c) $T_2 > T_3 > T_1$ (d) $T_3 > T_1 > T_2$

20 At the moment of moving the magnet in either of the two directions (1) and (2) with the same speed, e.m.f of 0.5 V_B is induced in the coil. Which of the following choices is correct at the moment of moving the magnet? *(1st session 23)*



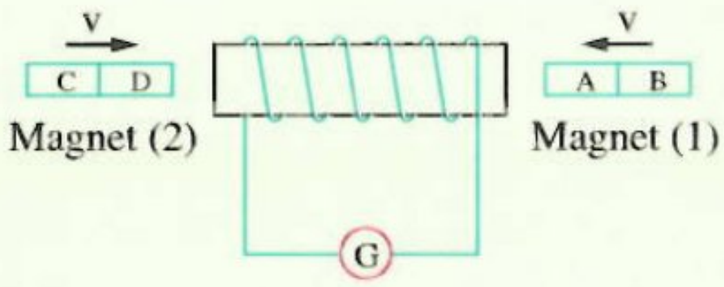
(a) The brightness of the lamp vanishes momentarily on moving the magnet in the direction (2).
(b) The brightness of the lamp increases on moving the magnet in the direction (2).
(c) The brightness of the lamp remains constant on moving the magnet in both directions (1) or (2).
(d) The brightness of the lamp increases on moving the magnet in the direction (1).

21 The figure illustrates a magnet moving with velocity (v) to the left towards a solenoid connected to a galvanometer. Despite this no current is induced in the coil, because the solenoid moves ..... . *(1st session 23)*



(a) with velocity (v) to the left (b) with velocity (2 v) to the left
(c) with velocity (v) to the right (d) with velocity (2 v) to the right

22 Two identical magnets (1), (2) were placed at the same distance from a solenoid as shown in the figure. On moving each of the magnets with the same speed simultaneously towards the two terminals of the solenoid and along its axis, it is noticed that the pointer of the galvanometer doesn't deflect. This is due to ..... . *(2nd session 23)*



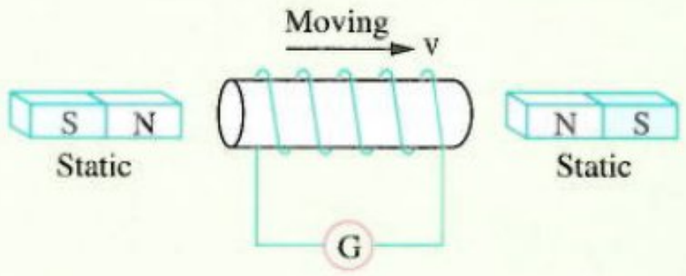
(a) pole (A) is north and pole (D) is north (b) pole (A) is north while pole (D) is south
(c) pole (A) is south while pole (D) is north (d) pole (B) is south and pole (D) is south

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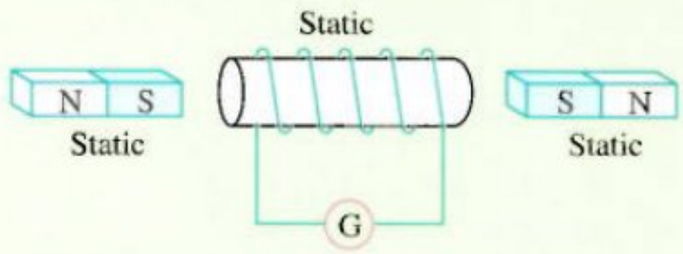
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23 Each figure of the following shows a solenoid which is connected to a galvanometer where two magnets are placed such that each is close to one of its two ends. If the solenoids are similar and the magnets are similar and have the same distances from the solenoid at the instant of measuring the emf.

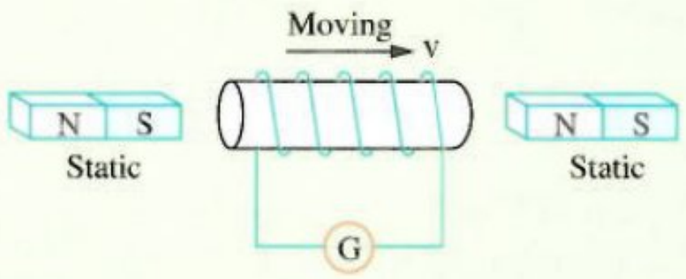
(1)



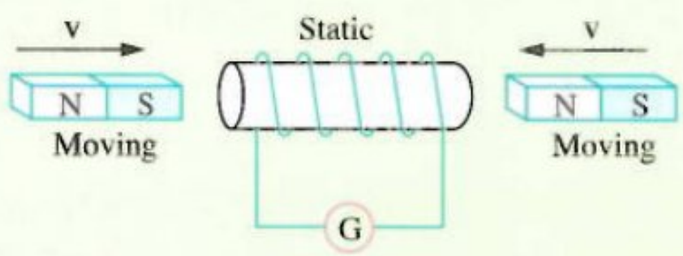
(2)



(3)



(4)



What is the correct order for the magnitude of the induced emf in each solenoid according to its state of motion which is shown in each figure? *(1st session 24)*

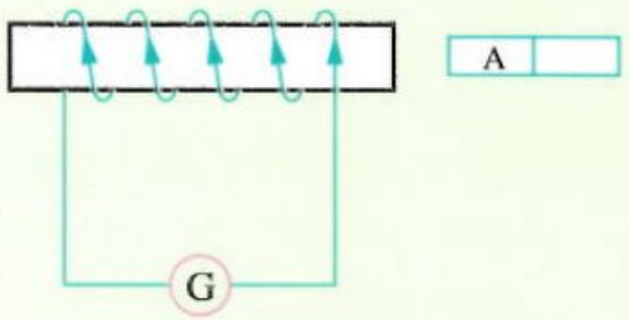
(a) $(emf)_2 = (emf)_4 > (emf)_1 = (emf)_3$

(b) $(emf)_1 = (emf)_4 > (emf)_2 = (emf)_3$

(c) $(emf)_4 = (emf)_2 > (emf)_1 > (emf)_3$

(d) $(emf)_1 = (emf)_3 > (emf)_2 = (emf)_4$

24 The following table shows multiple probabilities for getting induced electric current in the solenoid by moving the shown magnet in the figure:

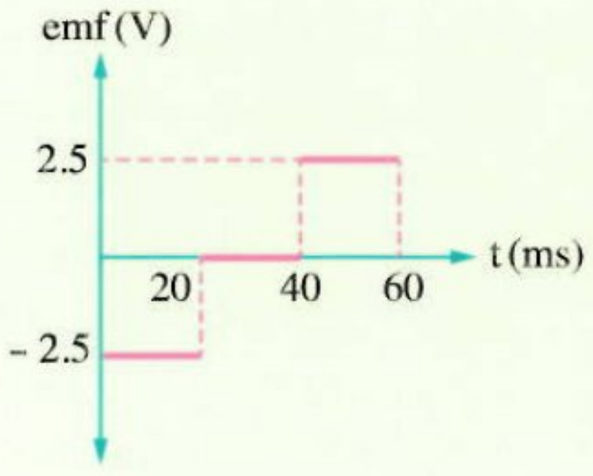


Probability	Pole A	The magnet movement
1	South	Approaches the solenoid
2	South	Gets away from the solenoid
3	North	Approaches the solenoid
4	North	Gets away from the solenoid

Which two probabilities lead to generating electric current in the shown direction in the figure? *(1st session 24)*

(a) 1, 2 (b) 1, 4 (c) 3, 4 (d) 2, 3

25 The opposite graph represents the relation between the induced emf in a square metal frame that enters a region of magnetic field of flux density 0.2 T with a uniform velocity till it gets out of it and the time (t), so the area of the metal frame equals ..... . *(1st session 24)*



(a) 0.5 cm^2 (b) 0.5 m^2

(c) 0.25 cm^2 (d) 0.25 m^2

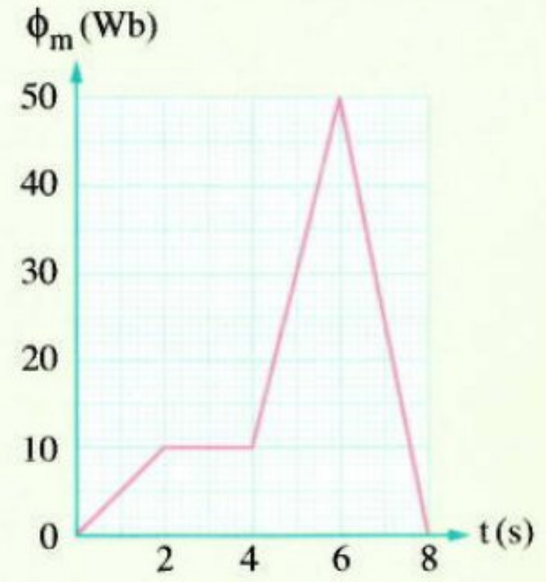
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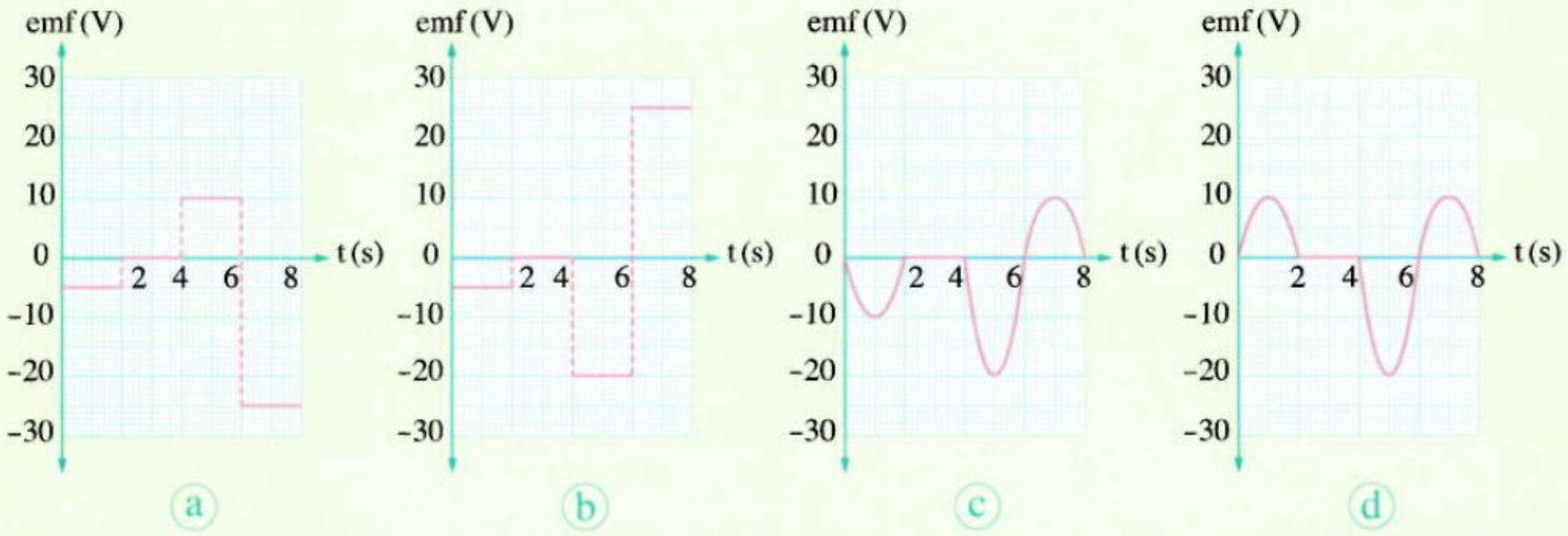
26 A circular coil with 60 turns and a face area of 36 cm^2 is penetrated by a perpendicular magnetic flux of density 10^{-6} T. The average induced electromotive force in the coil during its rotation for half a cycle from the vertical position in a time of 400 ms equals ..... .

(1st session 24)

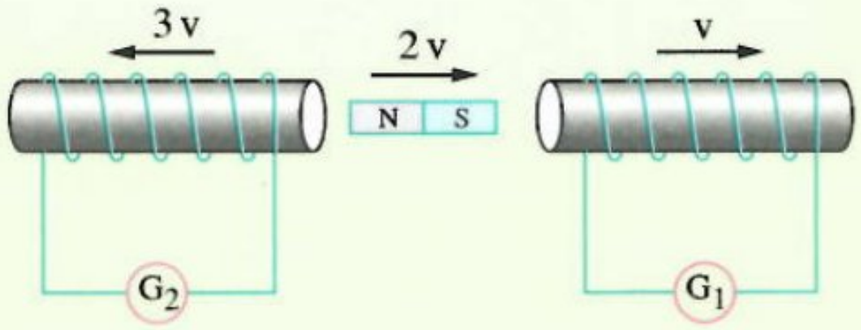
(a) 1.08 nV (b) 0.54 μV (c) 1.08 μV (d) 0.54 nV

27 The opposite graph shows the change in the magnetic flux that penetrates a circular coil consisting of a single turn. Which of the following graphs represents the relationship between the induced electromotive force (emf) in the coil and time (t) during the same time interval? *(2nd session 24)*





28 Two identical galvanometers connected to two identical coils where a magnet is placed at the mid-distance between the coils. If the magnet and the coils move at the speeds shown in the opposite figure, then ..... .



	The relation between the readings of the galvanometers	The currents
(a)	$G_2 > G_1$	in the same direction
(b)	$G_2 > G_1$	in opposite directions
(c)	$G_1 > G_2$	in opposite directions
(d)	$G_1 > G_2$	in the same direction

(2nd session 24)

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29 A magnetic flux penetrates a coil with 10 turns. If the magnetic flux decreases by 0.3 mWb within 0.02 s, then the induced electromotive force in the coil equals ..... .

(a) 0.15 V (b) 15 V (c) 150 V (d) 1.5 V *(2nd session 24)*

30 A circular coil with 200 turns and a face area of 5 cm^2 rotates inside a magnetic field of density 6×10^{-4} T around a fixed axis perpendicular to the flux direction. This motion generates an average induced electromotive force of 0.3 mV over a time interval of 400 ms. Which of the following options will produce the given electromotive force? *(2nd session 24)*

(a) The coil rotates $\frac{1}{2}$ cycle from the perpendicular position to the magnetic flux.

(b) The coil rotates $\frac{1}{4}$ cycle from the perpendicular position to the magnetic flux.

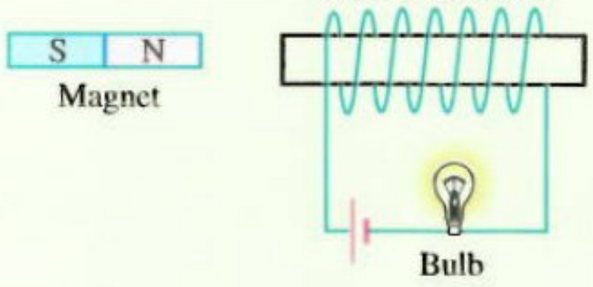
(c) The coil rotates $\frac{1}{2}$ cycle from the parallel position to the magnetic flux.

(d) The coil rotates $\frac{3}{4}$ cycle from the parallel position to the magnetic flux.

31 Two circular coils are placed perpendicular to a magnetic field that its density is changing uniformly where area of the first coil is double that of the other, and number of turns of the first coil is twice that of the other, so the ratio between the generated emf in the first coil and that generated in the second equals ..... . *(Practice 25)*

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{4}$ (d) $\frac{4}{1}$

32 In the opposite figure, if the induced emf in case of being generated is less than that of the battery, what happens to the luminosity of the electric bulb during approaching the magnet towards the coil? *(Practice 25)*



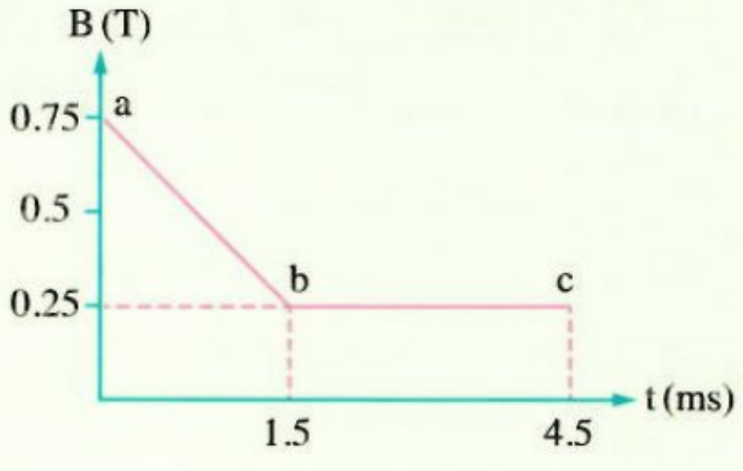
(a) The luminosity of the bulb increases.

(b) The luminosity of the bulb decreases.

(c) The luminosity of the bulb does not change.

(d) The luminosity of the bulb vanishes.

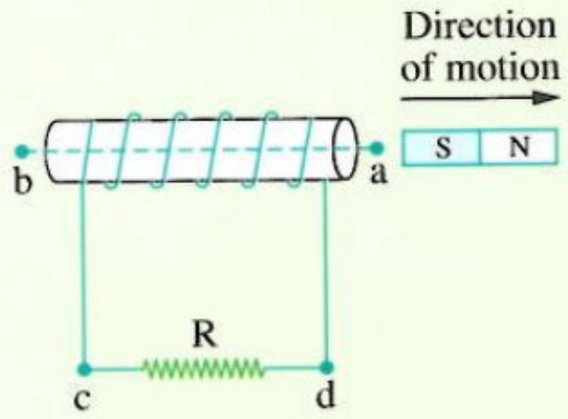
33 The opposite graph represents the relation between the magnetic flux density (B) inside which a coil of 200 turns each of area 30 cm^2 is placed and the time (t). If the direction of the magnetic flux is parallel to the coil axis, so the average induced emf in the coil during the interval (bc) equals ..... . *(Practice 25)*



(a) 0 (b) 25 V (c) 35 V (d) 50 V

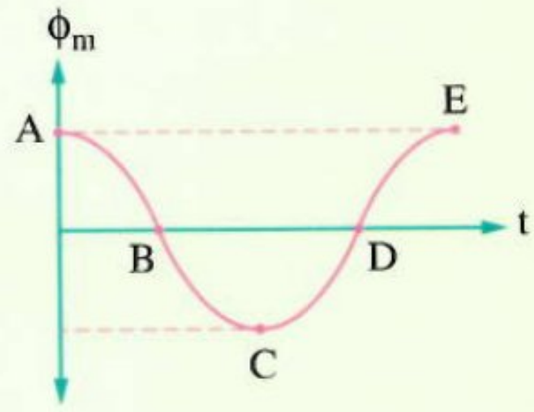
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34 In the opposite figure, when moving the magnet away from the solenoid, *(Practice 25)*



	Direction of the induced magnetic flux in the solenoid is:	Direction of the current passing in the resistor (R) is:
(a)	b ⟶ a	c ⟶ d
(b)	a ⟶ b	d ⟶ c
(c)	b ⟶ a	d ⟶ c
(d)	a ⟶ b	c ⟶ d

35 The opposite figure represents the change in the magnetic flux which cuts the coil through a certain time interval, so the induced emf in the coil becomes maximum at the points ..... . *(Practice 25)*

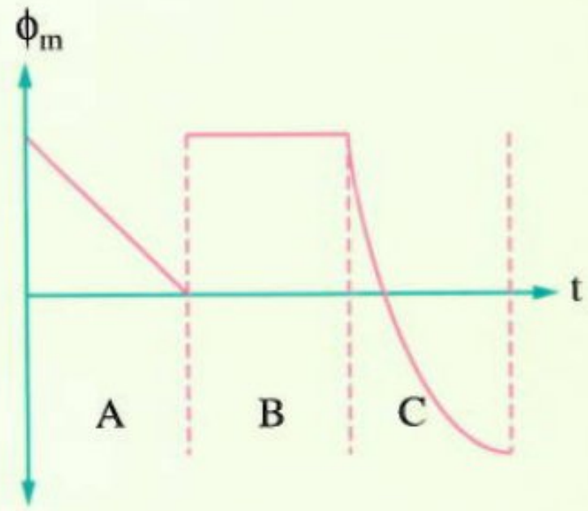


(a) A, E (b) C, D
(c) D, B (d) A, E, C

36 Which of the following actions does not lead to generating an induced emf in a coil placed perpendicular to a magnetic flux? *(Practice 25)*

(a) Changing the coil area which is exposed to the magnetic flux.
(b) Keeping the coil static inside a variable magnetic flux.
(c) Keeping the coil static inside a uniform magnetic flux.
(d) Changing the angle between the coil and the magnetic flux.

37 A variable magnetic flux is penetrating a metal ring where the opposite graph represents the relation between the magnitude of the magnetic flux and time, so the time interval in which a regularly decreasing electric current passing in the ring is the interval ..... . *(Practice 25)*

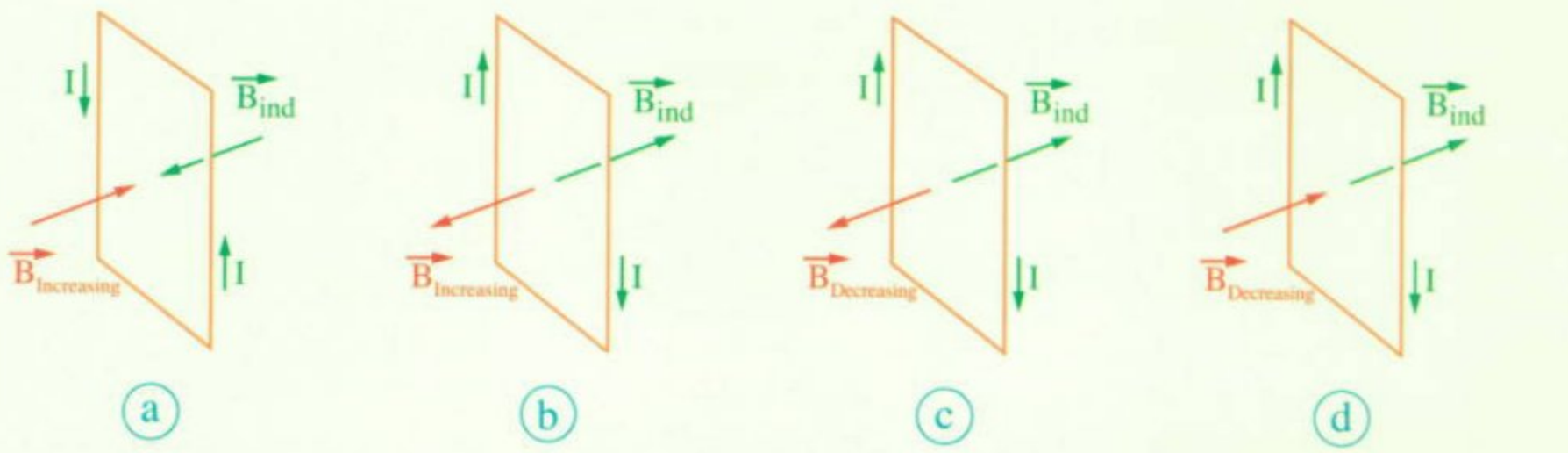


(a) A (b) B
(c) C (d) A, B

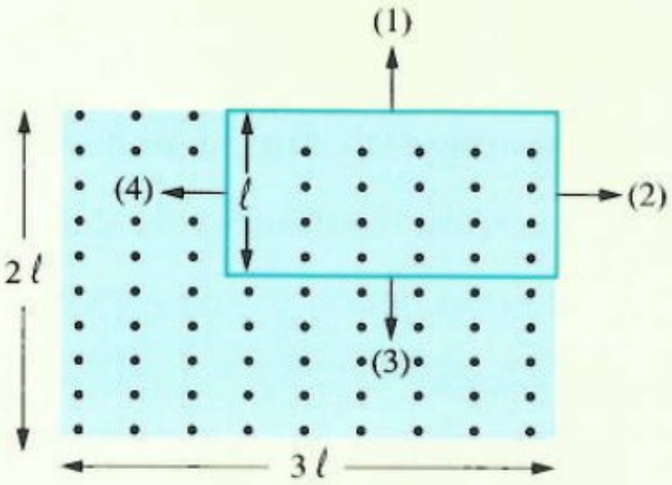
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38 The following figures show generating an induced current (I) in a coil due to increasing or decreasing the density of the magnetic flux (B) which is acting on the coil. Which of the figure based on Lenz's rule is incorrect? *(Practice 25)*



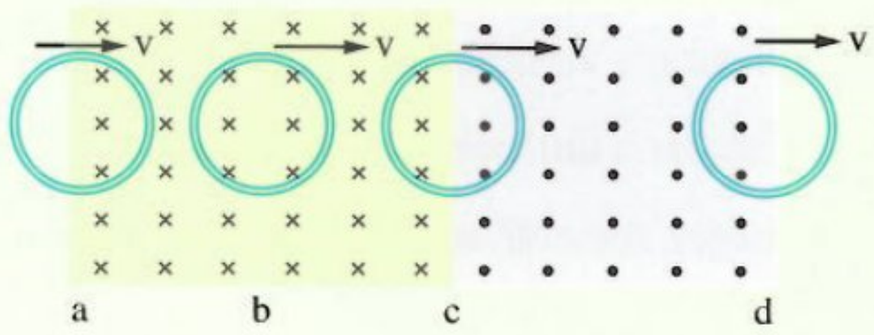
39 A metal frame of width ℓ and length 2ℓ is placed perpendicular to a uniform magnetic field limited to the shaded area as shown in the figure. If the frame moves a distance ℓ over a time t, in the direction that generates the maximum average induced electromotive force (emf), then that direction is ..... . *(Practice 25)*



(a) direction 1 (b) direction 2

(c) direction 3 (d) direction 4

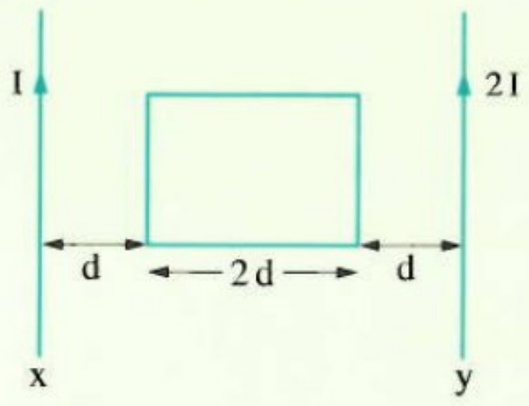
40 A metal ring moves with a uniform velocity v in the shown direction.



So, an induced electric current passes in the ring in clockwise direction at the moment it passes the position ..... . *(Practice 25)*

(a) a (b) b (c) c (d) d

41 In the shown figure, a metal frame and two long parallel wires (x), (y) are all in the same plane and carrying electric current. So, the direction of the induced current in the metal frame if within a time t: *(Practice 25)*



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	The current of wire (x) vanished	The current of wire (y) vanished
a	In the clockwise direction	In the clockwise direction
b	In the counter-clockwise direction	In the counter-clockwise direction
c	In the clockwise direction	In the counter-clockwise direction
d	In the counter-clockwise direction	In the clockwise direction

42 In figure (1), as the magnet approaches a solenoid, the galvanometer pointer deflects by an angle (θ) in a certain direction. In figure (2), the experiment is repeated using two identical magnets, similar to the one in figure (1),

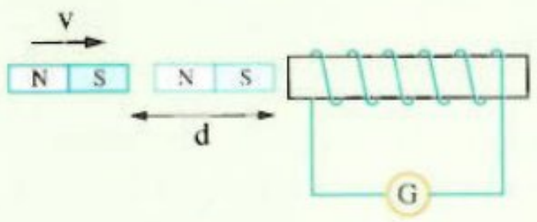


Figure (1)

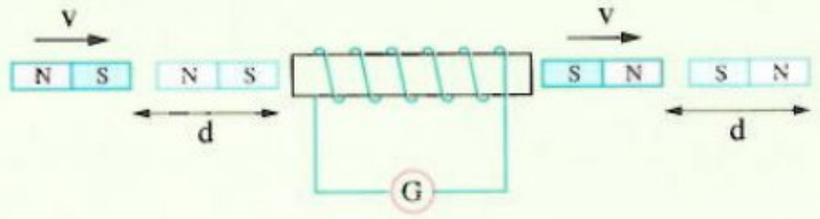


Figure (2)

which of the following choices is correct?

(a) The galvanometer pointer deflects by an angle (2θ) in the opposite direction.

(b) The galvanometer pointer deflects by an angle (2θ) in the same direction.

(c) The galvanometer pointer does not deflect.

(d) The galvanometer pointer deflects by an angle (θ) in the same direction. *(Practice 25)*

43 In figure (1) a solenoid of aluminum is moved towards a strong magnet, so induced electromotive force of value (emf) is generated at its ends, if the aluminum solenoid is replaced with another made of copper having the same number of turns and cross-sectional area, then the experiment is repeated in the same velocity as shown in figure (2), then the value of the induced emf in the copper coil is ..... .

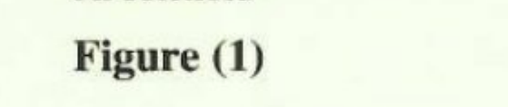


Figure (1)

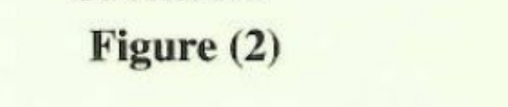


Figure (2)

(Practice 25)

(a) greater than emf

(b) less than emf

(c) equal to emf

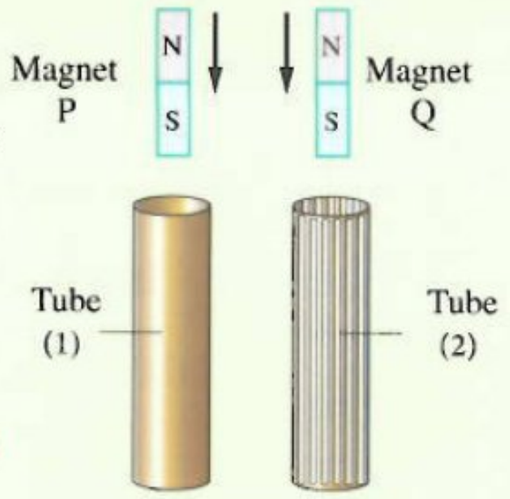
(d) equal to zero

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44 Two identical magnets are suspended vertically above two tubes (1), (2) respectively. The tubes are made of copper where tube (2) is divided into insulated longitudinal sheets as in the opposite figure. when the magnets are dropped at the same moment to pass through the tubes without touching them, two induced currents I_1, I_2 are generated respectively. Which of the magnets gets out completely from the tube first? And why? *(Practice 25)*



(a) Q, the reason $I_1 > I_2$

(b) Q, the reason $I_1 < I_2$

(c) P, the reason $I_1 > I_2$

(d) P, the reason $I_1 < I_2$

45 Which of the following action does not decrease the eddy currents in the metal core inside a coil carrying an electric current? *(Practice 25)*

(a) Making the metal core in the form of lamina insulated from each other.

(b) Making the metal core from a material of high conductivity.

(c) Using a current of low frequency.

(d) Making the metal core from a material of high resistivity.

46 Each of the following figures shows the relative motion between a magnet and a coil connected to a galvanometer during a certain period of time. Which figure shows the case in which the largest average electromotive force is induced in the coil? *(1st session 25)*

(a)

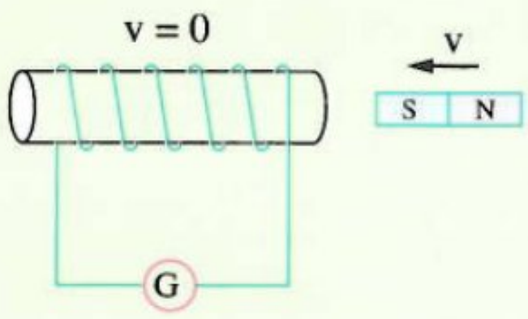


Figure (1)

(b)

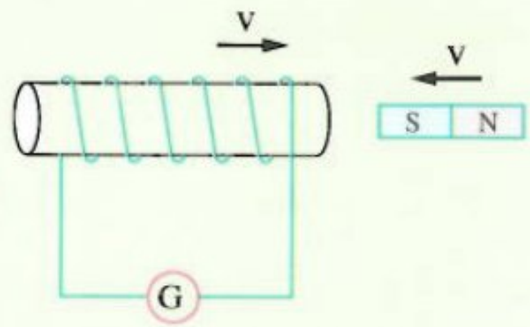


Figure (2)

(c)

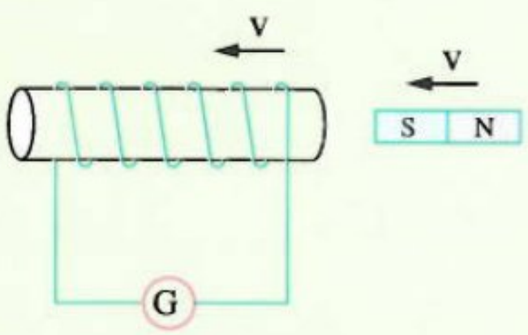


Figure (3)

(d)

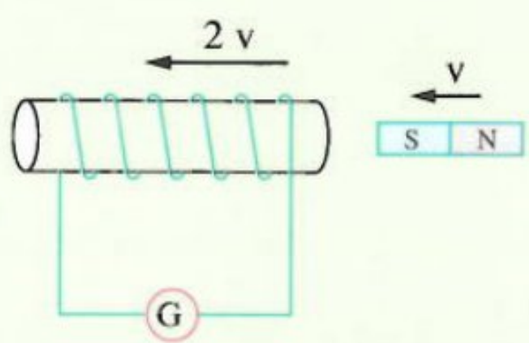
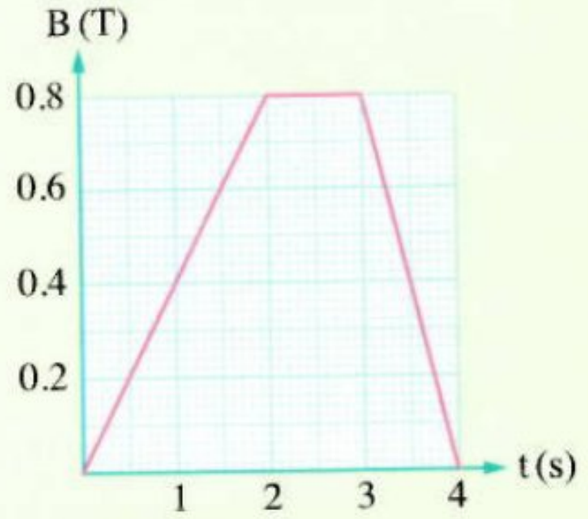


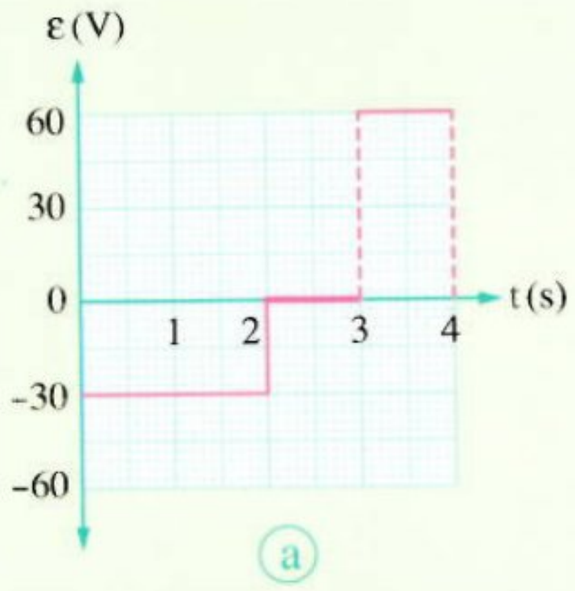
Figure (4)

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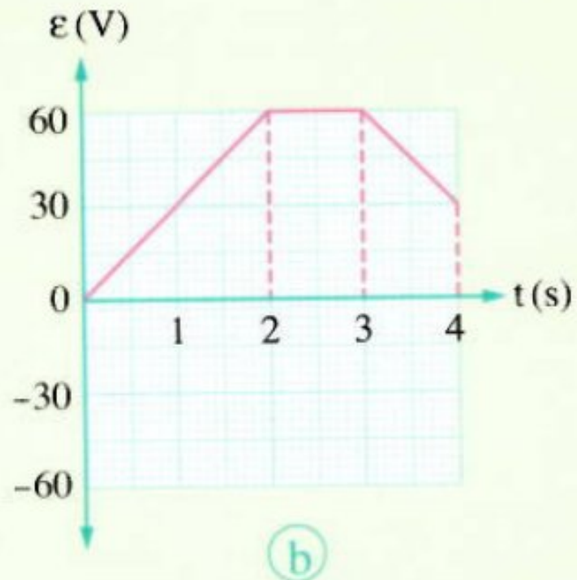
47 A circular coil of 100 turns, each has a cross-sectional area of 0.75 m^2, is placed perpendicular to the direction of magnetic flux whose density can be varied. The opposite graph represents the relationship between the magnetic flux density (B) and the time (t).



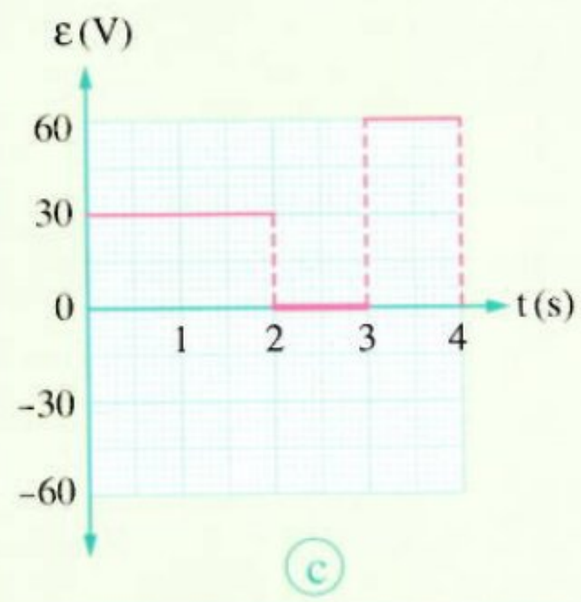
Which of the following graphs represents the relation between the induced electromotive force (ε) in the coil and time (t)? *(1st session 25)*



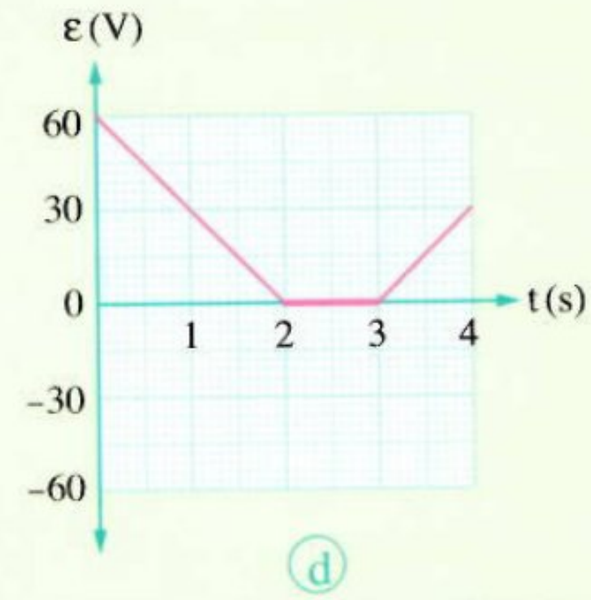
(a)



(b)



(c)



(d)

48 ✳ You have a thick metal piece placed in the core of an insulated solenoid through which an alternating current is passing. The intensity of eddy currents induced in the metal piece increases as ..... .

(1st session 25)

(a) the frequency of the alternating current in the coil increases

(b) the frequency of the alternating current in the coil decreases

(c) the resistance of the metal piece inside the coil increases

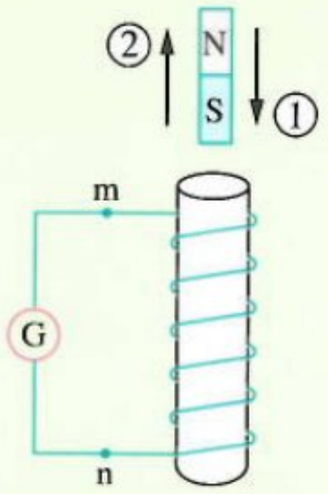
(d) the thickness of the metal piece inside the coil decreases

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49 The figure shows Faraday's experiment for electromagnetic induction. As the magnet is moved into the coil in direction ① and out of the coil in direction ② respectively.
The electric potential at point (m) shown in the figure compared to the electric potential at point (n) ..... . *(2nd session 25)*

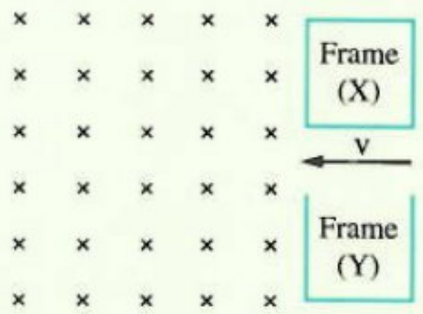


	When the magnet is moved into the coil in direction ①, the potential at point (m) is:	When the magnet is moved out of the coil in direction ②, the potential at point (m) is:
a	positive	negative
b	positive	positive
c	negative	positive
d	negative	negative

50 A solenoid of insulated wire is wound around a metal core. During the flow of a direct current through the solenoid, the metal core will have ..... . *(2nd session 25)*

a induced eddy currents in the same direction as that of the current in the coil
b induced eddy currents perpendicular to the direction of the current in the coil
c induced eddy currents in the opposite direction to that of the current in the coil
d no induced eddy currents

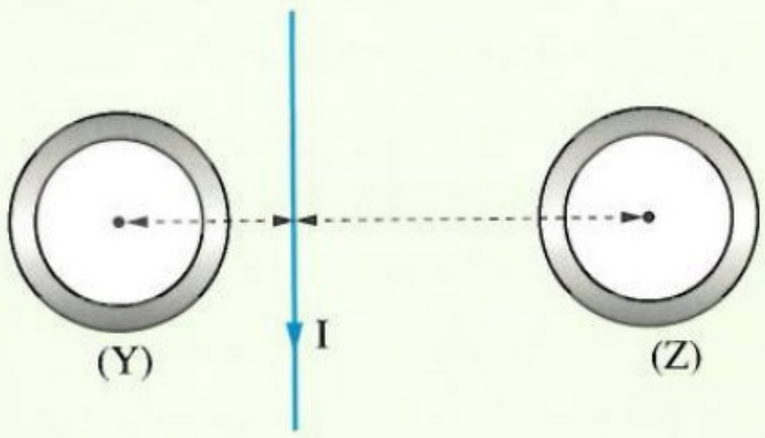
51 The figure shows two metal frames (X) and (Y), one is open and the other is closed. They move with velocity (v) on a smooth, frictionless surface toward a magnetic field as shown in the figure.
What happens to the velocity of the two frames as they enter the magnetic field? *(Practice 26)*



	Velocity of frame (X)	Velocity of frame (Y)
a	Remains constant	Increases
b	Remains constant	Decreases
c	Increases	Remains constant
d	Decreases	Remains constant

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52 The figure shows a part of a straight, current-carrying wire placed between two identical metal loops, all in the same vertical plane.



(1) Which loop (Y) or (Z) has a greater magnetic flux passing through it?

(2) Determine the direction of the induced current in the two loops when the current flowing through the wire increases steadily.

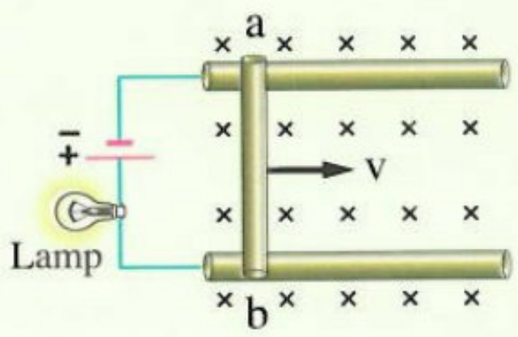
ⓐ Direction of the current in loop (Y).

ⓑ Direction of the current in loop (Z).

(2nd session 25)

The generated emf in a straight wire

53 In the opposite figure, during the motion of the rod (ab) to the right with velocity (v) as shown, the brightness of the lamp ..... . *(Exp. May 21)*
(Where the ohmic resistance of the rod and the two rails is negligible)

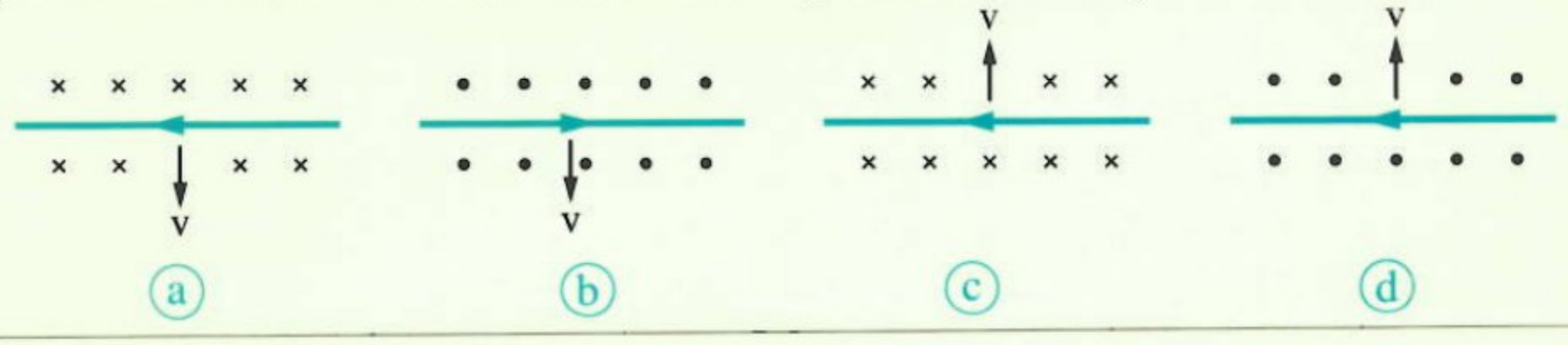


ⓐ vanishes
ⓑ increases
ⓒ doesn't change
ⓓ decreases

54 When a straight wire of unit length moves perpendicular to a magnetic field of flux density 0.4 T, an induced emf of 0.2 V gets generated between its terminals, so the velocity of the wire equals ..... . *(Exp. June 21)*

ⓐ 0.5 m/s
ⓑ 1 m/s
ⓒ 1.5 m/s
ⓓ 2 m/s

55 The following figures represent four straight wires. Each one is connected in a closed electric circuit while moving at velocity v in a uniform magnetic field. In which of these figures the direction of the induced current is represented correctly? *(Exp. June 21)*



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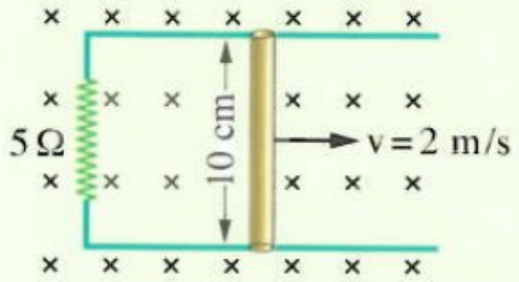
56 In the figure, a straight wire (AB) is placed in a uniform magnetic field that is perpendicular out of the page. To produce an induced emf such that the potential of point (A) becomes greater than the potential of point (B), the direction of motion of wire (AB) should be ..... .



(1st session 21)

(a) downwards of the page
(b) upwards of the page
(c) to the right of the page
(d) to the left of the page

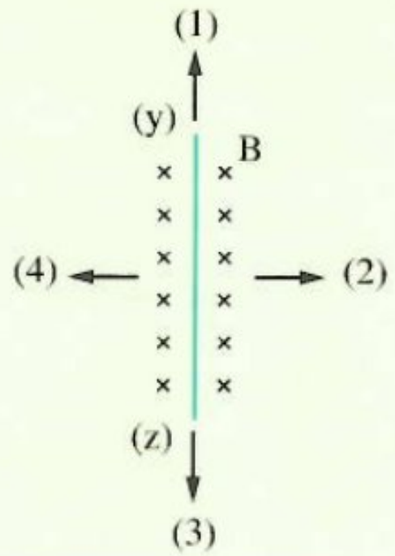
57 The figure represents a conductor moving perpendicular to a magnetic field of flux density 0.2 T, using the data on the figure, the passing current intensity in the resistor equals ..... .



(Where the ohmic resistance of each of the wire and the frame is negligible) *(1st session 21)*

(a) 4 mA
(b) 6 mA
(c) 8 mA
(d) 2 mA

58 The figure represents a straight wire (yz) that is moving in a uniform magnetic field (B) while being connected in a closed circuit, hence an induced current is produced in the direction from (z) to (y). To which direction (1), (2), (3) or (4) the wire (yz) has to be moved ? *(2nd session 21)*

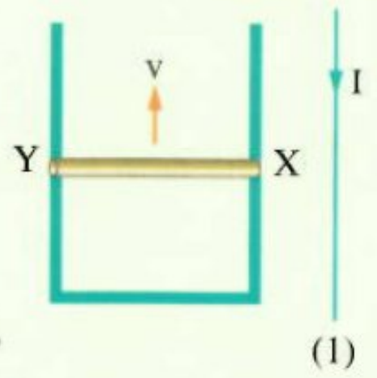


(a) (1)
(b) (2)
(c) (3)
(d) (4)

59 A straight wire of length 20 cm moves at velocity 0.5 m/s in a direction that makes angle (θ) with the direction of a magnetic field of flux density 0.4 T, so an induced emf of 20 mV gets generated between its terminals, hence the angle (θ) equals ..... . *(2nd session 21)*

(a) 60°
(b) 30°
(c) 45°
(d) 90°

60 The figure shows a straight current-carrying wire (1) through which a current of intensity (I) passes. When a wire (XY) is moved upwards with velocity (v) in the magnetic field of the electric current in wire (1), an induced electric current is generated in it from (X to Y).

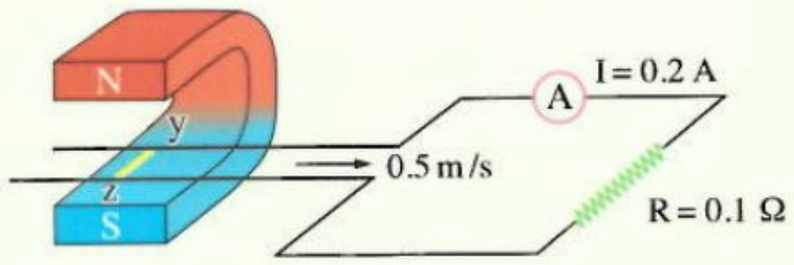


To decrease the intensity of the induced electric current to half its value, we must ..... . *(1st session 22)*

(a) increase the velocity of wire (XY) to its double
(b) decrease the electric current intensity in wire (1) to its quarter
(c) increase the velocity of wire (XY) to its four times
(d) decrease the electric current intensity in wire (1) to its half

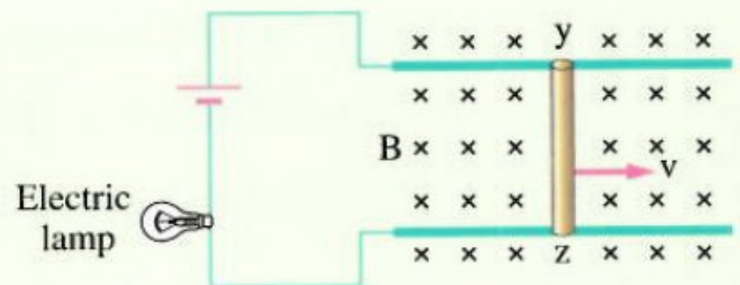
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61 The figure shows a metallic wire (yz) sliding on two metallic bars with velocity 0.5 m/s perpendicular to the direction of a magnetic field of flux density 2 T. If the reading of the ammeter is 0.2 A, so the length of the moving wire zy in the magnetic field equals ..... .



(a) 0.04 m (b) 0.02 m (c) 0.01 m (d) 0.03 m *(1st session 22)*

62 At moving the straight metallic wire (zy) toward right normal to the direction of a magnetic field (B) that is directed perpendicular into the page as shown in the figure where the induced emf in the wire is less than that of the battery.



Then : *(1st session 22)*

	The illumination of the electric bulb	The relation between the electric potential of points z and y
(a)	increases	the potential of point z is more than that of point y
(b)	increases	the potential of point z is less than that of point y
(c)	decreases	the potential of point z is less than that of point y
(d)	decreases	the potential of point z is more than that of point y

63 The figure represents a part of closed circuit that contains a straight wire (XY) that is placed in the same plane of the paper and moving upward with velocity v producing an induced current that passes from (X) to (Y). Which of the following figures represents the direction of magnetic flux acting on the wire according to the plane of the paper? *(2nd session 22)*

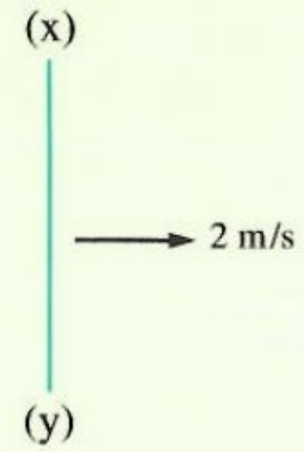
(a) (b) (c) (d)

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64 The figure shows a part of closed circuit that contains a straight wire (xy) of length 20 cm moving at velocity of 2 m/s perpendicular to the direction of a uniform magnetic flux, that produces an induced electromotive force of 0.02 V between the two terminals of the wire where the electric potential of point (x) is more than that of point (y). Then, the magnitude and the direction of the magnetic flux density are ..... .



(2nd session 22)

(a) 0.05 T, perpendicular into the page

(b) 0.5 T, perpendicular into the page

(c) 0.05 T, perpendicular out of the page

(d) 0.5 T, perpendicular out of the page

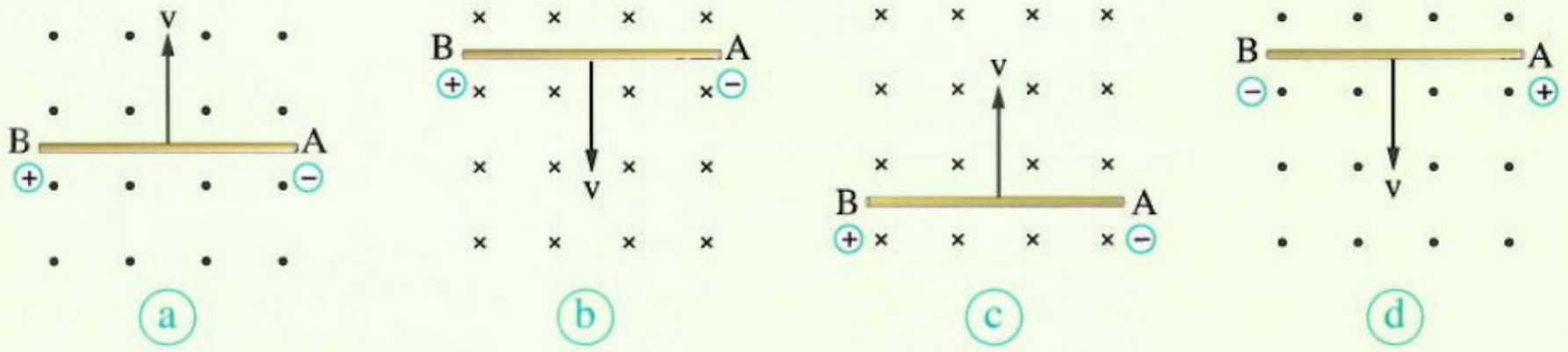
65 A wire of length 0.2 m is moved with velocity 2 m/s in a direction that makes an angle 30° with magnetic flux lines of density 0.4 T, so the instantaneous induced electromotive force is ..... .

(Exp. 23)

(a) 0.16 V (b) 0.32 V (c) 0.08 V (d) 0.24 V

66 A copper wire AB of length (ℓ) is placed in the same plane of the page and then it is moved perpendicular to a uniform magnetic field, which of the following figures correctly represents the polarity of the wire terminals?

(Exp. 23)



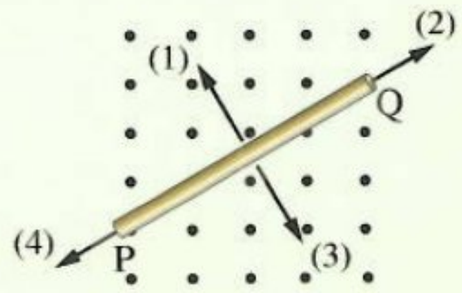
67 A copper wire of length ℓ is connected to a galvanometer. If the wire is moved with velocity v perpendicular to a magnetic field of flux density B, the pointer of the galvanometer deflects momentarily with an angle θ and when the velocity of the wire is increased to 2 v and the flux density is increased to 2 B, then the pointer of the galvanometer will deflect momentarily with an angle equals to ..... .

(Exp. 23)

(a) 2 θ (b) 4 θ (c) 6 θ (d) θ

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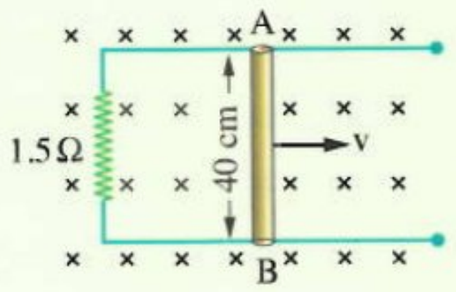
68 The opposite figure represents a uniform magnetic field acting on wire (PQ) that is placed in the same plane of the page and embedded in a closed circuit. If the direction of the induced current is from (Q) to (P), so the direction of motion of the wire is towards ..... . *(1st session 23)*



(a) (1) (b) (3) (c) (2) (d) (4)

69 The figure represents a wire (AB) that has a resistance of (0.5 Ω) moving perpendicular to a magnetic flux of density (0.2 T). In order to obtain an induced current of 0.1 A at the moment of the wire motion, the wire must move with a velocity equal to ..... .

(Neglect the resistance of the connecting wires) *(1st session 23)*



(a) 1.5 m/s (b) 1.875 m/s (c) 2.5 m/s (d) 0.625 m/s

70 A straight wire of length (ℓ) moves with velocity (v) inclined to a magnetic field of flux density (B) with an angle (30°) to the direction of the field, so an induced electromotive force is generated in it. To increase the electromotive force generated in it to the double ..... . *(2nd session 23)*

(a) the length of the wire must increase to (4 ℓ)

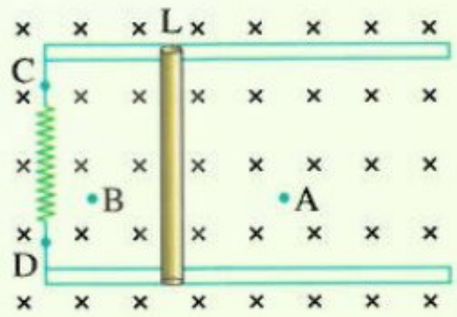
(b) the wire must move with velocity (3 v)

(c) the wire must move in a magnetic field of flux density = $\left(\frac{1}{2}B\right)$

(d) the wire must move perpendicular to the magnetic field

71 In the figure, the wire (L) is able to move in the same plane of the page through a magnetic field perpendicular into the page.

Which of the following choices is correct? *(2nd session 23)*



(a) If the wire is moved towards point "A", the potential of the point "C" will be greater than that of "D".

(b) If the wire is moved towards point "A", the potential of the point "C" will be less than that of "D".

(c) If the wire is moved towards point "B", the potential of the point "C" will be greater than that of "D".

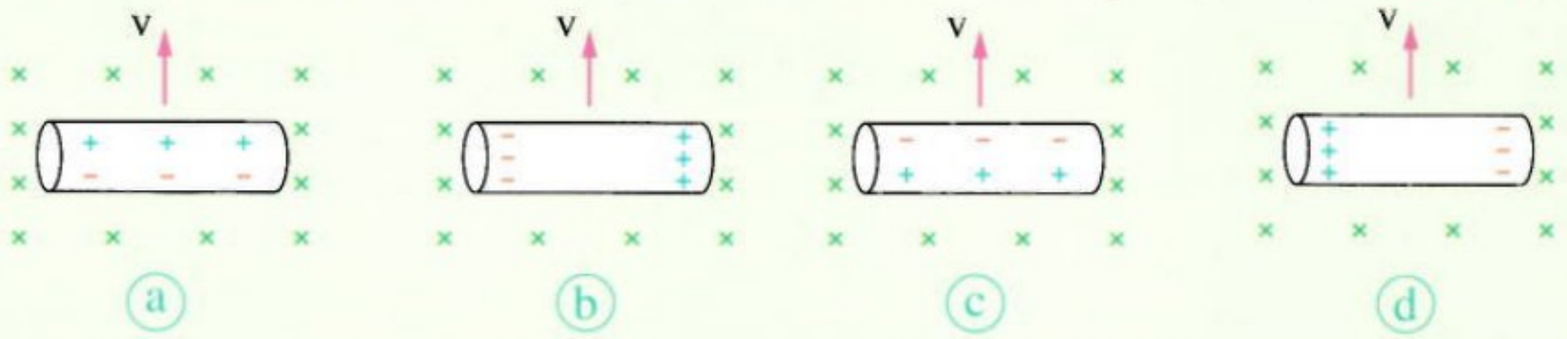
(d) If the wire is moved towards point "B", the potential of the point "C" will be equal to that of "D".

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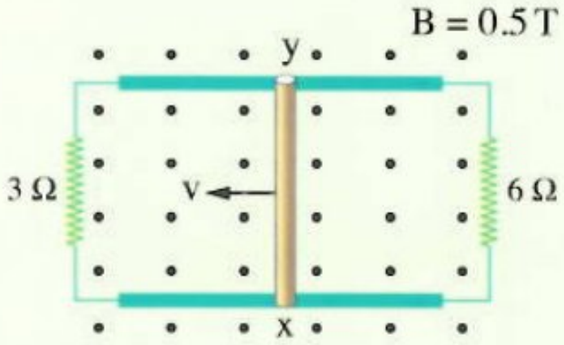
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72 A metallic wire moves with constant velocity (v) upwards in the plane of the page and is affected by a magnetic field perpendicular into the plane of the page. Which figure illustrates the drift of the electric charges inside the wire during its motion? *(2nd session 23)*



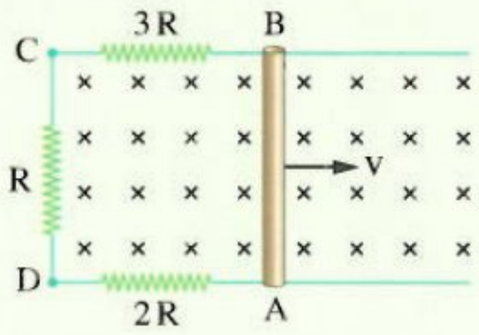
73 A metallic straight wire (yx) of length 0.2 m whose electric resistance is 1 Ω moves at velocity of 3 m/s perpendicular to the direction of magnetic field of flux density 0.5 T on two rails of negligible resistance that are connected with two resistors 6 Ω, 3 Ω as shown in the figure. Then, the potential difference across the resistor (3 Ω) during moving the wire equals ..... .



(2nd session 23)

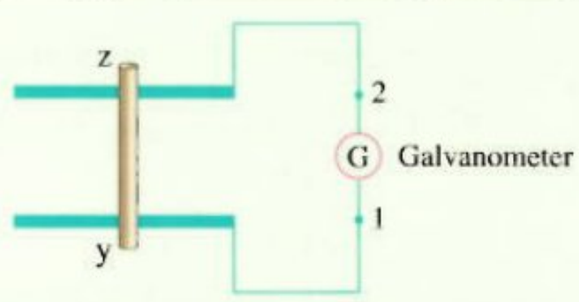
(a) 0.2 V (b) 0.3 V (c) 0.1 V (d) 0.4 V

74 The opposite figure shows a conductor AB which is moved rightwards perpendicular to a uniform magnetic flux, so during the motion of the conductor (AB) which of the following statements is correct? *(1st session 24)*



(a) Potential (C) equals (D).
(b) Potential (A) equals (B).
(c) Potential (C) less than (D).
(d) Potential (C) greater than (D).

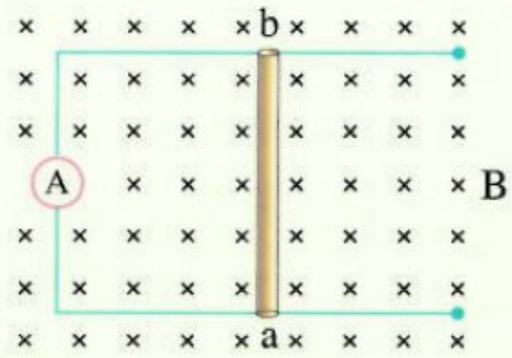
75 The opposite figure shows a movable wire yz that is affected by a uniform magnetic flux, so which of the following directions allows the flow of the current from point 1 to point 2? *(1st session 24)*



	Direction of the wire's motion	Direction of the magnetic flux
(a)	Leftwards	Perpendicular to the plane of the page outwards
(b)	Rightwards	Perpendicular to the plane of the page outwards
(c)	Rightwards	In the plane of the page leftwards
(d)	Leftwards	In the plane of the page rightwards

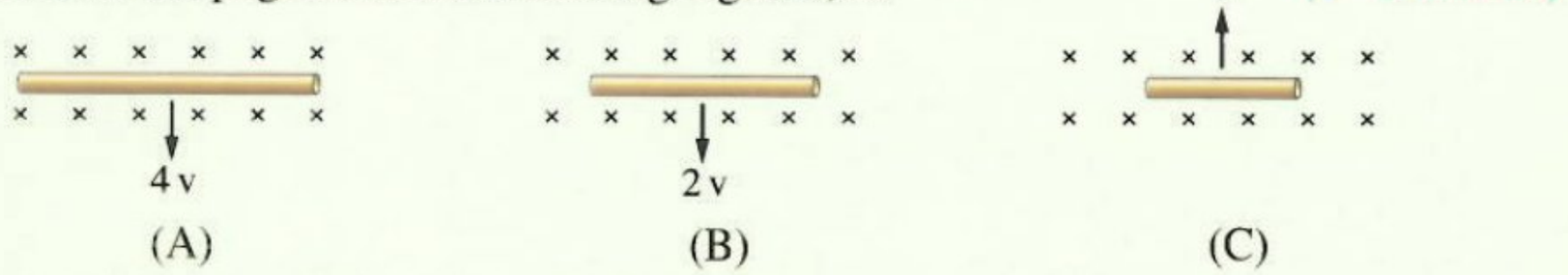
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76 The opposite figure represents a metal wire (ab) which moves perpendicular to a uniform magnetic field of flux density (B) generating in the wire induced electric current such that potential of point (a) is greater than that of point (b), so the direction of motion of the wire is ..... . *(2nd session 24)*



(a) to the left of the page
(b) to the right of the page
(c) upwards of the page
(d) downwards of the page

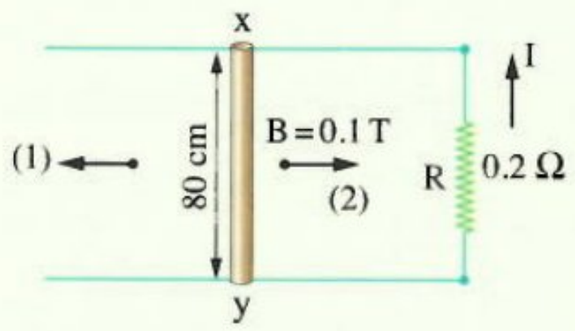
77 Three straight wires A, B, C of lengths ℓ, 1.5 ℓ, 2 ℓ respectively are moving with velocities v, 2 v, 4 v respectively, perpendicularly to a uniform magnetic flux of density (B) and its direction is into the page as in the following figures, so ..... . *(2nd session 24)*



Which of the following choices is correct?

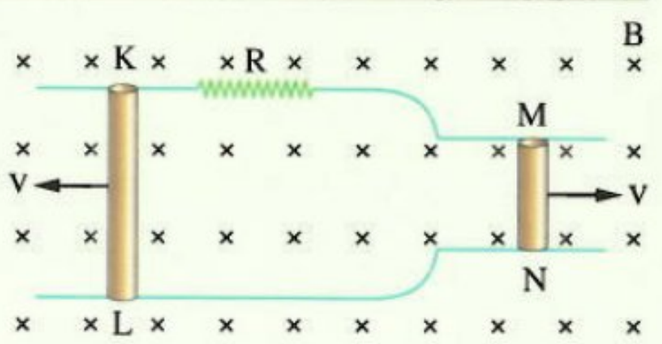
(a) $emf_{(C)} > emf_{(B)}$ (b) $emf_{(A)} > emf_{(C)}$ (c) $emf_{(B)} > emf_{(A)}$ (d) $emf_{(C)} > emf_{(A)}$

78 Depending on the data in the opposite figure and by ignoring the resistance of each of the conductor xy and the metal rails, on which the conductor xy is sliding, the two conditions for generating induced current of 2 A in the resistor R in the shown direction are ..... .
(Supposing that the whole figure is completely immersed in the magnetic field and its direction is into the page) *(Practice 25)*



	Direction of motion of the conductor	The uniform speed of the conductor
(a)	Direction 1	5 m/s
(b)	Direction 2	10 m/s
(c)	Direction 1	10 m/s
(d)	Direction 2	5 m/s

79 Two copper conductors KL, MN having the same cross-section and negligible resistance, length of MN is ℓ and that of KL is 2ℓ. The conductors are moving on a metal frame of negligible resistance with a velocity v in the direction shown in the opposite figure, so the intensity of the induced current which passes in the resistor R equals ..... . *(Practice 25)*

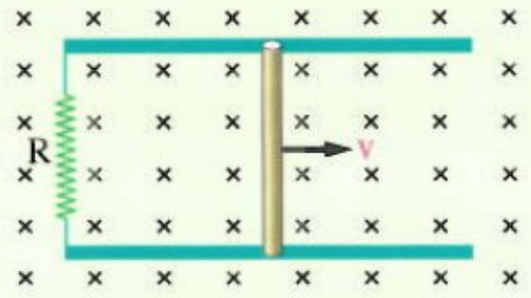


(a) $\frac{3\,B\ell v}{R}$ (b) $\frac{B\ell v}{R}$ (c) $\frac{2\,B\ell v}{R}$ (d) $\frac{4\,B\ell v}{R}$

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80 A straight metal rod of resistance (r) moves in contact with a frame of negligible resistance at a constant velocity (v) perpendicular to a uniform magnetic flux of density (B) in the direction shown in the figure.



Which of the following actions will increase the potential difference between the two terminals of the resistor (R)? *(1st session 25)*

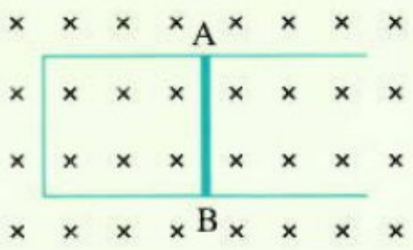
(a) Increasing the angle between the direction of the rod motion and the direction of the magnetic flux by 30°

(b) Moving the rod in a direction opposite to the direction of the magnetic flux.

(c) Moving the rod toward the resistor (R) at a constant velocity (v).

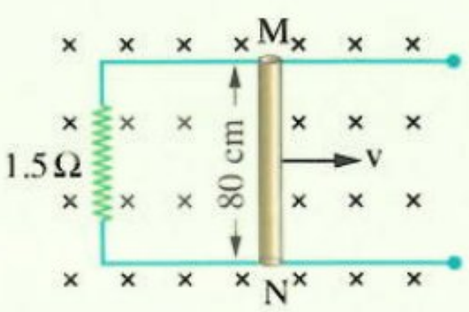
(d) Moving the rod at a speed greater than (v).

81 The figure shows a metallic rod (AB) of length 0.15 m sliding along a metallic frame of resistance (R). If the rod is moving perpendicular to a uniform magnetic flux of density 0.4 T, an emf of 0.03 V is induced between its terminals and a current flows through the rod from terminal (B) to terminal (A).
The magnitude and the direction of the uniform velocity of the rod motion are ..... . *(2nd session 25)*



(a) 0.5 m/s, to the left of the page
(b) 0.5 m/s, to the right of the page
(c) 2 m/s, to the right of the page
(d) 2 m/s, to the left of the page

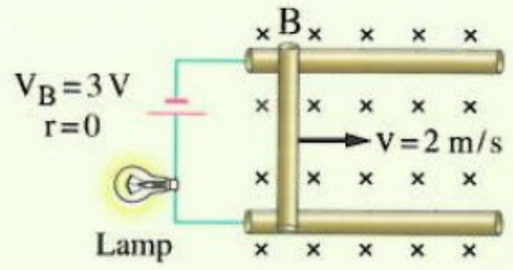
82 A metal conductor (MN) with a length of 80 cm and a resistance of 0.5 Ω moves in the direction shown in the figure at a uniform speed (v) perpendicular to a uniform magnetic field with a flux density of 0.5 T, and if the magnetic force acting on the conductor during its motion is 0.16 N, then the correct choice that correctly represents the value of the induced electromotive force (emf) generated in the conductor and the direction of the induced current in the circuit is ..... . *(Practice 26)*



	The induced electromotive force	The direction of the induced current
(a)	0.6 V	Clockwise
(b)	0.8 V	Clockwise
(c)	0.6 V	Anticlockwise
(d)	0.8 V	Anticlockwise

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83 The figure shows a two frictionless metal rods of negligible resistance, connected to an electric lamp of resistance of 3 Ω and a battery of 3 V and of negligible internal resistance. If a conductor of 40 cm long has a resistance of 1 Ω, moves at a constant speed of 2 m/s in the direction shown perpendicular to a uniform magnetic field of flux density B, it is noticed that the consumed electric power by the lamp during the conductor motion is 1.92 W, then the value of B is .....



(Practice 26)

(a) 0.2 T (b) 0.25 T (c) 0.3 T (d) 0.35 T

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